Ref. Ares(2021)4352011 - 05/07/2021

AMENDMENT Reference No AMD-790157-23

Grant Agreement number: 790157 — Integrated cascades of PROcesses for the extraction and valorisation of proteins and bioactive molecules from Legumes, Fungi and Coffee agro-industrial side streams (Prolific)

The parties agree to amend the Grant Agreement as follows ('**Amendment**'):

1 . Addition of a linked third party

The following new linked third party is added:

- ASSOCIAZIONE NAZIONALE COOPERATIVEDI PRODUZIONE LAVORO E SERVIZI (AN-CPLS), affiliated or linked to INNOVA — as from 1 September 2018

This implies the **following changes** to the Grant Agreement:

- **Article 14** becomes applicable.

  The new linked third party and the 'starting date of participation' are added to **Article 14.1**:

  "ASSOCIAZIONE NAZIONALE COOPERATIVEDI PRODUZIONE LAVORO E SERVIZI (AN-CPLS), affiliated or linked to INNOVA — as from 1 September 2018"

  The other options for linked third parties:

  - are added (**Preamble**, **Articles 4.1**, **5.2**, **5.3.1**, **5.3.3**, **6.4**, **17.2**, **18.1.2**, **20.3**, **20.4**, **20.6**, **21.3.1**, **41.2**, **44.1**, **44.1.1(b)**, **44.1.2**, **44.1.2(b)(i)**, **44.1.3**, **44.1.3(b)(i)**, **48.2**, **50.2.2**, **50.3.1**, **50.3.3**, **56.1**, **Annex 3a**) and

  - become applicable (**Article 6.3, 50.3.1(n)**).

2. Change of Annex 1 (description of the action)

Annex 1 is changed and replaced by the Annex 1 attached to this Amendment.

3 . Changes of Annex 2 (estimated budget)

Annex 2 is changed and replaced by the Annex 2 attached to this Amendment.

All other provisions of the Grant Agreement and its Annexes remain unchanged.

This Amendment **enters into force** on the day of the last signature.

This Amendment **takes effect** on the date on which the amendment enters into force, except where a different date has been agreed by the parties (for one or more changes).

Please inform the other members of the consortium of the Amendment.

SIGNATURES

For the coordinator

For the JU

Enclosures:

Annex 2
Annex 1

ANNEX 1 (part A)

Bio-based Industries Research and Innovation action

NUMBER — 790157 — Prolific

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1.1. The project summary

Project Number [1]	790157	Project Acronym [2]	Prolific

One form per project

General information	
Project title [3]	Integrated cascades of PROcesses for the extraction and valorisation of proteins and bioactive molecules from Legumes, Fungi and Coffee agro-industrial side streams
Starting date [4]	01/09/2018
Duration in months [5]	48
Call (part) identifier [6]	H2020-BBI-JTI-2017
Topic	BBI.2017.R4 Proteins and other bioactive ingredients from side streams and residues
Fixed EC Keywords	Bioproducts (products that are manufactured using biological material as feedstock) biomaterials, bioplastics, biofuels, bioderived bulk and fine chemicals, bio-derived novel materials
Free keywords	Legumes, fungi, coffee, prototypes, extraction, enzymatic hydrolysis, bio-active, ingredients

Abstract [7]

Agro-industrial residual biomass, side streams and food production by-products like legumes, fungi and coffee are potential sources of valuable ingredients even though the routes for their exploitation are still at an early stage. Pursuing the ambition of achieving prolific valorization of untapped biomass streams, the project R%D&I activities and partners of Prolific have been positioned around a central innovation cycle that is mainly driven by industrial end-users who exactly know the needs of their customers and the technical constraints and industrial environment in their respective sector.
The PROLIFIC project will apply a range of processing technologies to recover significant amounts of proteins/ peptides and other value added compounds (e.g. carotenoids, phenols, caffeine and fibers) from industrial processing residues of legumes (seeds of peas, beans and chickpea), fungi (cuttings and mycelia of different species) and coffee (silver skin residue and not compliant roasted seeds). The economically and environmentally sustainable extraction, enzymatic modification, and conditioning techniques will be upscaled in industrially relevant environment. This will enable the production of consequent amounts of compounds and fractions necessary for the production of 16 product prototypes for the food, feed, packaging and cosmetic sectors.
The Prolific project will assess the environmental, societal, ethical, safety, and regulatory issues at each step of the targeted value chains. SME and large companies strive at achieving competitive biomass exploitation and attracting when needed additional investors. The industrial partners business strategies for the future market penetration starts with the clear identification of the customer needs and goes up to the preparation of give-aways of food and cosmetics prototyped products as a reward for filling in questionnaires about the Prolific products along with intensive dissemination and communication.

1.2. List of Beneficiaries

Project Number [1]	790157	Project Acronym [2]	Prolific

List of Beneficiaries

No	Name	Short name	Country	Project entry date[8]	Project exit date
1	FACHHOCHSCHULE NORDWESTSCHWEIZ	FHNW	Switzerland		
2	ALMA MATER STUDIORUM - UNIVERSITA DI BOLOGNA	UNIBO	Italy		
3	IRIS TECHNOLOGY SOLUTIONS, SOCIEDAD LIMITADA	IRIS	Spain		
4	UNIVERSITA DEGLI STUDI DI PARMA	UNIPR	Italy		
5	STAZIONE SPERIMENTALE PER L'INDUSTRIA DELLE CONSERVE ALIMENTARI	SSICA	Italy		
6	CELABOR SCRL	CELAB	Belgium		
7	IGV INSTITUT FUR GETREIDEVERARBEITUNG GMBH	IGV	Germany		
8	STOLZENBERGER REINER ERICH	SB	Germany		
9	BIO BASE EUROPE PILOT PLANT VZW	BBEPP	Belgium		
10	CONSERVES FRANCE	CONS	France		
11	RTD TALOS LIMITED	TALOS	Cyprus		
12	TSATSOS GEORGIOS	COSM	Greece		
13	ILLYCAFFE S.P.A	ILLY	Italy		
14	NUTRITION SCIENCES	NS	Belgium		
15	PLEURETTE	PLEUR	France		
16	FEMTO ENGINEERING SRL	FTEN	Italy		
17	INNOVACOOP SRL	INNOVA	Italy		
18	UNIVERSITA DI PISA	UNIPI	Italy	01/06/2020	

1.3. Workplan Tables - Detailed implementation

1.3.1. WT1 List of work packages

WP Number[9]	WP Title	Lead beneficiary[10]	[illegible]	Start month[12]	End month[13]
WP1	Inventory and characterization of the initial feedstocks	2 - UNIBO	[illegible]	1	7
WP2	Flexible and scalable protocols for feedstock pre-treatment and extraction of the fractions of interest	4 - UNIPR	[illegible]	3	21
WP3	Conversion of fractions and first steps of extracts valorisation	2 - UNIBO	[illegible]	6	36
WP4	Evaluation and selection of the best performing isolated fractions according to foreseen applications	6 - CELAB	[illegible]	13	46
WP5	Industrial scale up of the processes and final product prototyping	7 - IGV	[illegible]	13	46
WP6	Life-Cycle Assessment (LCA) and Cost-Effectiveness Analysis (CEA)	1 - FHNW	[illegible]	1	46
WP7	Dissemination, exploitation and commercialisation	11 - TALOS	[illegible]	1	48
WP8	Project coordination and management	1 - FHNW	[illegible]	1	48
WP9	Ethics requirements	1 - FHNW	[illegible]	1	48
		Total	[illegible]		

1.3.2. WT2 list of deliverables

Deliverable Number[14]	Deliverable Title	WP number[9]	Lead beneficiary	Type[15]	Dissemination level[16]	Due Date (in months)[17]
D1.1	Inventory and preliminary assessment on availability and sustainability of feedstocks	WP1	2 - UNIBO	Report	Confidential, only for members of the consortium (including the Commission Services)	5
D1.2	Report on technical and regulatory specifications of foreseen final extracts	WP1	7 - IGV	Report	Confidential, only for members of the consortium (including the Commission Services)	7
D1.3	Review(s) on the technical and commercial state of the art of selected residues and foreseen final extracts	WP1	2 - UNIBO	Report	Public	7
D1.4	Results on biochemical profile and safety characteristics of the selected residues	WP1	5 - SSICA	Report	Confidential, only for members of the consortium (including the Commission Services)	6
D2.1	Report on the pre-treatment of raw materials and protein extraction from selected feedstocks	WP2	3 - IRIS	Report	Confidential, only for members of the consortium (including the Commission Services)	18
D2.2	Characterisation of protein fractions as related to the extraction technologies applied	WP2	4 - UNIPR	Report	Confidential, only for members of the consortium (including the Commission Services)	18
D2.3	Extraction of other valuable non protein compounds from solid fibre-rich fractions	WP2	6 - CELAB	Report	Confidential, only for members of the consortium (including the Commission Services)	21
D3.1	Results of concentration and further fractionation of extracts	WP3	7 - IGV	Report	Confidential, only for members of the consortium (including the Commission Services)	28

Deliverable Number[14]	Deliverable Title	WP number[9]	Lead beneficiary	Type[15]	Dissemination level[16]	Due Date (in months)[17]
D3.2	Results of biochemical/ biocatalytic conversion of protein extracts	WP3	4 - UNIPR	Report	Confidential, only for members of the consortium (including the Commission Services)	28
D3.3	Results on extract stabilization	WP3	7 - IGV	Report	Confidential, only for members of the consortium (including the Commission Services)	36
D3.4	Results of polymer formulation for packaging	WP3	2 - UNIBO	Report	Confidential, only for members of the consortium (including the Commission Services)	36
D4.1	Results on safety compliance of the extracted/converted fractions	WP4	6 - CELAB	Report	Confidential, only for members of the consortium (including the Commission Services)	34
D4.2	Results on fractions tested for food applications	WP4	7 - IGV	Report	Confidential, only for members of the consortium (including the Commission Services)	36
D4.3	Results on fractions tested for cosmetic applications	WP4	12 - COSM	Report	Confidential, only for members of the consortium (including the Commission Services)	36
D4.4	Results on formulations tested for packaging applications (polymers and composites)	WP4	18 - UNIPI	Report	Confidential, only for members of the consortium (including the Commission Services)	42
D4.5	Results on fractions tested for feed applications	WP4	14 - NS	Report	Confidential, only for members of the consortium (including the Commission Services)	36
D5.1	Dataset of validated pre-production	WP5	9 - BBEPP	Report	Confidential, only for members	38

Deliverable Number[14]	Deliverable Title	WP number[9]	Lead beneficiary	Type[15]	Dissemination level[16]	Due Date (in months)[17]
	prototypes and best technological procedures scaled-up				of the consortium (including the Commission Services)	
D5.2	Eight food product prototypes	WP5	7 - IGV	Demonstrator	Confidential, only for members of the consortium (including the Commission Services)	46
D5.3	Two cosmetic product prototypes	WP5	12 - COSM	Demonstrator	Confidential, only for members of the consortium (including the Commission Services)	46
D5.4	Four packaging prototypes	WP5	18 - UNIPI	Demonstrator	Confidential, only for members of the consortium (including the Commission Services)	46
D5.5	Two feed product prototypes	WP5	14 - NS	Demonstrator	Confidential, only for members of the consortium (including the Commission Services)	46
D6.1	Intermediate LCA/ CEA results for final products	WP6	1 - FHNW	Report	Confidential, only for members of the consortium (including the Commission Services)	18
D6.2	Report on product regulations related to final products	WP6	7 - IGV	Report	Confidential, only for members of the consortium (including the Commission Services)	40
D6.3	Final LCA/CEA results for final products	WP6	1 - FHNW	Report	Confidential, only for members of the consortium (including the Commission Services)	46
D7.1	Dissemination and communication strategy and action plan	WP7	11 - TALOS	Report	Confidential, only for members of the consortium (including the	5

Deliverable Number[14]	Deliverable Title	WP number[9]	Lead beneficiary	Type[15]	Dissemination level[16]	Due Date (in months)[17]
					Commission Services)	
D7.2	Website publicly available	WP7	11 - TALOS	Other	Public	6
D7.3	Data Management Plan	WP7	1 - FHNW	Report	Confidential, only for members of the consortium (including the Commission Services)	6
D7.4	Interim report on performed dissemination, communication and networking activities	WP7	11 - TALOS	Report	Public	24
D7.5	Final report on performed dissemination, communication and networking activities	WP7	11 - TALOS	Report	Public	48
D7.6	Draft plan for the exploitation of results	WP7	17 - INNOVA	Report	Confidential, only for members of the consortium (including the Commission Services)	24
D7.7	Final plan for the exploitation of results including Exploitation Agreement among the partners	WP7	17 - INNOVA	Report	Confidential, only for members of the consortium (including the Commission Services)	48
D7.8	Preliminary business plan for the product ingredients and prototypes	WP7	11 - TALOS	Report	Confidential, only for members of the consortium (including the Commission Services)	48
D8.1	Minutes of the kick-off meeting	WP8	1 - FHNW	Report	Confidential, only for members of the consortium (including the Commission Services)	2
D8.2	Management Report M30	WP8	1 - FHNW	Report	Confidential, only for members of the consortium (including the	30

Deliverable Number[14]	Deliverable Title	WP number[9]	Lead beneficiary	Type[15]	Dissemination level[16]	Due Date (in months)[17]
					Commission Services)	
D8.3	Management Report M42	WP8	1 - FHNW	Report	Confidential, only for members of the consortium (including the Commission Services)	42
D8.4	KPI & Impact Questionnaire	WP8	2 - UNIBO	Report	Confidential, only for members of the consortium (including the Commission Services)	37
D8.5	Final KPI & Impact Questionnaire	WP8	2 - UNIBO	Report	Confidential, only for members of the consortium (including the Commission Services)	48
D9.1	H - Requirement No. 1	WP9	1 - FHNW	Ethics	Confidential, only for members of the consortium (including the Commission Services)	1
D9.2	HCT - Requirement No. 2	WP9	1 - FHNW	Ethics	Confidential, only for members of the consortium (including the Commission Services)	1
D9.3	A - Requirement No. 3	WP9	1 - FHNW	Ethics	Confidential, only for members of the consortium (including the Commission Services)	1
D9.4	EPQ - Requirement No. 4	WP9	1 - FHNW	Ethics	Confidential, only for members of the consortium (including the Commission Services)	1

1.3.3. WT3 Work package descriptions

Work package number [9]	WP1	Lead beneficiary [10]	2 - UNIBO
Work package title	Inventory and characterization of the initial feedstocks		
Start month	1	End month	7

Objectives

- Map the availability and sustainability of the selected feedstocks
- Produce an updated state-of-the art review on existing processing technologies and commercial products
- Determine the foreseen technical and regulatory specifications of the final extracts
- Determine the molecular and safety characteristics of the feedstocks
- Provide process data for the elaboration of the LCA and CEA performed in WP6

Description of work and role of partners

WP1 - Inventory and characterization of the initial feedstocks [Months: 1-7]
UNIBO, FHNW, UNIPR, SSICA, IGV, SB, CONS, COSM, ILLY, NS, PLEUR, FTEN
Description of work
Leader: Annalisa Tassoni (UNIBO, Italy), Co-leader: Georgios Tsatsos (COSM, Greece)

Task 1.1. Inventory, availability and sustainability of selected feedstocks (Month 1-5)
Leader: UNIBO. Partners involved: IGV, CONS, ILLY, PLEUR
The selected residues can have an extremely variable composition resulting from e.g. seasonal fluctuations, different farming practices and post-harvest processing technologies across EU. Therefore the first step will be to map the selected agro-industrial residues across Europe in terms of amounts, geographical provenience, regional distribution, seasonality, quality variation, current residue handling practices, economic data. In particular, these data will be collated (UNIBO) and made available by CONS for legume residues (non-compliant fresh pea seeds and rehydrated beans and chickpeas seeds), by PLEUR (processing cuttings, left overs and mycelium of Agaricus bisporus, Pleurotus ostreatus and Lentinus edodes) and by ILLY for coffee residues (silver skin and not compliant roasted coffee seeds). A preliminary assessment of logistic and sustainability aspects (amounts of the residues distributed, quality check, stability during storage and transport, cost of shipment) and of the availability of the feedstocks during and after the project will be provided. Data for LCA and CEA (WP6) will be provided in agreement with partners (IGV).

Task 1.2. State of the art on processing technology and valorisation review. Specification requirements on target compounds by producers/end-users and compliance with existing regulations (Months 1-7)
Leader: IGV. Partners involved: UNIBO, SSICA, COSM, SB, NS, FEMTO, ILLY
An up-to-date review of the literature and patents on existing processing technologies for the selected residues, will be carried out. All relevant research and recent developments will be taken into account for the project implementation. An in depth survey (at EU and world level) of available commercial extracts/ingredients derived from the 3 selected feedstocks, will be also performed. The latter will include a market survey and benchmarking of existing and in development state solutions. The results will be summarized in review paper(s) to be published in high reputation open access scientific journals (IGV, UNIBO). Producers and End-users (IGV, SSICA, COSM, SB, NS, FEMTO, ILLY) will list the physico-chemical and biochemical specifications of the target extracted compounds in order to comply with specific final products technical requirements. The limits for specific substances such as heavy metals, aflatoxins, pesticides etc. for each specific application will be set. Producers and End-users will also advise regarding the compliance of the foreseen extraction/conversion processes (WP2 and WP3) and of the final products (WP5) with existing regulation (e.g. REACH, Novel Foods) and customer needs. An inventory of legal aspects related to the possible use of target compounds will be prepared. Current bottlenecks will be highlighted and “easy” short-term goals and “harder” long-term legal issues will be identified (desktop study) (IGV).

Task 1.3. Molecular and safety characterization of initial feedstocks (Month 1-6)
Leader: SSICA. Partners involved: FHNW, UNIBO, UNIPR, IGV

Subtask 1.3.1. Molecular characterization
Representative samples of each type of residue will be collected for the determination of the biochemical composition in terms of dry residue, proteins, fibres, sugars and lipids (SSICA, UNIBO, UNIPR). Detailed analysis of protein

content of the different streams will be evaluated by Kjeldahl method and total amino acid analysis by LC. The results will be compared with those of the starting material in order to determine the degree of amino acid extraction from the stream. Protein integrity will be determined by measuring free amino groups by o-phtaldialdehyde test and calculating the degree of hydrolysis. Chiral amino acid analysis will be performed in order to evaluate protein damage by assessing the racemization induced by thermal treatments, extreme pH or fermentation. The amount of D- amino acid contained will be the molecular marker for assessing protein integrity. Intact proteins, if present, will be analysed by SDS-PAGE and MS of the most prominent bands (UNIPR). Lipid analyses will be carried out both with a soxhlet extraction using hexane will and also with an Accelerated Solvent Extractor using a mix of solvent (hexane and acetone); sugars will be analysed using a GC/MS method which allows the quantification of the dissolved non-cellulosic neutral sugars and the sugars obtained by acid hydrolysis (IGV). Soluble and insoluble fibres will be analysed by means of conventional methods (UNIBO). Total polyphenols, flavonoids, flavanols, tannins and alkaloids will be analysed by means of spectrophotometric and HPLC-DAD analyses (UNIBO).

Subtask 1.3.2. Safety characterization
The different selected residues will be analysed in order to assess their safety degree before being processed in the WP2 pipelines. In particular, toxins, biogenic amines, pesticides and other contaminants, heavy metals/metalloids, microbial contaminations, will be carefully investigated. Aflatoxins (G1, G2, B1, B2) and Patulin will be analysed by HPLC (SSICA). Ochratoxins, thricothecens (in legumes) will be determined with specific immunofluorometer, HPLC, LC/MS methods (SSICA). Up to 16 aromatic hydrocarbons polycyclic (PAH) (among which benzo[a]anthracene (BaA), benzo[a]pyrene (BaP), benzo[b]fluoranthene (BbF) and chrysene (CHR)) by GC/MS. Biogenic amines (monoamines, diamines and polyamines of plant and/or microbial origin) in free and conjugated forms will be analysed by HPLC fluorometer (UNIBO). The agro-chemical residues will be determined by chromatographic methods. In particular, common pesticides, fungicides and herbicides will be determined by LC and by GC methods (IGV). Metals and metalloids will be determined by state-of-the-art analytics regarding total element concentrations using qqq-ICP-MS. The presence of heavy metals, such as lead, cadmium, nickel, chromium, arsenic and mercury, will be determined using ICP-MS after an acidic digestion of raw materials (IGV, FHNW). In case of sufficient concentrations, metal/metalloids can be analysed species specifically by LC-qqq-ICP-MS to distinguish potentially more toxic from species of less concern (e.g. non-toxic arsenobetaine vs. trivalent arsenic) (FHNW). The presence of pathogenic microorganisms such as Salmonella sp., Listeria monocytogenes, Escherichia coli O157:H7, Clostridium botulinum, will be investigated by means of both traditional and molecular techniques (SSICA).

Participation per Partner

Partner number and short name	WP1 effort
1 - FHNW	
2 - UNIBO	
4 - UNIPR	
5 - SSICA	
7 - IGV	
8 - SB	
10 - CONS	
12 - COSM	
13 - ILLY	
14 - NS	
15 - PLEUR	
16 - FTEN	
Total	

List of deliverables

Deliverable Number[14]	Deliverable Title	Lead beneficiary	Type[15]	Dissemination level[16]	Due Date (in months)[17]
D1.1	Inventory and preliminary assessment on availability and sustainability of feedstocks	2 - UNIBO	Report	Confidential, only for members of the consortium (including the Commission Services)	5
D1.2	Report on technical and regulatory specifications of foreseen final extracts	7 - IGV	Report	Confidential, only for members of the consortium (including the Commission Services)	7
D1.3	Review(s) on the technical and commercial state of the art of selected residues and foreseen final extracts	2 - UNIBO	Report	Public	7
D1.4	Results on biochemical profile and safety characteristics of the selected residues	5 - SSICA	Report	Confidential, only for members of the consortium (including the Commission Services)	6

Description of deliverables

D1.1 Inventory and preliminary assessment on availability and sustainability of feedstocks (Lead: UNIBO, participants: IGV, CONS, ILLY, PLEUR). (R, CO, Month 5)
D1.2 Report on technical and regulatory specifications of foreseen final extracts (Lead: IGV, participants: UNIBO, SSICA, SB, COSM, NS, FEMTO, ILLY). (R, CO, Month 7)
D1.3 Review(s) on the technical and commercial state of the art of selected residues and foreseen final extracts (Lead: UNIBO, participants: IGV, SSICA, SB, COSM, NS, FEMTO, ILLY). (R, PU, Month 7)
D1.4 Results on biochemical profile and safety characteristics of the selected residues (Lead: SSICA, participants: FHNW, UNIBO, UNIPR, IGV). (R, CO, Month 6)

D1.1 : Inventory and preliminary assessment on availability and sustainability of feedstocks [5]

Inventory and preliminary assessment on availability and sustainability of feedstocks

D1.2 : Report on technical and regulatory specifications of foreseen final extracts [7]

Report on technical and regulatory specifications of foreseen final extracts

D1.3 : Review(s) on the technical and commercial state of the art of selected residues and foreseen final extracts [7]

Review(s) on the technical and commercial state of the art of selected residues and foreseen final extracts

D1.4 : Results on biochemical profile and safety characteristics of the selected residues [6]

Results on biochemical profile and safety characteristics of the selected residues

Schedule of relevant Milestones

Milestone number[18]	Milestone title	Lead beneficiary	Due Date (in months)	Means of verification
MS1	Assessed biochemical and safety characteristics, availability and exploitability of the selected residues also on the basis of the updated state of the art review(s)	2 - UNIBO	7	Assessed biochemical and safety characteristics, availability and exploitability of the selected residues also on the basis of the updated state of the art review(s)

Work package number [9]	WP2	Lead beneficiary [10]	4 - UNIPR
Work package title	Flexible and scalable protocols for feedstock pre-treatment and extraction of the fractions of interest		
Start month	3	End month	21

Objectives

Starting from the results obtained in WP1 the objectives of WP2 are to:
● Set up and optimise flexible and scalable extraction protocols for proteins and other bioactive compounds
● Improve protein stability and availability for extraction by means of pre-treatment of the agri-food residues
● Recover and fractionate the protein content of the raw materials
● Characterize at molecular and functional level the protein fractions recovered
● Recover other valuable compounds (cellulose/hemicellulose, chitosan, -glucans, polyphenols, carotenoids, phytosterols, caffeine) still available in the raw material after protein extraction.
● Provide process data for the elaboration of the LCA and CEA performed in WP6

Description of work and role of partners

WP2 - Flexible and scalable protocols for feedstock pre-treatment and extraction of the fractions of interest
[Months: 3-21]
UNIPR, IRIS, SSICA, CELAB, IGV, BBEPP
Description of work
Leader: Tullia Tedeschi (UNIPR, Italy), Co-Leader: Simona Neri (IRIS, Spain)

In this WP flexible pre-treatment and extraction protocols will be set up and optimised in order to isolate proteins and bioactive compounds from selected agro-industrial residues in a cascading approach. Only fully scalable technologies (to be scaled up in WP5) will be applied.

Task 2.1. Pre-treatment, protein extraction, fractionation and characterization (Months 3-18)
Leader: UNIPR. Partners involved: IRIS, SSICA, BBEPP
Subtask 2.1.1. Pre-treatment of the raw material
Prior to protein extraction, legumes by-products, i.e. peels, pods, broken grains, grains with different size or colours, seeds etc. will be pre-treated by washing, inert materials (residues of leaves, soil, small stones, etc.) removal and milling (SSICA). Dry mechanical fractionation and sieving system will be employed to obtain concentrated protein-rich fractions from coffee and mushroom residues (IGV).
Subtask 2.1.2. Protein extraction
Four different approaches will be applied to extract proteins, i.e. environmentally friendly aqueous extraction, ultrasound-assisted extraction, microwave-assisted extraction and enzyme-assisted extraction. All approaches will be evaluated for their efficiency, sustainability and possibility to scale up (BBEPP).
Pre-treated legumes by-products will be mixed and homogenized in suitable buffer and, after a first separation step, a solid fibres-rich fraction and a liquid proteins-rich fraction will be obtained. The fibres-rich fraction will be freeze-dried, milled and sieved in order to obtain a fine powder (to be used for subsequent compounds extraction in Task 2.2). Proteins will be isolated from the liquid proteins-rich fraction by isoelectric precipitation and subsequent centrifugation. The final protein extract will be stabilized by freeze-drying or other suitable techniques (WP3, Task 3.3). The same procedure will also be performed at pilot scale (WP5) with suitable equipments and minor modifications (SSICA). Ultrasound-assisted extraction (UAE) will be applied in order to improve the protein extraction efficiency in the case of recalcitrant substrates such as coffee and fungi residues. Lab scale trials (up to 4.0 L) will be performed varying sonication time, output power and pulse mode in order to increase performances of protein extraction. The selected reaction conditions will be used for upscaling the intensification process using a dynamic sonication system capable of treating 1.0 L/min (IRIS). Microwave-assisted extraction will also be evaluated for improving the protein extractability of the selected raw materials applying a response surface methodology (RSM). Parameters such as extraction time, type of solvent (acid/alkali/water) and solid to liquid ratio will be optimised. Previous pre-treatment of the material involving grinding and sieving will also be considered (IRIS).
Proteic fraction of legume, fungi and coffee residues will be extracted by enzyme-assisted extraction using specific proteases (alcalase, trypsin, pepsin, papain, or combination of them) finally yielding peptides. This will actually allow achieving a two-fold target: first proteins recalcitrant to solubilisation will be extracted in form of peptides, and second

anti-nutritional proteic material (trypsin inhibitor and lectins) will be degraded. The proteases will also enable to tailor the degree of hydrolysis of the extracts and to enrich the extracts with the desired proteins and amino acids, according to enzymes specificity (UNIPR).

Subtask 2.1.3. General and detailed molecular composition of the protein fractions
In order to verify the purity degree of the fractions isolated from legumes by-products, their chemical composition will be determined in terms of water content, proteins percentage and fibres percentage (SSICA). Protein fractions will be characterized in-depth using SDS-PAGE and MS, in order to identify the major proteins extracted. Total and free amino acid analysis and racemization degree of amino acids will be performed by means of LC and GC techniques in order to evaluate the protein integrity after extraction. These analyses are also useful to assess the nutritional value of the fraction. o-phtaldialdehyde analysis (OPA) will also be used to evaluate the hydrolysis degree, particularly in hydrolyses resulting from protease treatment. Hydrolysates will also be characterized in terms of peptides using LC/MS. The potential beneficial (in terms of bioactive peptides) or negative (allergens) health effects will be evaluated in silico by database search on the identified proteins and peptides, and also considering the degree of hydrolysis. Simulated gastrointestinal digestion will be performed on the extracted fraction in order to assess digestibility in relation with the applied extraction procedure. A specific focus will be devoted to the analysis of anti-nutrients in the fractions, and more specifically lectins and trypsin inhibitors, also in relation with the extraction processes applied (UNIPR).

Task 2.2. Recovery of valuable non protein compounds from residual solid fibre-rich fractions (Months 9-21)
Leader: CELAB. Partners involved: IRIS, UNIPR, BBEPP
Subtask 2.2.1. Polysaccharide fractions
Fibres (cellulose/hemicellulose)
Extraction of hemicellulose and cellulose will be performed from legumes and coffee from fibrous-rich milled solid residues after protein extraction (Task 2.2.) using conventional process (acidic hot extraction, alkaline hot extraction) as reference methods. A special effort will be devoted to the development and the optimization of subcritical water extraction applied to lignocellulosic materials to extract hemicellulose and cellulose, as an innovative, cost-effective and eco-friendly process. Subcritical water will be used to hydrolyse lignocellulosic material without adding chemical reagents. The starting fibrous-rich milled solid residues, containing lignocellulosic material (Subtask 2.2.1), will be firstly characterized to determine their particle size and density which represent key parameters for the development of subcritical water process. If needed, the extraction of hemicellulose and cellulose will also be performed by subcritical Accelerated Solvent Extraction (ASE) at a laboratory scale and using small amounts of ethanol (CELAB). Extraction parameters will be studied (temperature, pressure, time, pH and liquid-to-solid ratio, ethanol ratio) and optimized in terms of total extraction yields and composition (purity) of the desired carbohydrate polymers with the help of statistical design of experiments. Extraction pilot-plants and reactors will be used to upscale the processes from gram-scale to kilogram-scale in batch systems (CELAB). Data for LCA and CEA will also be collected and the inputs will be delivered to WP6 and feedbacks from regulatory compliance studies will be used to further tailor the processes. All approaches will be evaluated for their efficiency, sustainability and possibility to scale up (BBEPP).
Chitin/chitosan and ß-glucans
Conventional extraction and separation methods (alkaline hot water extraction) will be used to extract ß-glucan and chitin/chitosan from mushroom fibrous-rich milled solid residues (Task 2.2) after protein extraction. A special focus will be devoted to the development of innovative and green techniques, such as enzymatic pre-treatment and subcritical water extraction. Both ß-glucan and chitin/chitosan fractions will be dried (freeze-drying or spray-drying) and characterized by Fourier Transform Infrared Spectroscopy (FTIR) and Ultra-Performance Liquid Chromatography coupled to mass spectrometry (UPLC-MS). The glucosamine content in the chitin/chitosan fraction will be determined by UPLC-MS technique after hydrolysis (CELAB). The conversion of chitin to chitosan through deacetylation process will be developed and optimized by controlling reaction parameters (temperature, duration, alkaline concentration). Subcritical water will also be used for this purpose by applying high temperature and high pressure. To determine the quality and then the possible use of the obtained chitosan (e.g. cosmetic, polymer) it will be characterized by FTIR for structure profiling and UPLC-MS for glucosamine content (CELAB). The final acetylation degree will be determined by Nuclear Magnetic Resonance (UNIPR). Gel Permeation Chromatography coupled to mass spectrometry (GPC-MS) will be used for the determination of the average molecular weight of the obtained chitosan (UNIPR) Extraction pilot-plants and reactors will be used to upscale the processes from gram- to kilogram-scale per batch (CELAB). Data for LCA and CEA will also be collected and the inputs will be delivered to WP6 and feedbacks from regulatory compliance studies will be used to further tailor the processes. All methods will be evaluated for their efficiency, sustainability and possibility to scale up (BBEPP).

Subtask 2.2.2. Extraction of other added-value compounds
Other compounds (carotenoids, polyphenols, phytosterols, caffeine) having valuable activities (e.g. antioxidant and/or anti-microbial) and remaining in the fibre rich solid-residues (Subtask 2.2.1) after protein extractions will be recovered.

This additional extraction process will start with characterizing the particle size of the powder which represent a key parameter for the development of extraction process and also to evaluate the content in high valuable molecules. Extraction will be optimized at bench scale using the following parameters: type and concentration of solvents, liquid-solid ratio, temperature and time of extraction. Various processes will be investigated and improved in terms of (1) recovery percentage, (2) productivity of the extraction processes (litre or kg of starting material treated per hour) and (3) selectivity of the extraction operation for specific target compounds. Conventional extraction (maceration) will be compared to green extraction (sub-critical water, supercritical CO2) and assisted extraction techniques (microwave and ultrasounds). Only solvent complying with the regulations of cosmetic, food and feed sectors will be used (CELAB). In addition, different samples of solid-residues, resulting from the different protein extraction processes (Task 2.2) will also be sent to IRIS facility in order to perform UAE of bioactive compounds, including caffeine. The initial tests will be performed at bench scale in a static system up to 4.0 L capacity. This system will be used to assess the residual content of active molecules of the raw material and evaluate how the protein extraction processes affect their maximum recovery yield. Subsequently, the setup of the UAE will be scaled up into a dynamic system capable of processing up to 1.0 L/min. The flow cell design, the power output, the number of transducers, the piping arrangement, will be tailored to the targeted compounds and source of biomass residues (IRIS). Detailed biochemical characterization of extracted compounds (polyphenols, carotenoids, phytosterols, caffeine) will be performed by chromatographic techniques (GC-MS, GC-FID, UPLC-MS-DAD) and using spectrophotometric techniques (CELAB). Data for LCA and CEA will also be collected and the inputs will be delivered to WP6 and feedbacks from regulatory compliance studies will be used to further tailor the processes. All methods will be evaluated for their efficiency, sustainability and possibility to scale up (BBEPP).

Subtask 2.2.3. Final residue analysis and valorisation
The final residues coming out from Task 2.2 extraction procedures will characterised for their general residual chemical composition (residual fibres and other compounds) (CELAB, SSICA) and used for composites formulation (WP3, Subtask 3.4.2) and prototype production (WP5, Task 5.4) so to have a zero-waste biorefinery extraction process.

Participation per Partner

Partner number and short name	WP2 effort
3 - IRIS	
4 - UNIPR	
5 - SSICA	
6 - CELAB	
7 - IGV	
9 - BBEPP	
Total	

List of deliverables

Deliverable Number[14]	Deliverable Title	Lead beneficiary	Type[15]	Dissemination level[16]	Due Date (in months)[17]
D2.1	Report on the pre-treatment of raw materials and protein extraction from selected feedstocks	3 - IRIS	Report	Confidential, only for members of the consortium (including the Commission Services)	18
D2.2	Characterisation of protein fractions as	4 - UNIPR	Report	Confidential, only for members of the consortium (including	18

List of deliverables

Deliverable Number[14]	Deliverable Title	Lead beneficiary	Type[15]	Dissemination level[16]	Due Date (in months)[17]
	related to the extraction technologies applied			the Commission Services)	
D2.3	Extraction of other valuable non protein compounds from solid fibre-rich fractions	6 - CELAB	Report	Confidential, only for members of the consortium (including the Commission Services)	21

Description of deliverables

D2.1 : Report on the pre-treatment of raw materials and protein extraction from selected feedstocks [18]
Report on the pre-treatment of raw materials and protein extraction from selected feedstocks

D2.2 : Characterisation of protein fractions as related to the extraction technologies applied [18]
Characterisation of protein fractions as related to the extraction technologies applied

D2.3 : Extraction of other valuable non protein compounds from solid fibre-rich fractions [21]
Extraction of other valuable non protein compounds from solid fibre-rich fractions

Schedule of relevant Milestones

Milestone number[18]	Milestone title	Lead beneficiary	Due Date (in months)	Means of verification
MS2	Recovery of protein content from the selected feedstocks	4 - UNIPR	15	Recovery of protein content from the selected feedstocks
MS3	Recovery of non-protein compounds from solid fibre-rich fractions	6 - CELAB	18	Results of extraction trials documented in D2.3., datasets shared with consortium in meeting presentations and in the PROLIFIC intranet

Work package number [9]	WP3	Lead beneficiary [10]	2 - UNIBO
Work package title	Conversion of fractions and first steps of extracts valorisation		
Start month	6	End month	36

Objectives

● Set up procedures for converting the extracts from biomass streams of WP2 into valuable ingredients and components (fillers, reinforcements) tailored to their targeted final applications in food and cosmetic packaging.
● Provide data for the elaboration of the LCA and CEA performed in WP6.

Description of work and role of partners

WP3 - Conversion of fractions and first steps of extracts valorisation [Months: 6-36]
UNIBO, FHNW, UNIPR, SSICA, IGV, BBEPP, COSM, FTEN, UNIPI
Description of work
Leader: Annamaria Celli (UNIBO, Italy), Co-leader: Giuseppe Amato (FTEN)
All approaches will be evaluated for their efficiency, sustainability and possibility to scale up (BBEPP).
Task 3.1. Concentration and further fractionation of extracts (Months 6-36)
Leader: IGV.
In order to improve final yields, extracts of WP2, after gentle drying, will be tested for further fractionation to ensure a maximized valorisation of these extracts. In order to maximise the concentration of proteins, polysaccharide extracts (cellulose/hemicellulose, chitosan, and glucans) and other bioactive molecules (carotenoids, polyphenols, phytosterols, caffeine), a combination of dry-mechanical fractionation methods (air-sifting, high rotational fractionation, sieving system) will be tested to obtain the desired yields of final extracts. Purity analysis of the products obtained will be performed and an optimal process design will be established for large-scale production of the targeted pure fractions. If deemed necessary, an additional combination of enzyme-assisted dry and wet processing, will be tested to produce the desired purified protein and fibre food ingredients. The developed protein ingredients will be partly used in product prototypes (WP5) as such or after being further converted in the Task 3.2, to obtain fractions enriched in peptides and amino acids. All other extracted fractions (fibrous and other added value compounds) will be used as such in the product prototyping (WP5).

Task 3.2. Biochemical/biocatalytic conversion of protein extracts (Months 6-28)
Leader: FHNW. Partners involved: UNIPR, BBEPP
Subtask 3.2.1. Tailoring protein extracts to peptide and modified peptide mixtures
Protein rich extracts will be treated with selected proteolytic enzymes cocktails, possibly followed by ultrafiltration, in order to obtain tailored hydrolysates containing peptides having specific length (longer vs. shorter peptides) and specific amino acid composition. Some hydrolysates will be further tailored by reaction with fatty acid derivatives in order to obtain long chain acyl peptides and acyl amino acids for specific technological functionalities such as emulsifiers of surfactants (e.g. cosmetic applications). All the obtained mixtures will be deeply characterized by LC/ESI-MS (UNIPR).

Subtask 3.2.2. Immobilized and nano-protected proteases hydrolytic treatment of protein fractions
The protein-rich extracts will be submitted to hydrolysis by immobilized proteases selected jointly by UNIPR and FHNW. Spherical nanoparticles of silica (Stöber synthesis) or (3-aminopropyl)triethoxy silane-coated magnetic particles will be used as immobilization support. The selected protease(s) and combination thereof will be covalently linked to these supports using glutaraldehyde. A few nanometre enzyme protection layer will be grown around the immobilized proteases by adding self assembling mixture of various organosilanes. The protected enzymes will be applied in MBR system (continuous) or magnetic reactor (semi-continuous) to produce protein hydrolysates free of enzymes. The application of these nanobiocatalysts will mainly be dedicated to applications requesting enzyme-free extracts. All the obtained mixtures will be deeply characterized by LC/ESI-MS. Larger batches of nanobiocatalysts will be subcontracted to a Contract Manufacturing Organisation that is acquainted with the synthesis process for large-scale experiments of protein hydrolysis to be carried out in WP5.

Subtask. 3.2.3. Characterization of converted peptides/amino acids fractions
The protein hydrolysates obtained Subtasks 3.2.1 and 3.2.2 will be characterized in terms of total nitrogen content, total and free amino acid composition, peptide composition, degree of hydrolysis, using the same methods reported in WP2. Protein digestibility will also be assessed by performing simulated gastrointestinal digestion following published

and validated consensus protocols, which use simulated gastrointestinal fluids with proper enzymes. The degradation of anti-nutritional proteins (lectins and trypsin inhibitors) eventually carried over in the protein fraction will also be checked (UNIPR).

Task 3.3. Stabilisation of extracts (Months 12-36)
Leader IGV. Partners involved: UNIBO, COSM, FTEN, BBEPP
Stabilisation and preservation of the functional properties of the compounds during the different extraction steps will be performed by working for instance under oxygen-free conditions (to prevent oxidation of antioxidants like polyphenols) and/or at lower temperatures (to prevent denaturation of proteins), and by the use of membrane- based separation techniques. The microbiological preservation options will be examined (e.g. solvent, water content, use of environmentally friendly preservatives, etc.). The common practices in use and regulatory limitations for each sector will be taken into account after advise of end users and/or producers (IGV, COSM, FTEN, BBEPP). The limited stability and/or solubility, often combined with low initial concentrations require an innovative recovery strategy to answer growing demands in cosmetics, nutrition, health and material science. Encapsulation techniques and spray-drying method coupled with an innovative co-extrusion technique will be applied to extracts/ingredients. In the field of plastic materials, the possibility of protecting molecules with antioxidant properties during processing at high temperatures through their encapsulation in inorganic structures, such as layered double hydroxide nanoclays (LDHs), will be investigated (UNIBO).

Task 3.4. Utilization of extracted bioactive fractions and extraction residues in polymer formulation for packaging (Months 15-36)
Leader: UNIPI. Partners involved: UNIBO, SSICA, FTEN
Subtask 3.4.1. Preparation of improved formulations of films for food coating (FTEN, UNIPI, SSICA)
UNIPI in cooperation with SSICA will define a formula based on chitosan (extracted in Subtask 2.2.1), solvent (acetic acid) and biobased plasticizers (glycerol, gallic acid, etc) for the preparation of a thin layer to be coated on PLA films (by a Zehntner ZUA 2000.200 coater). FTEN has already identified PLA films provider (Planet bioplastics, etc.), or will produce specifically PLA films. UNIPI will give Particular attention to set proper conditions for coating (solution density, temperature, etc.) and to chitosan concentration in order to obtain homogeneous coating with improved barrier properties and anti-fungal and anti-mites properties when used in packaging with direct contact to food.
Moreover, SSICA will evaluate appropriate modifications of the developed formula based on chitosan to be used for the application on food product by means of dipping technique. Principal coating parameters, will be considered, chitosan concentration, immersion time and drying temperature, in order to obtain a uniform coating on the surface of the food with antifungal properties.

Subtask 3.4.2. Preparation of improved polymeric materials by addition of bioactive molecules (UNIBO, FTEN, UNIPI)
Bioactive molecules, such as polyphenols, caffeine, phytosterols and carotenoids, recovered in Subtask 2.2.2, will be added to commercial biopolymers (mainly PLA and PHAs) in order to assign specific characteristics, such as antioxidant properties. First tests will be performed using a Haake MiniJet II mini injection molder composed of a mini extruder equipped with MiniLab Injection Moulding Press for bar specimens (size: 25x5x1.5 mm) (UNIPI). Then the selected formulation will be tested in pilot scale using a Comac co-rotating twin-screw extruder in the amount of 25 Kilos per batch (UNIPI). Some tests will be carried out using bioactive molecule encapsulated in inorganic nanostructures (layered double hydroxides, LDHs, produced in Task 3.3), that can protect molecules and avoid their migration (UNIBO). The formulation of the polymeric materials will be characterized in WP4 (Task 4.4) and optimised in WP5 (Subtasks 5.4.2 and 5.4.3) for the preparation of cosmetic packaging (jars) and coffee capsules.

Subtask 3.4.3. Exploitation of fibrous biomass residual from extract production in composites. (UNIBO, FTEN, UNIPI)
The residues, obtained after extraction procedures in WP2 (Task 2.2) and final residues will be mixed with a PLA, PHA and/or PBS matrix at small scale in a Brabender Plasticorder mixer (50 grams), in order to verify the improvement in polymer performances, e.g. mechanical properties (UNIBO). UNIPI will cooperate to the task by producing blends in Haake mini lab extruder, and producing specimen by lab scale injection moulding for test of mechanical properties by tensile tests. A characterization of the materials will be carried out in WP4 (Task 4.4) in order to study the possible use of these materials in the production of a rigid packaging for cosmetic (WP5, Subtask 5.4.2) (FTEN).

Data for LCA and CEA will also be collected and the inputs will be delivered to WP6 and used to further tailor the processes.

Partner number and short name	WP3 effort
1 - FHNW	
2 - UNIBO	
4 - UNIPR	
5 - SSICA	
7 - IGV	
9 - BBEPP	
12 - COSM	
16 - FTEN	
18 - UNIPI	
Total	

List of deliverables

Deliverable Number[14]	Deliverable Title	Lead beneficiary	Type[15]	Dissemination level[16]	Due Date (in months)[17]
D3.1	Results of concentration and further fractionation of extracts	7 - IGV	Report	Confidential, only for members of the consortium (including the Commission Services)	28
D3.2	Results of biochemical/ biocatalytic conversion of protein extracts	4 - UNIPR	Report	Confidential, only for members of the consortium (including the Commission Services)	28
D3.3	Results on extract stabilization	7 - IGV	Report	Confidential, only for members of the consortium (including the Commission Services)	36
D3.4	Results of polymer formulation for packaging	2 - UNIBO	Report	Confidential, only for members of the consortium (including the Commission Services)	36

Description of deliverables

D3.1 : Results of concentration and further fractionation of extracts [28]

Results of concentration and further fractionation of extracts

D3.2 : Results of biochemical/ biocatalytic conversion of protein extracts [28]

Results of biochemical/ biocatalytic conversion of protein extracts

D3.3 : Results on extract stabilization [36]

Results on extract stabilization

D3.4 : Results of polymer formulation for packaging [36]
Results of polymer formulation for packaging

Schedule of relevant Milestones

Milestone number[18]	Milestone title	Lead beneficiary	Due Date (in months)	Means of verification
MS4	Achieved protocol for biochemical/biocatalytic conversion of protein extracts to enhance properties related to the target applications	4 - UNIPR	18	Achieved protocol for biochemical/biocatalytic conversion of protein extracts to enhance properties related to the target applications
MS5	Selection of 2 soluble proteases and 2 nanobiocatalysts for the effective hydrolysis of proteins	4 - UNIPR	18	Experiments with a set of enzymes finalised and results assessed in terms of yield
MS6	Achieved protocol for polymer formulation for packaging	2 - UNIBO	36	Result of protocol development and application presented on meeting and shared in intranet, documented in D3.4

Work package number [9]	WP4	Lead beneficiary [10]	6 - CELAB
Work package title	Evaluation and selection of the best performing isolated fractions according to foreseen applications		
Start month	13	End month	46

Objectives

● Assessment of safety compliance of the WP2 and WP3 final extracted/converted fractions
● Assessment of the biochemical and functional characteristics and properties of final extracted/converted fractions (WP2 and WP3) in order to identify the most promising fractions for the applications foreseen by end-users (WP5). The fraction selection criteria will be a favourable safety and techno-economic analysis, a positive LCA and CEA, and/or an improved functionality compared to the benchmark product currently in the market.
● Data for LCA and CEA evaluation in WP6 will be provided.

Description of work and role of partners

WP4 - Evaluation and selection of the best performing isolated fractions according to foreseen applications
[Months: 13-46]
CELAB, FHNW, UNIBO, UNIPR, SSICA, IGV, SB, COSM, NS, UNIPI
Description of work
Leader: Job Tchoumtchoua (CELAB, Belgium), Co-leader: Chiara Zurlini (SSICA, Italy)

Task 4.1. Safety compliance of the extracts (Months 13-34)
Leader: CELABOR. Partners involved: FHNW, UNIBO, IGV, UNIPR, SSICA, NS
All the analyses concerning the safety assessment (mycotoxins, pathogens, heavy metals, biogenic amines, pesticides, food allergens, etc.) performed on initial feedstocks (Subtask 1.3.2) will be repeated on fractions extracted (WP2) and/or further converted (WP3) in case risk was detected in the raw material (FHNW, UNIBO, UNIPR, SSICA, IGV, CELAB, NS). This verification is necessary in order to certify the full usability of the extracts as ingredients in the foreseen food, cosmetic, packaging, feed products prototypes (WP5). In addition, specifically for cosmetic applications, the full compliance with specific regulations (e.g. REACH) will be assessed such as residual solvents, residues of digestion enzymes (IGV). The cosmetic efficiency of the fractions will be tested by in vitro and/or in vivo tests allowing us to validate their activity mechanisms and to justify their guidelines for dermo-cosmetic cares. In the manufacture of different traditional products and cosmetic hygiene about 7,000 different substances are used. Some of them are highly toxic (and even recognized carcinogens), other can trigger allergic reactions or are irritating and most of its effects by accumulation and its interaction with other products are not known. The products obtained from PROLIFIC will represent suitable replacements for some of these chemicals, therefore the positive impact will be important. Allergenicity depending on dose, sensitization tests and skin irritation will be investigated for the compounds to be used in cosmetic formulations (IGV).

Task 4.2. Fraction selection for food (Months 13-36)
Leader IGV. Partners involved: UNIBO, CELAB, SSICA, SB
The extracted/converted compounds and fractions (WP2 and WP3) will be screened for composition (e.g. minerals, protein, starch, fibre, lipids, polysaccharides, polyphenols), anti-nutritional factors (e.g. trypsin inhibitor) and functionality (e.g. pasting properties, water absorption, oil absorption, emulsification). Once the initial screening is performed, a rigorous selection will follow to ensure the next steps will meet the needs of the food industry.

Subtask 4.2.1. Specific analysis for cereal-based and vegan and vegetarian food products (IGV, SB)
Wheat protein is unique in its ability to form a viscoelastic dough suitable for making bread and other bakery products. Gluten, the major storage protein of wheat, is a heterogeneous mixture of the proteins gliadins and glutenins. These proteins have a unique amino acid composition that makes possible the formation of viscoelastic dough, which can entrap carbon dioxide gas during yeast fermentation. Bread-making quality of wheat varieties will be tested by measuring dough strength after mixing wheat flour with water and then kneading the dough. The effect of other dough ingredients (proteins/peptides from legumes, carotenoids, phytosterols, polyphenols, etc.) on viscoelastic properties will be tested in the same way. Progressive rheological analysis for evaluation of prepared whole grain flour enriched with ingredients to be selected (using Mixolab) will be applied for all mixtures (we will use rye and spelt mixtures rich in dietary fibre, vitamins and minerals, enriched with ingredient and fractions to be selected). The set of technological analyses will include test weight (TW), content of crude protein (CP), content of starch (S), falling number (FN) and material

hardness based on particle size index (PSI) assessment. Content of wet gluten (WG) and gluten index (GI), including sedimentation test (SDS) will be assessed. The different fractions will be used for subsequent mini-baking analyses (bread, pastries) based on prepared flours of individual fraction samples. Mutual correlations and regressions among the technological parameters and characteristics of both intermediate and final products will be calculated.

Subtask 4.2.2. Specific analysis for meat products (SSICA, UNIBO, CELAB)
Preservative activity (antioxidant, antifungal) of the extracts (in particular fibre, chitosan and -glucans, polyphenols and carotenoids from Task 2.2) will be tested using numerous in vitro tests. Several spectrophotometric tests will be used to measure the total antioxidant capacity: 1). Hydrogen atom transfer (HAT) tests, such as the oxygen radical absorbance capacity (ORAC), apply a competitive reaction scheme in which antioxidants and substrates compete for thermally generated peroxyl radicals (UNIBO, CELAB); 2). Tests based on electron transfer (ET) that measure the ability of an antioxidant to reduce an oxidant, which changes colour when reduced. These include the 2,2-diphenyl-1-picrylhydrazyl (DPPH), 2,2'-azino-bis(3-ethylbenzothiazoline-6 -sulphonic acid) (ABTS), superoxide dismutase (SOD) and ferric reducing antioxidant power (FRAP) methods. DPPH will also be carried out on TLC plates for lipophilic extracts (e.g. carotenoids, Task 2.2) that cannot be dissolved in aqueous solution (spectrophotometric method) (UNIBO, CELAB). Phytochemical extracts will undergo tests of Fe-chelating activity to complete the antioxidant activity profile specific to meat products (SSICA). The chitosan extracts will be tested for their in vitro antifungal and anti-mites activities (SSICA). The selected extracts will be subjected to Comet assay test to measure the potential DNA damage in terms of mutagenic effect on human intestinal cells (SSICA).

Task 4.3. Fraction selection for cosmetics (Months 13-36)
Leader COSM. Partners involved: UNIBO, CELAB

Subtask 4.3.1. Fractions with emulsifying properties
These will be acyl-derivatised peptide fractions (Subtask 3.2.1). The derivatised fractions will be characterised in terms of Hydrophilic Lipophilic Balance (HLB) and Characteristic value (Cc), (HLD-NAC theory). Emulsions with different type of oils (e.g. polar/non-polar) and at different acyl-derivatised peptides/oils ratios, will be examined by optical microscopy, centrifugation, heat-cold cycles and Lumifuge tests in order to evaluate the suitability of the fractions to be used for cosmetic emulsions (COSM).

Subtask 4.3.2. Anti-microbial properties of chitosan
Chitosan (Task 2.2) will be assessed for use in toothpaste formulation. The minimum inhibitory concentration (MIC) test (COSM) and/or the bio-autographic method on TLC (CELAB) will be used to test the fraction activity against common mouth pathogens such as Enterococcus, Candida and Staphylococcus species. Based on the tests, the partner COSM will select the concentration of chitosan for the formulation of a prototype toothpaste produced in WP5.

Subtask 4.3.3. Film forming properties of chitosan and β-glucan
Chitosan and β-glucan capacities of film former, moisturising and soothing will be assessed in various cream formulations (COSM). The potential use of chitosan and β-glucan fractions in sunscreen products to improve sensorial aspects and water-resistance, will be examined. A preliminary formulation will be produced and sensorial parameters such as greasiness, spreading, tackiness etc. will be evaluated by a panel test (COSM). The water resistance will be measured against a benchmark waterproofing agent (e.g. ANTARON V-220, VP/EICOSENE COPOLYMER) and a control without any waterproofing agent.

Subtask 4.3.4. Biological activity of the oil-soluble fractions
The biological activity of the oil soluble fraction of polyphenol/carotenoid/phytosterol fractions (Subtask 2.2.2) in terms of anti-aging activity (anti-elastase and anti-collagenase), skin whitening (anti-tyrosinase) will be assessed for cosmetic formulations.
Anti-ageing activity (anti-elastase and anti-collagenase).
The anti-ageing assay is relevant for skin products formulation in cosmetic application. Spectrophotometric tests will be carried out to determine elastase and collagenase inhibition by polyphenols (CELAB, UNIBO). The first step of the screening will consist in the test of the extracts produced in WP2 and WP3. At this concentration, an inhibition higher than 50% must be reached in order to classify the extracts as having an anti-aging potential. In a second step, several concentrations will be tested to determine the concentration at which 50% of the enzyme is inhibited (IC50). The lower IC50 value is obtained, the higher the extract is effective.
Skin whitening (anti-tyrosinase).
Tyrosinase catalyses the initial step in the formation of the pigment melanin from tyrosine but also catalyses the oxidation of polyphenols in plant extracts. Spectrophotometric and TLC methods will be used (CELAB, UNIBO) to screen the extracts produced in WP2 and WP3 containing a polyphenol concentration of 500 µg/mL. At this concentration, an inhibition of tyrosinase > 50% must be reached in order to classify the extracts having an anti-aging potential. In a

second step, several concentrations will be tested to determine the concentration at which 50% of the enzyme is inhibited (IC50). The lower IC50 value is obtained, the higher the extract is effective.

Task 4.4. Selection of formulations of polymers and composites for packaging (Months 24-42)
Leader UNIPI. Partners involved: UNIBO, SSICA, CELAB, FTEN
Polymer formulations, prepared in WP3 (Task 3.4) based on a commercial biopolymer matrix and containing bioactive molecules (polyphenols, phytosterols, carotenoids, caffeine) as additives, polysaccharides (chitosan) as a coating, extraction residues, as reinforcing agents, will be characterized. Investigations focus on properties relevant for the targeted applications in WP5 (food and cosmetic packaging), before being tested successively in WP5 (Task 5.4) for the production of final prototypes:
• a coating for foodstuff and a film for food packaging (Subtask 5.4.1),
• a rigid jar for cosmetic packaging (Subtask 5.4.2) and
• coffee capsules (Subtask 5.4.3).
Materials produced in WP5 will also be tested in this task in order to eventually improve the final formulations. Therefore, a flux of information produced in this task is exchanged between WP3 and WP5 in order to optimize the production of prototypes.
In particular, for coating of dry-cured meat and flexible biodegradable packaging for raw and fermented meat products (Subtask 5.4.1) thermal and mechanical properties, barrier properties (oxygen and water vapour permeability), migration, anti-fungal and anti-mites properties will be analysed (SSICA, UNIBO, FTEN, UNIPI, CELAB).
For cosmetic packaging (Subtask 5.4.2), thermal and mechanical properties, barrier properties (oxygen and water vapour permeability), migration, antioxidant properties, will be analysed (UNIBO, FTEN, UNIPI, CELAB).
For coffee capsules production (Subtask 5.4.3), thermal and mechanical properties, barrier properties, migration, antioxidant properties, will be analysed (UNIBO, FTEN, UNIPI, SSICA, CELAB).

Task 4.5. Fraction selection for feed (Months 13-42)
Leader: NS. Partners involved: UNIBO, CELAB
Subtask 4.5.1. Trials on feed formulation
Target compounds (e.g. proteins/peptides, fibres, carotenoids, polyphenols) with potential for feed applications will be evaluated by NS. In this task at least 6 fractions will be tested. The target compounds (100 g) will be characterised for nutritional and bioactivity values (antioxidant, anti-microbial, anti-agglutination, anti-inflammatory) according to standardised methods (NS, UNIBO, CELABOR, see previous Tasks for type of assays). Feedback on the results will be given to previous WPs for improvements of the process. Next, the compounds will be integrated in new feed formulations (functional feed ingredients). These new functional feed ingredients will be formulated in animal feed compositions mainly for monogastric animals (in casu pigs and poultry). Formulation will be based on linear programming, by using WinMix™ software (NS). Animal feed will be formulated as iso-nutritional and iso-caloric diets, in order to test the net effect of the project outcomes. For poultry, animal diets will be formulated according requirements defined by CentraalVoederBureau (The Netherlands) which are internationally accepted. The aim is to formulate feed products that may (1) reduce raw material depletion and (2) decrease use of antibiotics. Larger quantities of selected feed formulations (with input of primary compounds) will be produced in WP5.

Subtask 4.5.2. Application tests
To get insight into the potential of target compounds as feed ingredients, gastrointestinal simulation models for pigs and poultry will be performed. This task simulates the gastrointestinal tract of monogastric animals – in casu pigs and poultry, the two target livestock species for the project -and consist of a sequential stomach, small intestine and large intestine simulation. It was developed within the Eureka E!2452 project, and this model was validated by 2 independent parties by cross-checking results collected from monogastric livestock animals. This model needs larger amounts of substances (500 g) and one gastrointestinal simulation run takes 5 working days (NS).

Data for LCA and CEA will also be collected and the inputs will be delivered to WP6 and used to further tailor the processes.

Participation per Partner

Partner number and short name	WP4 effort
1 - FHNW	
2 - UNIBO	

Partner number and short name	WP4 effort
4 - UNIPR	
5 - SSICA	
6 - CELAB	
7 - IGV	
8 - SB	
12 - COSM	
14 - NS	
18 - UNIPI	
Total	

List of deliverables

Deliverable Number[14]	Deliverable Title	Lead beneficiary	Type[15]	Dissemination level[16]	Due Date (in months)[17]
D4.1	Results on safety compliance of the extracted/converted fractions	6 - CELAB	Report	Confidential, only for members of the consortium (including the Commission Services)	34
D4.2	Results on fractions tested for food applications	7 - IGV	Report	Confidential, only for members of the consortium (including the Commission Services)	36
D4.3	Results on fractions tested for cosmetic applications	12 - COSM	Report	Confidential, only for members of the consortium (including the Commission Services)	36
D4.4	Results on formulations tested for packaging applications (polymers and composites)	18 - UNIPI	Report	Confidential, only for members of the consortium (including the Commission Services)	42
D4.5	Results on fractions tested for feed applications	14 - NS	Report	Confidential, only for members of the consortium (including the Commission Services)	36

Description of deliverables

D4.1 : Results on safety compliance of the extracted/converted fractions [34]

Results on safety compliance of the extracted/converted fractions

D4.2 : Results on fractions tested for food applications [36]

Results on fractions tested for food applications

D4.3 : Results on fractions tested for cosmetic applications [36]

Results on fractions tested for cosmetic applications

D4.4 : Results on formulations tested for packaging applications (polymers and composites) [42]

Results on formulations tested for packaging applications (polymers and composites)

D4.5 : Results on fractions tested for feed applications [36]

Results on fractions tested for feed applications

Schedule of relevant Milestones

Milestone number[18]	**Milestone title**	**Lead beneficiary**	**Due Date (in months)**	**Means of verification**
MS7	Selection of fractions for food applications	7 - IGV	32	Assessment and ranking of fraction according to defined criteria and benchmarks presented and discussed with partners at monthly or biannual meeting
MS8	Selection of fractions for cosmetic applications	12 - COSM	34	Assessment and ranking of fraction according to defined criteria and benchmarks presented and discussed with partners at monthly or biannual meeting
MS9	Selection of fractions for polymer and composite formulations for packaging applications	18 - UNIPI	40	Assessment and ranking of fraction according to defined criteria and benchmarks presented and discussed with partners at monthly or biannual meeting
MS10	Selection of fractions for feed applications	14 - NS	34	Selection of fractions for feed applications

Work package number [9]	WP5	Lead beneficiary [10]	7 - IGV
Work package title	Industrial scale up of the processes and final product prototyping		
Start month	13	End month	46

Objectives

● Scale up of processes for pre-treatment, fractionation, (bio-)chemical modification, and product purification and formulation of previously selected bioactive fractions
● Technological transfer of novel food and non-food uses and quality and safety assessments of the final products
● Food (cereal-based, vegetarian, vegan, meat), cosmetic, packaging (films for foods, cosmetic packaging, coffee capsules), feed product prototypes formulated by including the most promising new extracted/converted fractions as ingredients/additives will be developed, based on the inputs from WP2, WP3 and WP4.
● Delivery of process data to be used as input for refined LCA and CEA analysis in WP6.

Description of work and role of partners

WP5 - Industrial scale up of the processes and final product prototyping [Months: 13-46]
IGV, FHNW, UNIBO, IRIS, SSICA, CELAB, SB, BBEPP, CONS, COSM, ILLY, NS, UNIPI
Description of work
Leader: Dennis Schaller (IGV, Germany), Co-leader: Jeroen Van Malderen (BBEPP, Belgium)

Task 5.1. Set up of scale up processes for the production of selected fractions (Months 13-38)
Leader BBEPP. Partners involved: CELAB, IGV, FHNW, UNIBO, SSICA, IRIS, CONS
Product prototypes formulated by including the most promising new extracted/converted fractions will be developed, based on the inputs from WP2, WP3 and WP4. All the processes will be scaled up, starting with agro-residue pre-treatment, fractionation, (bio-)chemical modification/conversion, product purification and formulation. All process techniques chosen in PROLIFIC are scalable to industrial levels, either by upscaling or outscaling. Scale-up will be performed at BBEPP (freeze and air-oven drying followed by grinding; acid, alkaline, enzymatic treatment and solvent extraction, sub-critical water extraction depending on the conditions, downstream processing through centrifugation, filtration, evaporation, flocculation, etc.; chemical modification), CELAB (ultrasound assisted extraction and supercritical CO2 extraction) and SSICA for legumes by-products treatment. FHNW will scale up the proteolysis with protected nanobiocatalysts. The progression of the processes will be monitored by analysing extracts taken at regular intervals with the previously utilised methodologies (WP2, WP3, WP4) (CELAB, IGV, FHNW, UNIBO, SSICA, IRIS). The scale-up volume will depend on the target fractions (proteins/peptides, polysaccharides, polyphenols, carotenoids, phytosterols and caffeine) and on the nature of the used technology (classical technique: maceration tanks , open columns, open extraction tanks; or pressurized techniques: subcritical water extractor, supercritical CO2 extractors). Protein and polysaccharide fractions (cellulose/hemicellulose, chitin/chitosan, -glucans), that will be used as the main ingredients in the final formulation, will be produced by using successive batches of a minimum 40-50 kg of pre-treated feedstock and will be processed using classical technologies. In the case of secondary metabolites (polyphenols, carotenoids, phytosterols, caffeine) that are present in low percentages in the feedstocks and that will be used as additives at low concentration in the final formulation, successive batches of 5-6 L/batch (0.5 kg to 2 kg/batch depending on the density of the material) will be carried out in order to produce sufficient quantity of target enriched-fractions (200-500 g) for further formulations and applications. These promising biomass/fractions/ingredients resulting from the scale up processes will be evaluated in suitable final product applications (food, cosmetic, packaging and feed sectors), while investigators will remain in contact with end users throughout the trials to ensure targets are being met in terms of functionality demands and final product quality. The aim is obtain extracts with high protein contents and increased amounts of essential amino acids. Furthermore, special nutritional values will be set with extracts enriched in fibres, chitosan, glucans, polyphenols, carotenoids, phytosterols and caffeine. Final extraction residues will be also analysed regarding components above mentioned and utilised for composite production aiming at a zero waste total valorisation process of the initial feedstocks, to be achieved by energy- and cost-efficiency approaches in order to target optimised qualitative and quantitative throughputs of the products. The extracts will be provided in dried and/or concentrated condition for the subsequent product prototyping.
Besides providing sufficient sample material for the subsequent Tasks in WP5, this task will also deliver process data that will be used as input for refined LCA and CEA analysis in WP6.

Task 5.2. Food product prototyping (Months 27-46)

Leader: IGV. Partners involved: SB, SSICA

Subtask 5.2.1 Cereal-based product prototyping (IGV, SB)

The Partners involved will follow-up with marketing the technology and/or products to the food industry (with strong involvement from SB). Essential technological inputs will be necessary to reach the main objective of the project. The production process for protein-products, especially food ingredient grade extracts and basic products as intermediate products have to be upgraded and improved. A product range containing protein (i.e. bread, buns, pasta, and specific breakfast cereal-based snacks such as Flakes, Nuggets, Crispies, low-calories Chocolate-Flake and protein Loops) will be developed for the muesli industry. The raw material portfolio for the extracts will be broadened and the extracts and extraction residues themselves will be integrated as ingredients for extrusion for reaching special nutritional values, aim being to considerably improve the organoleptic features. For the extraction stage, new methods have been integrated in order to gain higher protein contents in the extracts and selected components. A technological task for the extrusion will be to preserve special nutritional values during processing, without losing functionality. The extrusion for gaining Flakes, Crispies, Nuggets and Snacks (extended range with Loops and Choco-Flakes) will be adjusted by the addition of phytochemical-extracts and other biomass extraction residues (such as residual biomass generated regularly by IGV from microalgae) gaining a total protein value for the daily diet. The modules of the Planetary Roller Extruder will be configured for each product and data clustered for technological transfer and validation procedures. Amino acid ratios will be tailored - based on WHO recommendations – for each of the target group (elderly, children, sportspeople). New baked products will be developed on basis of selected protein and peptide fractions and mixtures thereof. The most suitable ratios of components regarding demanded technological quality and nutritive benefit of final product will be determined and recommended. Further development activity will include new intermediate and final products based on ancient wheat flours and rye flours enriched with isolated fractions with specific (higher) content of compounds with health benefits (e.g. peptides, essential amino acids, polyphenols, other antioxidant compounds and caffeine). IGV food technologists and Sensory Panelists will support the piloting demonstration tests of SB for baked products, pasta and breakfast cereals. Prototypes will be prepared as give-aways for marketing activities.

Subtask 5.2.2. Vegan and vegetarian food prototyping (IGV, SB)

Prototyping for meat surrogate products, such as meatballs, vegetal hamburgers, Bolognese, sports nutrition products, will be performed at IGV facilities by means of formulation tests with fine ground, but already extruded protein ingredients. Recipes (i.e. high protein Bolognese with pasta) will be linked to nutritional facts about the initial biomasses involved and the functionalities they provide. This will aim to optimise commercial routes that widen the adoption of already popular “meatless meat” based dishes, and provide novel vegan, vegetarian dishes and menus. First batches of system-products (i.e. meatless meat products, consisting of extruded protein plus other clean label ingredients such as spices, seasoning, etc.) will be evaluated in the stores of SB for consumer acceptance by panels, which will evaluate the organoleptic properties of the final products. The clear aim is a pre-final benchmarking, since SMEs always wish to extend their product portfolio. The products will undergo continuous consumer acceptance studies for testing the “willingness to buy” over several months. Results will be clustered and feedback given to the key Stakeholders and IGV.

Subtask 5.2.3. Meat products prototyping (SSICA)

As to more conventional meat products (hamburger, cooked and fermented sausages), the phytochemical extracts selected and characterized in WP4 (e.g. enriched carotenoids and polyphenols fractions), will be used as functional ingredients preventing fat and protein oxidation, chelating and stabilizing meat iron, and exerting a protective role during processing, shelf-life, and after consumption. Besides, sausages will take advantage from extracts rich in soluble fibres (-glucans selected and characterized in WP4) to enhance cooking yields and texture, and to improve the nutritional profile by the addition of prebiotic molecules and partial fat replacement. Appropriate procedures will be set for the addition and solubilisation at different concentration of phytochemical extracts into the meat matrices (model systems), still preserving antioxidant and Fe-chelating activities. The sensory evaluation of colour and flavour acceptability, the colour stability will be assessed to verify the affinity with the meat matrix and the correct dose of each extract. Antioxidant capacity (ferric reducing ability (FRAP), diphenyl-picryl-hydrazyl-hydrate test (DPPH), Fe-chelating activity), total phenols (Folin analysis), oxidative stability, lipid peroxidation (Tiobarbituric acid reactive (TBAR’s)) and protein oxidation, will be assessed in enriched meat model systems. Technological parameters (drip, ripening and cooking losses of hamburger-, fermented sausages -, and cooked sausages-like model systems, respectively) will be recorded; nutritional profile, and textural analysis will be assessed in abovementioned model systems manufactured with soluble fibres. The formulations that gave good results will be scaled up in the pilot plant of SSICA by means of traditional and innovative technologies. Around 30 kg of raw meat/batch will be processed for each product. Technological parameters will be recorded in meat products with soluble fibres (-glucans). Final outcomes will be characterized by means of the same tests described above; in addition sensory profile, will be determined. After in vitro gastrointestinal digestion Comet assay will be performed to compare products enriched in extracted phytochemicals with the controls. All products will be eventually analysed for microbiological quality.

Task 5.3. Cosmetics formulation and prototyping (Months 28-46)
Leader: COSM. Partners involved: IGV
The two most promising fractions of Task 4.3 in terms of functional properties and commercial opportunity, will be selected for the manufacturing of prototype cosmetic products. In cosmetics, specific extracts act as thickening agents, water-binding agents, and antioxidants, which turns into applications for the treatment of skin problems, such as ageing, tanning and pigment disorders, and these will be also monitored. Moreover, there are also potential uses in the areas of skin-whitening and pigmentation reduction products. In particular, COSM will incorporate the selected fractions in prototype cosmetic formulations. These may be a cream (with sunscreen or whitening or anti-age properties) and/or a toothpaste. A prototype formulation will be created based on the physicochemical characteristics of the fractions. Relevant processing parameters for the upscaling such as addition sequence and time, mixing/homogenisation energy and time, mixing temperature and cooling rate will be investigated.

Task 5.4. Prototyping of polymers and composites based formula for packaging (Months 28-46)
Leader: FTEN. Partners involved: UNIBO, SSICA, UNIPI, COSM, ILLY
Subtask 5.4.1. Production of an anti-fungal coating for dry-cured meat and of a flexible biodegradable film for raw and fermented meat food packaging (, SSICA)
Chitosan based-coatings, optimized in WP3 (Subtask 3.4.1), will be used to protect ripened meat products in order to enhance their quality and safety. SSICA will apply and evaluate the effectiveness of chitosan coatings on dry-cured hams (30 hams/batch) during the process to control both Penicillium nordicum and Ocratoxin A (OTA) contamination and mite infestations, evaluating their impact on safety and sensory properties. Hamburgers and fermented sausage, produced at SSICA pilot plant in Task 5.2, will be packed under vacuum with the film PLA+chitosan developed in WP3. For food packaging application, hygienic-sanitary properties will be evaluated including global and specific migration in compliance with the EU legislation (SSICA).

Subtask 5.4.2. Production of a rigid biodegradable packaging with antioxidant properties for cosmetics (FTEN, UNIPI, UNIBO, COSM)
FTEN will produce prototypes of cosmetic jars by injection moulding on a specific mould designed and produced by FTEN in agreement with COSM needs. In detail, starting with a definite design, FTEN will carry out an industrialization activity with the complete definition of the 3D models of the object using 3D CAD system Pro/Engineer Creo, and realizing a first prototype using RP technology. After prototype approval, FTEN will build the mould necessary for the production of the final material. PLA and PHAs will be used as matrix. Materials will be prepared by UNIPI, using the formulations tested in WP3, Subtask 3.4.1, with bioactive molecules extracted in WP2 to achieve antioxidant properties, also intercalated in LDH nanostructures.
Reinforcement with fibrous materials obtained as by-products in WP2, according to the formulation carried out in WP3, Subtask 3.4.2 will also be tested at pilot scale (25-50 kg of pellets produced by UNIPI, injection moulded by FTEN), a content of fibres of at least 20% by weight and possibly 30-40% by weight is envisaged. The addition of natural fibres is expected to lower the weight and the cost of the material while improving the biodegradability. A full characterisation of the materials produced will be performed in terms of thermal, structural, mechanical and morphological properties (UNIBO and UNIPI). For cosmetic packaging applications, prototype hygienic-sanitary properties will be evaluated including global and specific migration in compliance with the EU legislation (COSM). Prototypes will be prepared as give-aways for marketing activities.

Subtask 5.4.3. Production of bio-based and compostable coffee capsules with antioxidant properties (UNIPI, FTEN, ILLY, UNIBO, SSICA)
PLA based formula will be optimised for coffee capsules production. Particular attention will be devoted to PLA crystallization, achieved by addition of nucleating agents such as PDLA or talc, in order to achieve resistance to stress in temperature as required during coffee brewing. PLA based formula will be functionalised by adding the antioxidant bio-molecules extracted in WP2, according to the protocols developed in WP3, Task 3.4.1. FTEN will design and build a mould for coffee capsule as described in Subtask 5.4.2 and in agreement with ILLY, and will produce demonstrators of coffee capsules. ILLY will test the final product. A full material characterization in terms of thermal, structural, mechanical and morphological properties will be carried out (UNIBO, UNIPI, FTEN). For food packaging application, prototype hygienic-sanitary properties will be evaluated including global and specific migration in compliance with the EU legislation (SSICA).

Task 5.5. Feed products prototyping (Months 27-46)
Leader: NS. Partners involved: IGV
Novel processes will be developed in order to process the extracted fibres, carotenoids and polyphenols fractions into feed ingredients and by-products animal feeds (pigs and poultry). In addition, an optimisation of the process for the production of feed by adding essential amino acids and protein hydrolysates from the downstream fractions will

be performed via planetary roller extrusion and their physicochemical, thermal and functional characteristics will be assessed (IGV). The nutritional value of the fractions will be analysed and shall comply with regulations in use for the supply of new, healthy animal products by selecting and matching bulk feed biochemistry in order to deliver feed that enhances the nutritional and health quality of animal products. Based on the nutritional profiles and safety assessments, we will identify the best blends of biomass sources aiming at high value nutritional combinations. In the case one of the ingredient produced in large-scale in Task 5.1 has some promising nutritional effect, in vivo tests including the following forms of studies will be carried out:

● small scale, controlled studies, in which individual/group housed animals will be fed a common basal diet (negative control)

● experimental diets containing specific feed ingredients at graded levels

In response to these treatments, feed acceptance (intake), growth rates, feed utilisation, and numerous specific parameters illustrating gut health as constituted by faecal consistency, bacterial enumeration/composition, intestinal secretions, mucosal immunity, integrity and functionality and some responses related to animal welfare, will be monitored.

Participation per Partner

Partner number and short name	WP5 effort
1 - FHNW	
2 - UNIBO	
3 - IRIS	
5 - SSICA	
6 - CELAB	
7 - IGV	
8 - SB	
9 - BBEPP	
10 - CONS	
12 - COSM	
13 - ILLY	
14 - NS	
18 - UNIPI	
Total	

List of deliverables

Deliverable Number[14]	Deliverable Title	Lead beneficiary	Type[15]	Dissemination level[16]	Due Date (in months)[17]
D5.1	Dataset of validated pre-production prototypes and best technological procedures scaled-up	9 - BBEPP	Report	Confidential, only for members of the consortium (including the Commission Services)	38
D5.2	Eight food product prototypes	7 - IGV	Demonstrator	Confidential, only for members of the consortium (including	46

List of deliverables

Deliverable Number[14]	Deliverable Title	Lead beneficiary	Type[15]	Dissemination level[16]	Due Date (in months)[17]
				the Commission Services)	
D5.3	Two cosmetic product prototypes	12 - COSM	Demonstrator	Confidential, only for members of the consortium (including the Commission Services)	46
D5.4	Four packaging prototypes	18 - UNIPI	Demonstrator	Confidential, only for members of the consortium (including the Commission Services)	46
D5.5	Two feed product prototypes	14 - NS	Demonstrator	Confidential, only for members of the consortium (including the Commission Services)	46

Description of deliverables

D5.1 : Dataset of validated pre-production prototypes and best technological procedures scaled-up [38]
Documentation of scaled-up extraction protocols and use of extracts and fractions in pre-prototypes

D5.2 : Eight food product prototypes [46]
Eight food product prototypes

D5.3 : Two cosmetic product prototypes [46]
Two cosmetic product prototypes

D5.4 : Four packaging prototypes [46]
Four packaging prototypes

D5.5 : Two feed product prototypes [46]
Two feed product prototypes

Schedule of relevant Milestones

Milestone number[18]	Milestone title	Lead beneficiary	Due Date (in months)	Means of verification
MS11	Technological transfer protocols scaled-up and validated	9 - BBEPP	38	Scale-up of extraction protocols carried out, documented including characterisation of extracts obtained
MS12	Feasible and low-carbon footprint product prototypes developed and technologies integrated	7 - IGV	46	product factsheets with KPIs for environmental impact of LCA tested products

Schedule of relevant Milestones

Milestone number[18]	Milestone title	Lead beneficiary	Due Date (in months)	Means of verification
				and outline of value chain development

Work package number [9]	WP6	Lead beneficiary [10]	1 - FHNW
Work package title	Life-Cycle Assessment (LCA) and Cost-Effectiveness Analysis (CEA)		
Start month	1	End month	46

Objectives

The objectives of this work package is to measure the financial and environmental benefits, costs and burdens associated with the new PROLIFIC bio-based products from a life cycle perspective. Life-Cycle Assessment (LCA) for environmental assessment and Cost-Effectiveness Analysis (CEA) for financial aspects will be applied at an early stage of technology development. This can directly support the improvement of the financial and environmental performance of the developed products by identifying main processes contributing to environmental impacts and help optimize those (hotspots). For selected products, an LCA comparison will be done between PROLIFIC bio-based products with conventional ones. The environmental assessment will be carried out following the standard ISO 14040 series. The specific objectives are:

- Perform the data collection process for the financial and environmental assessment of the products
- Implement, refine and validate LCA for the involved technologies
- Final report presenting LCA and CEA results of the new products and technologies developed in this project.

Description of work and role of partners

WP6 - Life-Cycle Assessment (LCA) and Cost-Effectiveness Analysis (CEA) [Months: 1-46]
FHNW, UNIBO, IRIS, UNIPR, SSICA, CELAB, IGV, SB, BBEPP, CONS, TALOS, COSM, ILLY, NS, PLEUR, FTEN, UNIPI
Description of work
Leader: Emmanuel Oertlé (FHNW, Switzerland); Co-leader: Xavier Joppin (CELAB, Belgium)

In WP6, the expected sustainability of the PROLIFIC products and technologies is assessed to foster overall performance and uptake of the PROLIFIC products through:

- Consideration of a realistic production scale right from the beginning of the assessment (including necessary estimations and forecasting)
- Environmental characterizations (Life Cycle Assessments) for proposed technologies and products to estimate environmental impacts, benefits and highlight crucial processes to foster environmental improvements of the technologies during its development phase;
- Cost and techno-economic analysis of the envisaged technologies and products to foster uptake;

Close links with WP1-WP5 partners are used to simultaneously integrate initial findings in the development and improvement process of the technologies and to support their uptake in cooperation with industrial partners.

Task 6.1. Products selection, data collection and estimations (Months 1-24)
Leader: FHNW. Partners involved: CELAB, IGV, UNIBO, IRIS, UNIPR, SSICA, SB, BBEPP, COSM, ILLY, NS, PLEUR, FTEN, UNIPI
Initial PROLIFIC product selection to be analysed in tasks 6.2 and 6.3: As many biobased products are foreseen in PROLIFIC project, only a selection of most promising products (high commercialisation potential) will be assessed with LCA and CEA (min. 1 LCA/CEA per product type). The selection that will be done in agreement with the consortium and WP-leaders will be based on the TRL, PROLIFIC experimental results and data availability for a realistic production scale. For each PROLIFIC product selected, a corresponding benchmark product will also be chosen in order to allow comparison and demonstrate the benefits of PROLIFIC products.
Life cycle inventory, data collection and estimations: For each selected product, the corresponding partners will be responsible to deliver data and perform estimations for qualitative and quantitative data required for the different assessments to be performed in this work package (LCA, CEA and product regulation). The data collection will be a continuous process in parallel to WP1, 2, 3, 4, 5 and 6 with well-structured questionnaires and communication. All analyses in this work package will focus on the assessment at full scale (for a potential production) and not at lab, prototyping or pilot scale. CELAB will support the work on LCI (Life cycle inventory), data collection and estimations by providing its field specific expertise.

Task 6.2. Feasibility assessment for LCA and CEA analysis (Months 24-46)
Leader: IGV. Partners involved: CELAB, FHNW, UNIBO, IRIS, UNIPR, SSICA, SB, BBEPP, COSM, ILLY, NS, PLEUR, FTEN, UNIPI

Data for LCA and CEA will be collected in Task 6.1 from each WP and will culminate under this task after scale-up procedures have been performed and prototypes have been validated. The inputs will be used to further tailor the processes. Regarding the advance beyond the state of the art and innovation in the final applications, a linear validation procedure will be designed based on standard validation protocols for each product category. In order to ensure that the entire processing chain (including all steps of up and downstream) is co-creating a feasible new alternative towards sustainable ingredients supply for food, feed and cosmetics industries, the technological transferability data will be collected from all processing Partners. Safety, quality and purity of the new products will be in line with EU legislation (see Section 2.2, Table 10) and will prove to meet end-market requirements in order to facilitate future market access and commercialization.

Task 6.3. Life-Cycle Assessment (LCA) of the foreseen products and technologies (Months 12-46)
Leader: FHNW. Partners involved: CELAB, IGV, UNIBO, IRIS, UNIPR, SSICA, SB, BBEPP, COSM, ILLY, NS, PLEUR, FTEN, UNIPI.
Life-cycle assessments (LCA) will be conducted for the foreseen products following the well-established phases: goal definition & scoping, life cycle inventory, impact assessment and interpretation. CELAB and FHNW will work in close collaboration in order to exactly determine the field of the study, hypothesis, assumptions, most relevant functional units according to the aim of the project and adequate indicators to assess the environmental impacts and benefits. LCA at an early stage of technology development faces several challenges, as the inventory will be based on technologies still under development and processes might not be well-defined yet. However, an early stage LCA can also directly foster and support the improvement of the environmental performance of the developed technologies by identifying main processes contributing to environmental impacts and help optimize those. As many data estimations will be required for the inventory, the LCA methodology will be streamlined in order to provide relevant information for possible optimization of the technologies' environmental impacts and demonstrate the environmental benefits of the implementation of the developed technologies and products compared with current ones (benchmark technologies defined in Task 6.1). FHNW, in collaboration with CELAB, will conduct the Life-cycle assessments using state-of-the-art software (Simapro) and database (Ecoinvent version 3). The partners will be responsible to deliver the required available data and to estimate the missing ones. In addition, CELAB will also compare different hypothesis and provide environmental information regarding specific processes and technologies.

Task 6.4. Cost-Effectiveness Analysis (CEA) of the foreseen products and technologies (Months 12-46)
Leader: FHNW. Partners involved: IGV, CELAB, UNIBO, IRIS, UNIPR, SSICA, SB, BBEPP, COSM, ILLY, NS, PLEUR, FTEN, UNIPI.
The foreseen bio-based products (selection from Task 6.2) will be evaluated in terms of projected costs by means of a CEA. This task consists of the following activities:
a) Determination of the framework for the CEA dataset
b) Collection and estimation of the cost components to be considered for the selected PROLIFIC products
c) Financial evaluation and comparison of the different technologies through the application of the CEA model
Similarly to task 6.3, a financial assessment at an early stage of technology development faces several challenges, as data will be based on technologies still under development and processes might not be well-defined yet. Nevertheless, this early assessment will aim at identifying and optimizing most critical processes. As many data estimations will be required for the inventory, the methodology will be streamlined in order to provide relevant information for possible optimisation. The partners will be responsible to deliver the required available data and to estimate the missing ones.

Task 6.5. Review of regulations for the foreseen products (Months 6-40)
Leader: IGV. Partners involved: CELAB, SSICA, SB, NS, COSM, ILLY
A preliminary assessment of the Regulations that might impact the projects has been carried out (Section 2.2, Table 8). The regulatory affairs Team of IGV will thoroughly monitor the compliance of the resulted intermediary and final products with the current Regulations for food, cosmetic, food and cosmetic packaging and feed. All the products and sub-products obtained from the whole value chain (individual compounds, blends and final products) and all related production processes will ensure the accomplishment of the EU regulatory framework in each sector. Moreover Regulation and Certification Bodies will be permanently consulted, allowing at quick adaptations in the overall R&D activities targeting the end-user sectors. Materials and articles intended to come into contact with food have to respect the requirements of EU Reg. 1935/2004 and in particular plastic materials are regulated by EU Reg. 10/2011. According to these regulations and related updates, overall migration tests and specific migration tests from plastics to food simulants have to be performed. The suitable food simulants will be selected as function of the final application of developed materials. Regulation (EC) N° 1223/2009 on cosmetic products is the main regulatory framework for finished cosmetic products on the EU. For each raw material and packaging used in an end product a separate safety assessment is required. More specifically the assessment is in accordance with the decision 2013/674/EU on Guidelines on Annex I

to Regulation (EC) No 1223/2009. Feed applications are developed under the Directive 2002/32/EC of the European Parliament and of the Council of 7 May 2002 and following regulations as summarised in Section 2.2, Table 10.

Participation per Partner

Partner number and short name	WP6 effort
1 - FHNW	
2 - UNIBO	
3 - IRIS	
4 - UNIPR	
5 - SSICA	
6 - CELAB	
7 - IGV	
8 - SB	
9 - BBEPP	
10 - CONS	
11 - TALOS	
12 - COSM	
13 - ILLY	
14 - NS	
15 - PLEUR	
16 - FTEN	
18 - UNIPI	
Total	

List of deliverables

Deliverable Number[14]	Deliverable Title	Lead beneficiary	Type[15]	Dissemination level[16]	Due Date (in months)[17]
D6.1	Intermediate LCA/CEA results for final products	1 - FHNW	Report	Confidential, only for members of the consortium (including the Commission Services)	18
D6.2	Report on product regulations related to final products	7 - IGV	Report	Confidential, only for members of the consortium (including the Commission Services)	40
D6.3	Final LCA/CEA results for final products	1 - FHNW	Report	Confidential, only for members of the consortium (including	46

List of deliverables

Deliverable Number[14]	Deliverable Title	Lead beneficiary	Type[15]	Dissemination level[16]	Due Date (in months)[17]
				the Commission Services)	

Description of deliverables

D6.1 : Intermediate LCA/CEA results for final products [18]
Intermediate LCA/CEA results for final products

D6.2 : Report on product regulations related to final products [40]
Report on product regulations related to final products

D6.3 : Final LCA/CEA results for final products [46]
Final LCA/CEA results for final products

Schedule of relevant Milestones

Milestone number[18]	Milestone title	Lead beneficiary	Due Date (in months)	Means of verification
MS13	Products to be analysed in Tasks 6.3 and 6.4 have been selected and qualitative data have been delivered by the partners	1 - FHNW	18	Products to be analysed in Tasks 6.3 and 6.4 have been selected and qualitative data have been delivered by the partners
MS14	LCA, CEA, safety & regulatory compliance assessment has been performed for at least one product prototype for each per industrial sector	1 - FHNW	46	LCA, CEA, safety & regulatory compliance assessment has been performed for at least one product prototype for each per industrial sector

Work package number [9]	WP7	Lead beneficiary [10]	11 - TALOS
Work package title	Dissemination, exploitation and commercialisation		
Start month	1	End month	48

Objectives

The objectives of WP7 are:
● To organise successful dissemination and communication activities to increase the visibility of the PROLIFIC project and the technologies and products developed by the consortium
● To create an exploitation roadmap consolidating commercialisation plans and business opportunities
● To deal with IPR issues during the project and establish the final IPR conditions as part of the Exploitation Agreement
● To monitor, assess and evaluate all activities performed and adjust action plan accordingly

Description of work and role of partners

WP7 - Dissemination, exploitation and commercialisation [Months: 1-48]
TALOS, FHNW, UNIBO, IRIS, UNIPR, SSICA, CELAB, IGV, SB, BBEPP, CONS, COSM, ILLY, NS, PLEUR, FTEN, INNOVA, UNIPI
Description of work
Leader: Nikos Gkouveris (TALOS, Cyprus), Co-Leader: Francesca Montalti (INNOVA, Italy)

Task 7.1. Developing dissemination strategy and action plan (Months 1-5)
Leader: TALOS. Partners involved: ALL
This task intends to specify actions that will drive and coordinate dissemination activities in the project. It is an activity defining experience sharing and starts with collecting relevant information and progress from the partners. The aim is to make the information available in different forms: press releases, newsletters, publications and presentations on national and EU specialised web reviews as well as international conferences. Deciding on the strategy and action plan will be the key of effective dissemination and will ensure frequent and up-to-date communications about ongoing activities and events, project state, advancements and results. The result of this task will define the goal, communication plan, target audience, media channels, conferences to be attended, etc. The action plan will contain clear and measurable Key Performance Indicators (KPIs) (number of publications per year, number of visitors on the website, number of audience reached, etc.) which will serve as the basis for the evaluation of dissemination activities.

Task 7.2. Dissemination and networking activities (Months 1-48)
Leader: TALOS. Key partners involved: INNOVA, FHNW, UNIBO; Contributors: ALL
Dissemination activities regarding the project will begin with the outset of the project and will continue throughout its whole life cycle. All dissemination and promotion activities will aim at raising awareness on the project and maximising its impact. The purpose of this task is to disseminate the project objectives and results to the target audience and stakeholders.
1) Execution of dissemination and communication activities: all partners will cooperate in these activities both at national and at European level. Dissemination activities will be undertaken by the consortium as a whole and by each partner individually. Dissemination activities will include the organisation of local dissemination events with the aim to involve and engage local stakeholders, as well as the organisation of workshops for more focused discussion with stakeholder groups. More specifically, the planned dissemination activities belong to the following categories:
a) Dissemination targeting general public: leaflets, brochures and posters will be developed and disseminated as well as press releases will be prepared every 6 months starting from project launch.
b) Dissemination targeting scientific community: Scientific partners will actively participate on scientific conferences and will publish scientific papers in peer-reviewed scientific journals.
c) Dissemination targeting industrial community: Additionally, partners will reach the wide public via industrial fairs and exhibitions as well as in-store boosts where PROLIFIC prototypes / ingredients will be distributed to the target groups (see more information in Section 2.2).
d) Dissemination via electronic means and social media: In order to reach a wide range of audiences, electronic communication tools, such as project website as well as social media will be actively used. It is planned that PROLIFIC project will have a presence at Facebook (for general public) and LinkedIn (a specialised group) which will give the necessary visibility for the project worldwide.

e) Workshops: The organisation of 2 workshops has already been agreed by the partners: one of them will be organised by ILLY in Trieste towards the end of the first year of the project. This workshop will include also a tour at their production plant and the workshop will be organised in the conference hall at their main production site. The main focus of this workshop will be the residual biomasses from agro-industrial processing. The second workshop will be organised by IGV in Potsdam towards the end of Year 3 of the project and will focus on products from ingredients extracted from residual biomasses and their exploitation and commercialisation potential.
2) Networking activities:
a) Networking with stakeholders / customers: the replication of the PROLIFIC initiative at European level will be a central task in WP7 where INNOVA will play a crucial role. Via the network of the different cooperative associations operating at EU level, the PROLIFIC project will have a direct access to a large number of SMEs, producers, biomass suppliers and also to consumer all over Europe. We will reach and involve these stakeholders on one hand via standard communication tools but also through all the associations' channels (newsletters, meetings, websites, social media, house organs), which will act as a multiplier and on the other hand via cooperation with the association in the involvement of targeted groups of companies for the organisation of workshops and in-store boosts.
b) Networking with business angels and potential investors: TALOS is the coordinator of the Cyprus Business Angels network (CYBAN) and CYBAN is a full member of the European Business Angels Network. These networks will be utilised via specialised workshop where PROLIFIC results (ingredients and prototypes) will be presented to potential investors with the aim to foster and enhance commercialisation potential of the results.
3) Monitoring and evaluation: dissemination, communication and networking activities will be monitored and evaluated on a regular basis. Each partner will be requested to prepare an internal dissemination report every 2 months about the dissemination and communication activities performed. These reports will be evaluated by the Dissemination Manager (WP7 leader) and benchmarked with the Key Performance Indicators defined in the Dissemination and Communication Plan. These indicators will be among others the number of dissemination activities performed, the number of persons reached by the dissemination activity, the type of audience reached as well as the geographical area targeted by the dissemination action.

Task 7.3. Communication activities (Months 1-48)
Leader: TALOS; Key partners: INNOVA, FHNW, UNIBO; Contributors: ALL
This task will raise awareness of the project to a larger public audience enabling PROLIFIC to be replicated in different contexts and in different application areas. This will develop an adequate communication strategy and its implementation and make plans for communication channels in order for the project to raise awareness and achieve a new and expansive engagement of relevant stakeholders. The communication activities will include: A) Establishing direct and indirect communication and sharing of knowledge, B) Giving visibility and access to internet website, C) Publications in digital and printed forms, D) Sharing of knowledge, tools and skills through workshops, webinars, lectures, papers and other similar activities to involve the stakeholder communities, etc.

Task 7.4. IPR and exploitation activities (Months 12-48)
Leader: INNOVA. Key partners: TALOS; Contributors: ALL
This task copes with the development of a clear business strategy for the commercial exploitation of the project results and will study the exploitation possibilities for these results.
1) Planning: The aim of this planning is to prepare the ground for further exploitation of the project results. The information gathered by the partners through dissemination, demonstration, piloting and prototyping activities as well as during the direct contacts with end-users will be used to prepare an Exploitation Plan as part of the Draft Plan for the Exploitation of the Results by month 24. This will outline a preliminary action plan for each partner describing their concept, strategy and related mechanisms on the use and exploitation of the knowledge generated during the project execution.
2) Execution: Through all the dissemination activities in Task 7.3 and the planning of the exploitation the partners will form a clear picture on the exploitation potential of the project results. Based on this knowledge the following activities will take place:
a) Knowledge management and IPR protection activities will focus on the identification, protection and exploitation of the generated knowledge. All scientific findings generated by the project will be assessed and evaluated by the Valorisation Committee for exploitation potential and this assessment will be made prior to the external dissemination of materials. The final aim of the IPR protection activities is to support the project partners in protecting and using the project results to their best advantage, leading to clear economic impact.
b) Plan for the Exploitation of Results: The final version of the document will describe the participants' actual achievements in dissemination and mainly focus on their plans for the future in terms of exploitation of project results. The document will follow the requested format in the contractual agreement with the EC and two versions (draft and final) will be delivered (at months 24 and 48).

c) Exploitation Agreement: The Exploitation Agreement will be a general agreement on project results exploitation, with identification of roles, rights and obligations of each consortium partners with respect to commercialisation of the project outcomes. The agreement will include: A) the list of key exploitable results, B) exploitation plans, including those that affect only a few partners, C) the market segment assigned to each partner that is interested in a portion of the exploitation opportunity, D) the definition of the business roadmap that the consortium plans to follow, E) the sanctions and penalties that a partner will incur in case of breach of contract. It will also cover exploitation restrictions, licensing agreements and protection of the intellectual property generated within the project.
d) Preliminary Business Plans: Preliminary business plans will be drafted for the different PROLIFIC ingredients and prototype products with the aim to help relevant partners to exploit commercially their technical solutions and improve the competitiveness of their organisation in the market. The preliminary business plans will include an analysis of the external and internal environment as well as the necessary financial, operational and risk analyses including sustainability plan for the future commercial viability of the technical solution. The business plans will be used to prepare pitches in front of business angels in the frame of the investment workshop.

Task 7.5 PROLIFIC Data Management Plan (DMP) (Months 1-48)
Task leader: FHNW Participants: all
PROLIFIC Project recognises the value of regulating research data management issues, and foresees the delivery of a Data Management Plan (DMP) at Month 6. The DMP will define how research data will be handled during the lifetime of the PROLIFIC project, and after it is completed. The DMP will be co-developed by the Administrative Coordinator together with the Scientific Coordinator and the WP leaders, followed by a plenary contribution by all partners involved in research and innovation activities. All partners will actively participate not only in the draft of the DMP at Month 6, but in its dynamic development over the time, consisting in its continuous update according to possible needs, evolving expectations and framework conditions. Status of the DMP will be discussed at GA meetings.

Participation per Partner

Partner number and short name	**WP7 effort**
1 - FHNW	
2 - UNIBO	
3 - IRIS	
4 - UNIPR	
5 - SSICA	
6 - CELAB	
7 - IGV	
8 - SB	
9 - BBEPP	
10 - CONS	
11 - TALOS	
12 - COSM	
13 - ILLY	
14 - NS	
15 - PLEUR	
16 - FTEN	
17 - INNOVA	
18 - UNIPI	
Total	

List of deliverables

Deliverable Number[14]	Deliverable Title	Lead beneficiary	Type[15]	Dissemination level[16]	Due Date (in months)[17]
D7.1	Dissemination and communication strategy and action plan	11 - TALOS	Report	Confidential, only for members of the consortium (including the Commission Services)	5
D7.2	Website publicly available	11 - TALOS	Other	Public	6
D7.3	Data Management Plan	1 - FHNW	Report	Confidential, only for members of the consortium (including the Commission Services)	6
D7.4	Interim report on performed dissemination, communication and networking activities	11 - TALOS	Report	Public	24
D7.5	Final report on performed dissemination, communication and networking activities	11 - TALOS	Report	Public	48
D7.6	Draft plan for the exploitation of results	17 - INNOVA	Report	Confidential, only for members of the consortium (including the Commission Services)	24
D7.7	Final plan for the exploitation of results including Exploitation Agreement among the partners	17 - INNOVA	Report	Confidential, only for members of the consortium (including the Commission Services)	48
D7.8	Preliminary business plan for the product ingredients and prototypes	11 - TALOS	Report	Confidential, only for members of the consortium (including the Commission Services)	48

Description of deliverables

D7.1 : Dissemination and communication strategy and action plan [5]

Dissemination and communication strategy and action plan

D7.2 : Website publicly available [6]

Website publicly available

D7.3 : Data Management Plan [6]

Data Management Plan

D7.4 : Interim report on performed dissemination, communication and networking activities [24]

Interim report on performed dissemination, communication and networking activities

D7.5 : Final report on performed dissemination, communication and networking activities [48]
Final report on performed dissemination, communication and networking activities

D7.6 : Draft plan for the exploitation of results [24]
Draft plan for the exploitation of results

D7.7 : Final plan for the exploitation of results including Exploitation Agreement among the partners [48]
Final plan for the exploitation of results including Exploitation Agreement among the partners

D7.8 : Preliminary business plan for the product ingredients and prototypes [48]
Preliminary business plan for the product ingredients and prototypes

Schedule of relevant Milestones

Milestone number[18]	**Milestone title**	**Lead beneficiary**	**Due Date (in months)**	**Means of verification**
MS15	Proper planning for dissemination, communication and networking activities prepared and execution of these activities launched	11 - TALOS	5	Proper planning for dissemination, communication and networking activities prepared and execution of these activities launched
MS16	Dissemination, communication, networking and exploitation activities successfully performed and relevant documents prepared	11 - TALOS	48	Dissemination, communication, networking and exploitation activities successfully performed and relevant documents prepared

Work package number [9]	WP8	Lead beneficiary [10]	1 - FHNW
Work package title	Project coordination and management		
Start month	1	End month	48

Objectives

This WP aims at monitoring and steering the project to produce the planned results with the required quality and on schedule. In short, management procedures are defined and implemented, interactions between the different bodies are stimulated and coordinated, and communication with the EC and reports are delivered in time. It consists of the following specific objectives: and consists in the achievement of the following specific objectives:

• to coordinate and supervise the project research and innovation activities to be carried out according to the work plan, also to monitor and ensure quality and timing of project deliverables;
• to carry out the overall administrative and financial management and reporting of the project;
• to manage contacts with the EU Commission and the BBI JU and establish effective internal and external communication;
• to resolve possible conflicts.

Description of work and role of partners

WP8 - Project coordination and management [Months: 1-48]
FHNW, UNIBO, IRIS, UNIPR, SSICA, IGV, SB, BBEPP, CONS, TALOS, NS, PLEUR, FTEN, INNOVA, UNIPI
Description of work
Leader: Philippe Corvini (FHNW, Switzerland), Co-Leader: Annalisa Tassoni (UNIBO, Italy)

The main work to be performed is related to the coordination and monitoring of the activities, the administrative and financial management, the coordination of the reporting and deliverables production and the resolution of possible conflicts. In addition, contacts with EC and compliance with procedures described in the signed versions of the Grant Agreement and the Consortium Agreement will be continually ensured through participation in EC project reviews and regular internal review of periodic reports. FHNW will be in charge of the administrative coordination of the PROLIFIC project and lead this WP. UNIBO as scientific coordinator will have a major contribution by assisting FHNW in the scientific and technological management of PROLIFIC. The WP leaders (UNIBO (WP1), UNIPR (WP2), UNIBO (WP3), CELAB (WP4), IGV (WP5), FHNW (WP6), TALOS (WP7)) will support the coordination and project management, e.g. organise and host meetings and will participate in the scientific and technical committee (STC). The Innovation Management Team (IMT) leader and the IPR Advisor as well as the members of the communication and dissemination will be nominated among the staff member of institutions of the executive board as mentioned in section 3.2.1.

Task 8.1 Coordination, project meetings and reporting (Months 1-48)
Leader: FHNW. Partners involved: ALL

Subtask 8.1.1 Coordination and reporting
The Administrative Coordinator and the Scientific Coordinator and WP leaders will monitor and follow up on the activities foreseen in the work plan and discussed in the periodic meetings (see Subtask 8.1.2). The progress and results achieved by the project will be reported in the periodic activity reports required by the Funding Agency under the supervision of the coordinators and WP leaders. If necessary, recovery actions will be identified and activities revised accordingly.

Subtask 8.1.2 Organization of Kick-off and periodic meetings
The Administrative Coordinator will organize, implement, and follow-up a kick-off meeting and yearly face to face meetings (two per year). The Administrative Coordinator, assisted by the Project Management Team (PMT) and the hosting Partner, will convene and manage the project meetings. The task will include preparation of related documentation (e.g., agendas, invitations, documents and reports, minutes, and action lists, etc.) and securing location of meeting places/catering. The kick-off meeting will be held early (possibly by Month 1) to familiarize all partners with the structure and operation of the consortium, their detailed activities for the following period. The periodic General Assembly, Steering Committee (SC) and Scientific and Technical Committee (STC) meetings will regularly bring members of project together to provide guidance on the project implementation. Additionally, they will represent the opportunity to solve any issues impeding the progress of the project and to approve the appropriate contingency plans.

Additional technical meetings through Skype, teleconference and other web-based means, as well as technical visits will be convened where necessary.

Subtask 8.1.3 Communication and exchange with the European Commission and BBI JU
Communication and exchange with the EC and the BBI JU will be guaranteed by the Administrative Coordinator, who will represent the only interface with it. He will act on behalf of and in agreement with the members of the Consortium and will report to the SC and the General Assembly any request by the Agency. He will inform and dialogue with the EC in relation to any opportunity, risk and problem, which may arise internally and during the project implementation.

Subtask 8.1.4 Operational management of project boards
The objectives and fields of action of each Governing body are described in section 3.2.

Task 8.2 Overall legal and contractual management (Months 1-48)
Leader: FNHW
The Administrative Coordinator, assisted by the Project Management Team, will supervise and guarantee coherence among project activities and procedures described in the Grant Agreement and the Consortium Agreement that have been signed. The Administrative Coordinator will lead discussion on amendments and revisions of these Agreements if necessary.

Task 8.3 Financial and Administrative Management (Months 1-48)
Leader: FNHW
The Administrative Coordinator, assisted by the Scientific Coordinator and the Project Management Assistant, will manage the EC contribution and distribute funds among partners according to the rules established in the Grant Agreement and the Consortium Agreement that have been signed. Financial statements, relevant certificates (if necessary) and project reports will be collected and submitted to the EC. The Project Management Team will guarantee assistance to individual partners on administrative and financial issues.

Task 8.4. Coordination of Knowledge and Innovation management activities (Months 1-48)
Leader: FHNW. Partners involved: ALL
The Administrative Coordinator and the Scientific Coordinator, in cooperation with the STC, will carefully review the results of the project and make them available to the competent project boards listed above and in section 3.2. Results will be communicated and shared in accordance with the provisions mentioned in Grant Agreement and Consortium Agreement that have been signed, the Plan for Exploitation and Dissemination of Results ensuring their complementarity. Actions under this task will be carried out in close cooperation with the IPR Manager, paying attention to monitoring, management and exploitation of those results having high innovation potential.

Participation per Partner

Partner number and short name	WP8 effort
1 - FHNW	
2 - UNIBO	
3 - IRIS	
4 - UNIPR	
5 - SSICA	
7 - IGV	
8 - SB	
9 - BBEPP	
10 - CONS	
11 - TALOS	
14 - NS	
15 - PLEUR	

Partner number and short name	WP8 effort
16 - FTEN	
17 - INNOVA	
18 - UNIPI	
Total	

List of deliverables

Deliverable Number[14]	Deliverable Title	Lead beneficiary	Type[15]	Dissemination level[16]	Due Date (in months)[17]
D8.1	Minutes of the kick-off meeting	1 - FHNW	Report	Confidential, only for members of the consortium (including the Commission Services)	2
D8.2	Management Report M30	1 - FHNW	Report	Confidential, only for members of the consortium (including the Commission Services)	30
D8.3	Management Report M42	1 - FHNW	Report	Confidential, only for members of the consortium (including the Commission Services)	42
D8.4	KPI & Impact Questionnaire	2 - UNIBO	Report	Confidential, only for members of the consortium (including the Commission Services)	37
D8.5	Final KPI & Impact Questionnaire	2 - UNIBO	Report	Confidential, only for members of the consortium (including the Commission Services)	48

Description of deliverables

D8.1 : Minutes of the kick-off meeting [2]

Minutes of the kick-off meeting

D8.2 : Management Report M30 [30]

Minutes of bi-anual meetings, incl. status and additional documents related to achieved MS, if different from deliverables, updated risk issue tracker,

D8.3 : Management Report M42 [42]

Minutes of bi-anual meetings, incl. status and additional documents related to achieved MS, if different from deliverables, updated risk issue tracker,

D8.4 : KPI & Impact Questionnaire [37]

KPI & Impact Questionnaire

D8.5 : Final KPI & Impact Questionnaire [48]

Final KPI & Impact Questionnaire

Schedule of relevant Milestones

Milestone number[18]	Milestone title	Lead beneficiary	Due Date (in months)	Means of verification

Work package number [9]	WP9	Lead beneficiary [10]	1 - FHNW
Work package title	Ethics requirements		
Start month	1	End month	48

Objectives

The objective is to ensure compliance with the 'ethics requirements' set out in this work package.

Description of work and role of partners

WP9 - Ethics requirements [Months: 1-48]
FHNW
This work package sets out the 'ethics requirements' that the project must comply with.

List of deliverables

Deliverable Number[14]	Deliverable Title	Lead beneficiary	Type[15]	Dissemination level[16]	Due Date (in months)[17]
D9.1	H - Requirement No. 1	1 - FHNW	Ethics	Confidential, only for members of the consortium (including the Commission Services)	1
D9.2	HCT - Requirement No. 2	1 - FHNW	Ethics	Confidential, only for members of the consortium (including the Commission Services)	1
D9.3	A - Requirement No. 3	1 - FHNW	Ethics	Confidential, only for members of the consortium (including the Commission Services)	1
D9.4	EPQ - Requirement No. 4	1 - FHNW	Ethics	Confidential, only for members of the consortium (including the Commission Services)	1

Description of deliverables

The 'ethics requirements' that the project must comply with are included as deliverables in this work package.

D9.1 : H - Requirement No. 1 [1]

The procedures and criteria that will be used to identify/recruit research participants must be submitted as a deliverable. The informed consent procedures that will be implemented for the participation of humans must be submitted as a deliverable. Templates of the informed consent/assent forms and information sheets (in language and terms intelligible to the participants) must be submitted as a deliverable. The applicant must clarify whether children and/or adults unable to give informed consent will be involved and, if so, justification for their participation must be included in the grant agreement before signature. In case children and/or adults unable to give informed consent are involved, details on how the consent of the legal representatives (and assent, when applicable) will be acquired must be included in the grant agreement before signature. The applicant must clarify whether vulnerable individuals/groups

will be involved, and the measures to protect them and minimise the risk of their stigmatisation must be included in the grant agreement before signature. The applicant must clarify whether invasive physical procedures will be used and what measures will be taken to minimise possible pain, and include this information in the grant agreement before signature. Details on incidental findings policy must be included in the grant agreement before signature. Copies of opinions/approvals by ethics committees and/or competent authorities for the research with humans must be kept on file.

D9.2 : HCT - Requirement No. 2 [1]

In case of use of human cells/tissues available commercially, details on cell/tissue types and provider must be kept on file. In case human cells/tissues are obtained within the project, details on cell/tissue types must be kept on file. In case human cells/tissues are obtained from another project, details on cell/tissue types must be kept on file. In case human cells/tissues are obtained from a biobank, details on the cell/tissue types and on the biobank and access to it must be kept on file. Copies of relevant documents for using, producing or collecting human cells or tissues (e.g., ethics approval, import licence, accreditation/designation/authorisation/licensing) must be kept on file.

D9.3 : A - Requirement No. 3 [1]

The beneficiary must confirm that relevant authorisations for animal experiments (covering also the work with genetically modified animals, if applicable) have been obtained, and kept on file. General information on the nature of the experiments and the procedures to ensure animal welfare and adherence to the Three Rs principle must be provided. The beneficiary must confirm that training certificates/personal licenses of the staff involved in animal experiments have been obtained and are kept on file.

D9.4 : EPQ - Requirement No. 4 [1]

Further information about the possible harm to the environment caused by the research and the measures that will be taken to mitigate the risks must be provided. The beneficiary must confirm that authorisations for relevant facilities (e.g. security classification of laboratory, GMO authorisation) have been obtained, and are kept on file. The beneficiary must confirm that appropriate health and safety procedures conforming to relevant local/national guidelines/legislation are followed for staff involved in this project.

Schedule of relevant Milestones

Milestone number[18]	**Milestone title**	**Lead beneficiary**	**Due Date (in months)**	**Means of verification**

1.3.4. WT4 List of milestones

Milestone number[18]	Milestone title	WP number[9]	Lead beneficiary	Due Date (in months)[17]	Means of verification
MS1	Assessed biochemical and safety characteristics, availability and exploitability of the selected residues also on the basis of the updated state of the art review(s)	WP1	2 - UNIBO	7	Assessed biochemical and safety characteristics, availability and exploitability of the selected residues also on the basis of the updated state of the art review(s)
MS2	Recovery of protein content from the selected feedstocks	WP2	4 - UNIPR	15	Recovery of protein content from the selected feedstocks
MS3	Recovery of non-protein compounds from solid fibre-rich fractions	WP2	6 - CELAB	18	Results of extraction trials documented in D2.3., datasets shared with consortium in meeting presentations and in the PROLIFIC intranet
MS4	Achieved protocol for biochemical/ biocatalytic conversion of protein extracts to enhance properties related to the target applications	WP3	4 - UNIPR	18	Achieved protocol for biochemical/biocatalytic conversion of protein extracts to enhance properties related to the target applications
MS5	Selection of 2 soluble proteases and 2 nanobiocatalysts for the effective hydrolysis of proteins	WP3	4 - UNIPR	18	Experiments with a set of enzymes finalised and results assessed in terms of yield
MS6	Achieved protocol for polymer formulation for packaging	WP3	2 - UNIBO	36	Result of protocol development and application presented on meeting and shared in intranet, documented in D3.4
MS7	Selection of fractions for food applications	WP4	7 - IGV	32	Assessment and ranking of fraction according to defined criteria and benchmarks presented and discussed with partners at monthly or biannual meeting
MS8	Selection of fractions for cosmetic applications	WP4	12 - COSM	34	Assessment and ranking of fraction according to defined criteria and benchmarks presented and discussed with partners at monthly or biannual meeting
MS9	Selection of fractions for polymer and	WP4	18 - UNIPI	40	Assessment and ranking of fraction according to defined

Milestone number[18]	Milestone title	WP number[9]	Lead beneficiary	Due Date (in months)[17]	Means of verification
	composite formulations for packaging applications				criteria and benchmarks presented and discussed with partners at monthly or biannual meeting
MS10	Selection of fractions for feed applications	WP4	14 - NS	34	Selection of fractions for feed applications
MS11	Technological transfer protocols scaled-up and validated	WP5	9 - BBEPP	38	Scale-up of extraction protocols carried out, documented including characterisation of extracts obtained
MS12	Feasible and low-carbon footprint product prototypes developed and technologies integrated	WP5	7 - IGV	46	product factsheets with KPIs for environmental impact of LCA tested products and outline of value chain development
MS13	Products to be analysed in Tasks 6.3 and 6.4 have been selected and qualitative data have been delivered by the partners	WP6	1 - FHNW	18	Products to be analysed in Tasks 6.3 and 6.4 have been selected and qualitative data have been delivered by the partners
MS14	LCA, CEA, safety & regulatory compliance assessment has been performed for at least one product prototype for each per industrial sector	WP6	1 - FHNW	46	LCA, CEA, safety & regulatory compliance assessment has been performed for at least one product prototype for each per industrial sector
MS15	Proper planning for dissemination, communication and networking activities prepared and execution of these activities launched	WP7	11 - TALOS	5	Proper planning for dissemination, communication and networking activities prepared and execution of these activities launched
MS16	Dissemination, communication, networking and exploitation activities successfully performed and relevant documents prepared	WP7	11 - TALOS	48	Dissemination, communication, networking and exploitation activities successfully performed and relevant documents prepared

1.3.5. WT5 Critical Implementation risks and mitigation actions

Risk number	Description of risk	WP Number	Proposed risk-mitigation measures
1	Perishability of selected residues, if improperly stored (LOW)	WP1, WP2, WP5	The feedstock residues will be transported under refrigeration conditions and stored at T= -20°C.
2	Extraction of protein and/or other components fractions fail to reach the desired target of efficiency and sustainability (MEDIUM)	WP2, WP5	Extraction methods, particularly of recalcitrant proteins, might be challenging, resulting in low efficiency non-economical process. To minimize this risk, different state-of-the-art technologies and innovative technologies for protein extraction will be tested. In particular, it will be assessed to which extent the extraction efficiency and economics are improved by initial innovative biomass pre-treatments. This will generate a set of different methods tailored to the different matrices, among which only the most efficient and performing methods, will be scaled up. Type of legume, fungi and coffee by-products with the best performance, will be selected. Added functionalities will be sought versus actual commercially available vegetable proteins. First techno economical assessment will support the decision-making process.
3	Extractions of protein with enzyme-assisted procedures fails to reach the desired target (MEDIUM)	WP2	To minimize the risk during enzyme-assisted procedures, the selection of the enzymes could be thought again by removing the poor yields reactants. This could finally allowed to obtain a particular enzymatic protocol for each substrate. In order to maximize recalcitrant proteins extraction, all the procedures will be optimized by modulating the temperature selection and the experimental conditions by applying proper equipment for stirring/homogenize the mixtures obtained.
4	UAE performance is below the expectations (LOW)	WP2	To minimize this risk, an optimization of the performance of the UAE will be carried out by means of a Surface Response Methodology. Most influential process parameters (such as amplitude, power or biomass ratio) will be identify and evaluated through a Design of Experiments (DoE). In addition, the use of solvents or heating will be also evaluated as enhancers of the UAE performance.
5	Subcritical Water Extraction (SWE) of de-proteinized mushroom for the extraction of Chitin/Chitosan may lead to clogging impacting the extraction yield (MEDIUM).	WP2	Pre-treatment of the raw material will probably lower the risk. The optimization of the particle size of the de-proteinized mushroom during the grounding step will indicate us the minimum particle size to be used to avoid the clogging of the filters of the pilot-scale SWE extractor. Other alternatives will be also used, such as, thermo- and pressure resistant plastic bags will be inserted to protect the filters of the extractor.

Risk number	Description of risk	WP Number	Proposed risk-mitigation measures
			Where appropriate, a metallic packing material will be used to minimise the clogging problems.
6	Degradation of bioactives (carotenoids, polyphenol, phytosterols, caffeine) may occur during protein extraction or concentration of fraction (MEDIUM)	WP2, WP3	Gentle pre-treatment and post-treatment, oxygen-free environment and low temperature will be set. An alternative route may also be considered in the entire process (extraction of sensitive compounds before protein extraction).
7	Residues from extractions are not miscible into the polymeric matrix and are not reinforcing agents in composites (MEDIUM)	WP3	Cellulose and hemicellulose fibres, extracted in Task 2.2.1, will be used to reinforce polymers
8	Biological activities (enzymes and microbial inhibitions) give non-repeatable results (MEDIUM)	WP4	Use standardised protocols and perform more replications to improve the reliability of the results
9	Pilot scale-up of proteins and fibres extraction from feedstock residues is not feasible from environmental point of view (LOW)	WP5	Environmental, energetic and cost efficiency together with performance versus cost ratio will be considered during scale-up. Methods to recovery and recirculate water, will be considered.
10	Processability of food grade ingredients is not economically feasible (MEDIUM)	WP5	Perform prototyping trials with close benchmark analysis of commercially available alternatives
11	Cereal-based and vegetarian/ vegan food products generated are at risk of microbial spoilage (LOW)	WP5	Adjust the standard baking and extrusion technologies for generating products with low humidity (planetary roller extrusion and sour dough technology)
12	Low dispersion and/or solubility of the new extracts in meat matrix (MEDIUM)	WP5	Change of one or more phases of the processing techniques (i.e. appropriate changes of the process times and temperatures and the use of new strategies during the mixing of the ingredients) to help the dispersion and/or the solubility of the new ingredients in meat products. The new manufacturing process will be applied without affecting the technological properties of the raw meats and the sensory profile of the final products.
13	Poor suitability of chitosan coatings on ham surface during process (MEDIUM)	WP5	Use of different technologies (i.e. dipping, spraying) and supports (i.e. nets) to improve the effectiveness of the new coatings during the critical phases of ham ripening. Different strategies will be combined (i.e. addition of molecules with antifungal activities) to ensure the performance of the new coating, without affecting the sensory profile of final products.
14	Cosmetics formulations are unstable with limited shelf-	WP5	Introduction of natural preservatives in the formulations. Repeated shelf-life trials

Risk number	Description of risk	WP Number	Proposed risk-mitigation measures
	life (MEDIUM)Introduction of natural preservatives in the formulations. Repeated shelf-life trials		
15	Solubility and accessibility of the fractions in feed (LOW)	WP5	Addition of feed grade emulsifiers or dispersants
16	No consumption of the feeds by animals during in vivo trials (MEDIUM)	WP5	Addition of pallatant or sweetener
17	Prototype production processes are not economically feasible for the application envisaged (MEDIUM)	WP5	Perform CEA and focus at application which are less cost sensitive (non commodity markets)
18	Incomplete delivery by a specific partner of the data required for performing the LCA or CEA of the corresponding technology (LOW).	WP6	Missing data will be estimated based on literature, similar technologies and expert knowledge from the consortium. Such estimations will have a lower accuracy but good level of consistency and the corresponding assumptions will be clearly indicated in the final deliverable.
19	Project activities and results are poorly disseminated/ lack of stakeholder interest (LOW)	WP7	The dissemination work plan (WP7) outlines a comprehensive set of dissemination activities, which range from networking to online communication, to workshops and international conferences. Regular reports on communication and dissemination activities will ensure that all activities stay in line and any indication of poor results (identified by monitoring activities) will be addressed on time.
20	Premature departure of skilled personnel (MEDIUM)	WP8	Hiring ambitious people. Stimulate commitment of personnel by offering career perspectives
21	A partner leaves the consortium due to e.g. bankruptcy, company take-over or budget revisions (MEDIUM)	WP8	The supervisory board will examine if the remaining partners can support the tasks. If not, a new partner will be attracted to replace the missing one.
22	Conflicts can arise among the participants (MEDIUM)	WP8	The Administrative Coordinator and the SC will take actions to overcome the disagreements. At first instance, the project coordinator will try to solve the problem by mediating and negotiating. The SC will discuss the problem striving for a cordial settlement. If necessary, they will vote on a resolution of the conflict according to the procedures described in the signed CA.

accessibility and an efficient quality assurance and quality control of the submitted data. By doing this, confidentiality and data protection will be carefully taken into account.

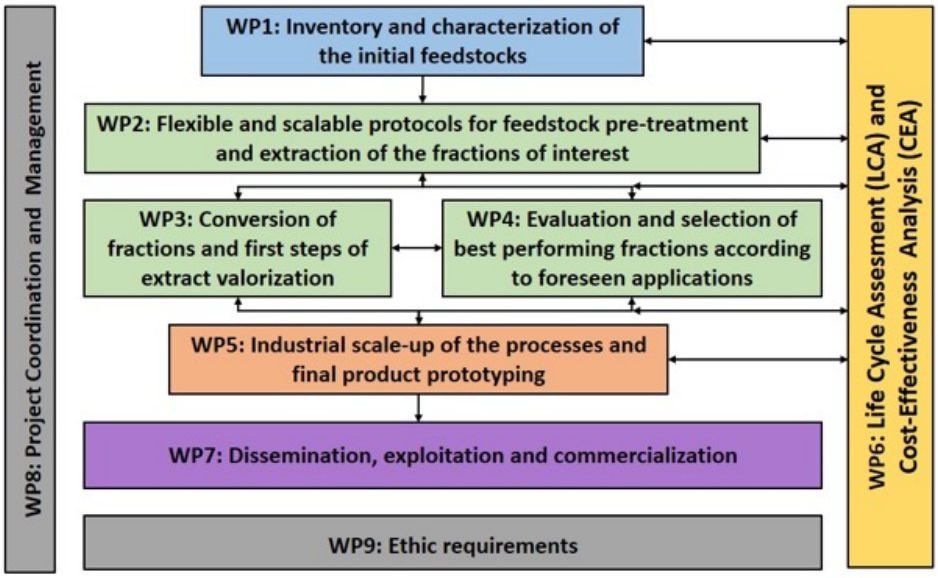
3 Implementation

3.1 Work plan

Overall structure of the work plan

A competitive work plan was designed with the aim of reaching the objectives of the PROLIFIC project, ensuring to fulfil its goals to reach a sustainable use of selected agro-industrial residues by optimising and validating flexible cascades of extraction/conversion processes, and by integrating the recovered protein and bioactive fractions into food, cosmetic, packaging and feed product prototypes. The ambition of the PROLIFIC project is to create sustainable value chains in the agriculture and food processing sectors, in order to assist Europe in the transition towards a green and circular economy. The project objectives have been used as the starting point for planning the work that needs to be implemented within the project (Section 1.1). All the actions that are required to meet the objectives of this Research and Innovation Action project have been collectively analysed by the interdisciplinary consortium and were driven by industrial developers and end users according to their market sectors and to actual EU and local regulatory constraints. The different partners contributed to draft the work plan on the basis of their relevant fields of knowledge, capacity, infrastructure, resources and industrial value chains. This industry-driven planning process has greatly facilitated the design of a clear, comprehensive and realistic work plan (see Gantt Chart and Pert Diagram below).

PROLIFIC project PERT diagram of different WPs showing their interdependencies



The present work plan is composed of a total of eight WPs to address the scope of the present BBI topic. All WPs are led by a WP leader and co-WP leader, at least one of them being an industrial partner, in order to ensure the clear focus on the targeted product prototypes of the PROLIFIC project.

The **technical/demonstration activities** are divided into 5 Work Packages (WPs 1-5), each with clear tasks that will enable the S&T objectives to be fulfilled through a coherent structure that will also facilitate project management. In summary:

WP1 will map the availability and sustainability of the selected feedstocks and produce an updated state-of-the art review on existing processing technologies and commercial products and on foreseen technical and regulatory specifications of the final extracts. This WP will also determine the molecular and safety characteristics of the feedstocks.

WP2 and WP3 are the technical core of the project dedicated to the optimisation of a flexible biorefinery cascading approach to extract, convert and valorise (by means of sustainable and economically viable laboratory techniques) the most abundant bioactive fractions (proteins/peptides, fibres, chitosan, carotenoids, polyphenols, phytosterols, caffeine) present in the different selected agro-industrial residues.

WP4 will assess the safety compliance and biochemical and functional characteristics of the WP2/WP3 final fractions, and will select, under the guide of end users, the most promising fractions for the foreseen food, cosmetic packaging and feed final product prototypes.

WP5 will bring **at pilot scale** the previously optimised (WP2 and WP3) extraction/conversion processes and include the selected fractions as ingredients additives in final product prototypes such food (cereal-based, vegetarian/vegan and meat), cosmetic (e.g. creams, toothpaste), packaging (food films, cosmetic jars, coffee capsules) and feed products for pigs and poultry. All these activities will be developed at least at TRL 5, while some will reach TRL 6-8 (Table 2). Before management, the work plan is complemented by **transversal activities** (WP6, 7):

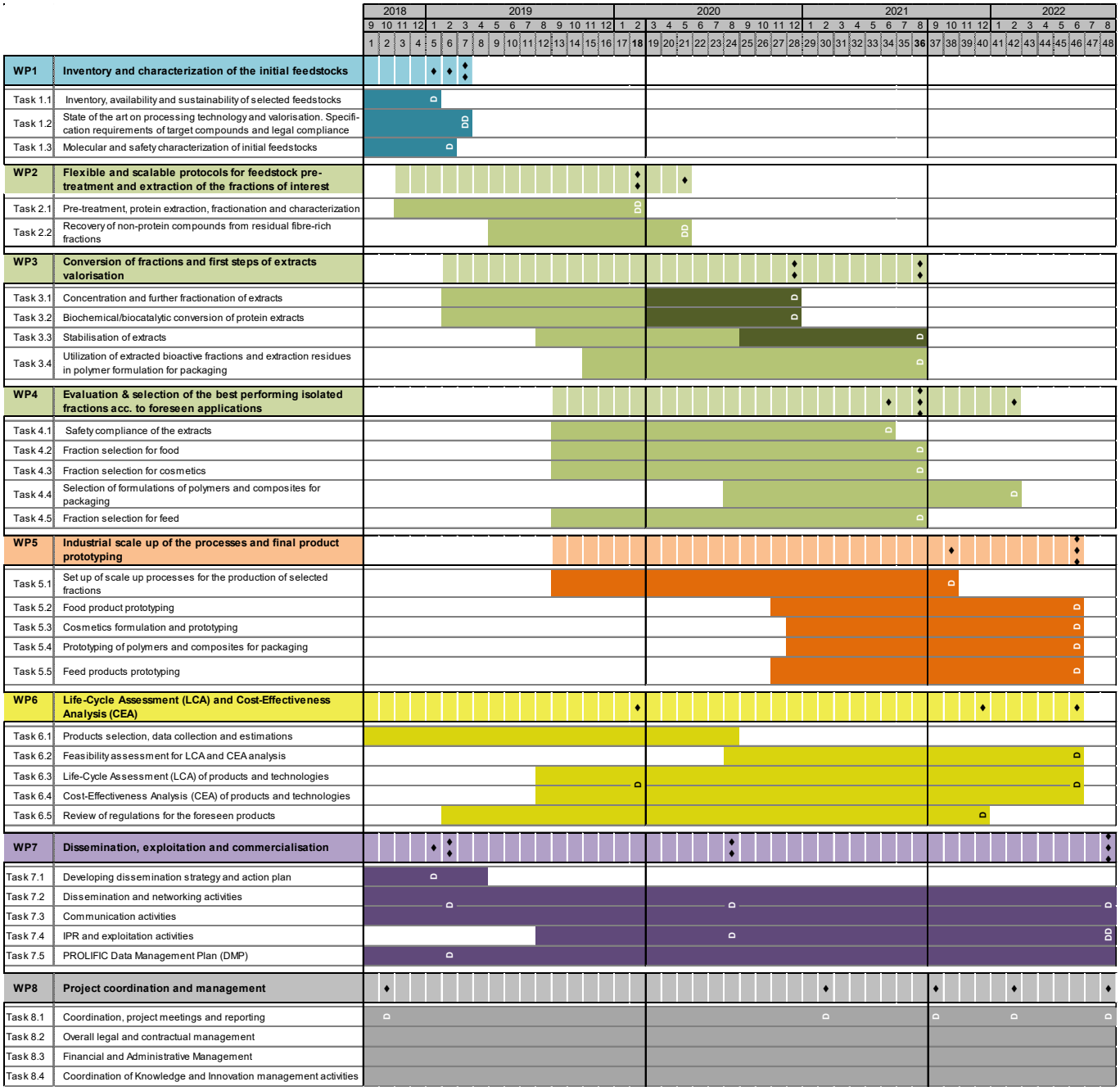
WP6 will collect data from WPs1-5 regarding for the elaboration of Life Cycle Assessment and Cost Effectiveness Analysis of the previously developed processes and products. A comprehensive review of regulations for the foreseen products will also be provided.

WP7 will ensure a broad dissemination and exploitation of project's end users and other stakeholders.

WP8 coordination and managing activities will ensure that all previous components and actors of the project will follow the work plan and stay in line with the EC requirements. Finally, **WP9** will ensure the fulfilment of project's **ethic requirements.**

The research and development work is complemented by a detailed risk assessment. The potential risks have been identified and contingency plans proposed (Table WT5 of Annex 1 Part A).

Table 11. Gantt chart showing the timing of the different WPs and their components outline



4 Members of the consortium

4.1 Participants

	1. Fachhochschule Nordwestschweiz FHNW	Switzerland

Partner Description

The School of Life Sciences (HLS) of the University of Applied Sciences and Arts Northwestern Switzerland (FHNW) is located in the center of the tri-national Basel area (CH – FR – DE). The HLS employs approximately 200 employees. The R&D activities of HLS encompasses a wide range of projects ranging from the development of new products and devices, and the optimization of production processes in terms of material and energy saving, to life cycle and risk assessments of such products. The four institutes of the HLS are Pharma Technology, Medical and Analytical Technologies, Chemistry and Bioanalytics (ICB) and Ecopreneurship (IEC). Approximately 50 and 60 scientists work at the two institutes involved in Prolific, IEC and ICB, respectively. The core activities of IEC include Environmental Technologies with a strong expertise in membrane-based filtration processes, Environmental Biotechnology (focus on immobilized biocatalysts) and Sustainable Resources Management. The activities of the ICB of the FHNW encompass modern chemical synthesis, analytics and bioanalytics, chemical engineering and nanotechnology. No Third party associated to FHNW is involved.

Main tasks attributed in the project

- Analysis of heavy metals (WP1 and WP4)
- Immobilization and protection of protease, set of proteolytic process (WP3 and WP5)
- LCA and CEA (WP6 coordinator)
- Administrative coordination and management (WP8, project coordinator)

Persons involved	Gender
Prof. Philippe F.-X. Corvini: Head of IEC and professor of environmental biotechnology has been involved in FP7 as coordinator of Minotaurus and manager of Kill Spill as well as main partner in Water4Crops, Sunflower, Bioclean. He is currently partner of H2020 Inmare and Mad4Water. He is also Vice-President of the European Federation of Biotechnology (EFB). P. Corvini's research field is microbial metabolism and biocatalytic processes. P Corvini acts as **administrative coordinator** of the Prolific project and will work on proteolytic treatment in WP3 and WP5.	M
Prof. Patrick Shahgaldian is professor of molecular nanotechnology and leads a research group devoted to the development of nanomaterials with molecular recognition and biocatalytic properties. He is partner in the H2020 Inmare and Mad4Water projects. He will be responsible for the supervision of activities dealing with the immobilization and protection of the proteases in WP3 and WP5.	M
Prof. Christoph Hugi is the leader of the Resource Management Group and has extensive experience in frameworks to assess the sustainability of existing and newly developed technologies, e.g. life cycle assessments (LCA) of organic photovoltaic cells and new wastewater treatment technologies, risk assessments (RAs) of new wastewater treatment technologies and cost-benefit analysis (CBAs) of solid waste treatment technologies.	M
Emmanuel Oertlé is a biochemical engineer with experience in sustainability and technology assessment. He is lecturer for Lifecycle Assessment (LCA) and his research activities are focused on sustainable resources management. Recent projects are dealing with the development of assessment tools, water reuse technologies and resource efficiency in industries. E Oertlé is leader of WP6.	M
Dr. Markus Lenz is a senior scientist specialized in the chemical speciation of heavy metals and selenium. He will be responsible for the analyses of heavy metals using IC-HPLC-ICP-qqqMS. He is partner of the H2020 project SCALE and has participated to several FP7 projects	M

Relevant publications

1. Gasser Christoph Andreas, Monika Čvančarová M, Ammann EM, Schäffer A, Shahgaldian P, Corvini PFX (2017). Sequential lignin depolymerization by combination of biocatalytic and formic acid/formate treatment steps. Applied Microbiology and Biotechnology 101: 2575-2588.
2. Correro MR, Moridi N, Schutzinger H, Sykora S, Corvini PF-X, Shahgaldian P (2016). Enzyme protection in an enzyme-thin and soft organosilica layer. Angewandte Chemie International Edition 55: 6285-6289

3. Moridi Negar, Corvini PF-X, Shahgaldian P (2015). Reversible supramolecular bioconjugation for the design of biocatalytic functional materials. Angewandte Chemie 127: 15013-15017.
4. Malinauskienė M, Kliopova I, Hugi C, Staniškis, JK (2017). Geostrategic Supply Risk and Economic Importance as Drivers for Implementation of Industrial Ecology Measures in a Nitrogen Fertilizer Production Company. Journal of Industrial Ecology. doi:10.1111/jiec.12561.
5. Gasser CA, Ammann EM, Schäffer A, Shahgaldian P, Corvini PF-X (2016). Production of superparamagnetic nanobiocatalysts for green chemistry applications. Applied Microbiology and Biotechnology 100(16): 7281-7296.

Relevant Projects:

1. FP7 KBBE Minotaurus (Coordinator, Immobilization of enzymes and microorganisms for wastewater treatment)
2. Materanet Delicate (Coordinator) Immobilization of enzymes for wastewater treatment
3. Swiss National Science Foundation (NRP 66) Depolymerization of lignin using lignin modifying enzymes
4. FP7 Water4Crop: Immobilization of laccase onto nanoparticles for wastewater reuse
5. H2020 Inmare: Immobilization of laccase of marine microbial origin for wastewater treatment

Relevant Patents

- P. Shahgaldian, P.F.-X. Corvini, Y. Dudal, N. Moridi; 2015 Patent (pending) EP15166312.7 “Method of reversibly functionalizing a solid surface”
- G. Hommes, P. F.-X. Corvini, Cumbo, A.; Shahgaldian, P.; Dudal, Y.; 2014 PCT (pending) WO2014118247A 1“Method of producing Nanobiocatalysts”
- P. Shahgaldian, A. Cumbo, P. Corvini, M. R. Correro, 2013 European Patent (pending) EP13178504 “Biocatalytical Composition”
- P. Shahgaldian, A. Cumbo, P. Corvini, 2010 European Patent (granted) EP10177028 “Preparation of a molecular recognition element”

Significant Infrastructure and Equipment

It possesses state of the art lab facilities such as pilot-scale lab, trace analysis lab for organics and inorganics, equipped with cutting-edge devices. Pilot scale MBRs and filtration systems for enzymatic treatment and fractionation as well as lab-scale fermenters for the production of enzymes. Advanced analytical equipments (GPC, LC-MS triple quad, GC-MS, IC-HPLC-ICP MS (triple quad), etc.) for analyzing products resulting from enzymatic treatment and inorganic contaminants.

	2. Alma Mater Studiorum - Università degli Studi di Bologna UNIBO	Italy

Partner Description

UNIBO has a considerable experience in managing EU-funded projects. Referring to JTI-BBI actions, UniBO is BBI-Associated member and participating to 3 projects as partner. In H2020-SC2 Food & Bioeconomy area, to date Unibo has been successful in more than 20 proposals, coordinating 3 of them. UNIBO also participated in FP7 with more than 230 funded projects in different specific programs (acting as coordinator in 47 of these). The *Plant and Algae Biology and Biotechnology Unit* of the Department of Biological, Geological and Environmental Sciences (BIGEA) offers wide expertise in the characterization (metabolomics, proteomics, mass-spectrometry and biological activity assays) and in the isolation of biomolecules from the plant bio-wastes/by-products. The *Polymer Science Unit*, belonging to the Department of Civil, Chemical, Environmental and Materials Engineering (DICAM), is expert in production of bio-based polymers and advanced materials, such as nanocomposites, for packaging, coating, biomedical applications.

Main tasks attributed in the project

The Unit of Plant and Algae Biology and Biotechnology (BIGEA) unit will co-coordinate the project and lead WP1 (dr. A. Tassoni) and take part to WP4 on the basis of its expertise in the extraction and use of bioactive compounds (such as polyphenols, fibers) from agrofood residues. The Unit of Polymer Science (DICAM) will lead WP3 and participate to WP5 on the basis of its expertise in the preparation of bio-based polymers and composites exploiting the extracted molecules/fibers from legume/coffee biowaste.

Persons involved	Gender
Dr. Annalisa Tassoni: PhD in cell Biology and Assistant Professor of Plant Biology. Plant biologist with a wide and qualified long-term expertise in the metabolomics of phytochemicals and their biotechnological and food applications as well as industrial production of plant phytochemicals via bioreactor technology from agro-industrial biomass. She was coordinator of FP7-Capacities project BIORICE and partner in BAKE4FUN and she's actually involved in H2020 projects NoAW and AgriMax (2016-2020). A. Tassoni will act as the **Scientific Coordinator** of the Prolific project.	F
Dr. Annamaria Celli: PhD in Materials Science Engineering, Assistant Professor in Polymer Chemistry, Coordinator of the Polymer Science research group at DICAM; scientific responsible for research contracts with National and International companies; co-author of many publications in peer-reviewed journals and some patents. She is the responsible of UNIBO for the H2020 projects NoAW and AgriMax (2016-2020).	F
Dr. Martino Colonna: PhD in Industrial Chemisty, Assistant Professor in Polymer Chemistry. His research interests are directed to the synthesis and characterization of new polymer materials (mainly polyesters and polycarbonates), preparation of composites and nanocomposites. Co-author of several publications on peer-reviewed journals and involved in H2020 projects NoAW and AgriMax (2016-2020). He is author or more than 20 patents mainly related to polyester synthesis and modification, also using monomers derived from renewable resources.	M
Dr Laura Sisti: PhD in Industrial Chemistry, Assistant Professor in Polymer Chemistry since 2008. Her research interests are directed to the synthesis and characterization of new polymer materials, preparation of composites and nanocomposites. She's co-author of several publications in peer-reviewed journals. She's involved in H2020 projects NoAW and AgriMax (2016-2020).	F
Dr. Maura Ferri: PhD in Cell Biology and Physiology and Researcher in Chemistry and Biotechnology. She is expert in plant secondary metabolism and in bioreactor production of phytochemicals from agro-industrial biomass, from the extraction to the assessment of several biological activities. She was involved in the FP7-Capacities project BIORICE and she is actually working in the H2020 project AgriMax (2016-2020).	F

Relevant publications, and/or products, services

1. Vannini M., Marchese P., et al. (2016) Block and random copolyamides of poly(m-xylylene adipamide) and poly(hexamethylene isophthalamide-co-terephthalamide): methods of preparation and relationships between molecular structure and phase behavior, Polym. Eng. Sc., 55, 1475-1484.
2. Gioia C., Banella M.B., et al. (2016) Advances in the synthesis of bio-based aromatic polyesters: novel copolymers derived from vanillic acid and ε-caprolactone, Polym. Chem., 7, 5396-5406.

3. Kalia S., Celli A., et al. (2013) Surface modification of plant fibers using eco-friendly methods for their application in polymer composites, textile industry and antimicrobial activities: A review, J. Environ. Chem. Eng., 1, 97–112.
4. Ferri M, Bin S, Vallini V, Fava F, Michelini E, Roda A, Minnucci G, Bucchi G, Tassoni A. (2016) Recovery of polyphenols from red grape pomace and assessment of their antioxidant and anti-cholesterol activities. New Biotechnology, 33: 338-344.
5. Ferri M, Graen-Heedfeld J, Bretz K, Guillon F, Michelini E, Calabretta MM, Lamborghini M, Gruarin N, Roda A, Kraft A, Tassoni A (2017). Peptide fractions obtained from rice by-products, by means of an environment-friendly process, show in vitro health-related bioactivities, PLoS ONE 12(1): e0170954.

Relevant Projects:

1. BIORICE (FP7 Capacities-SMEs, 2013-2015). The project aimed at recovering bioactive peptides from rice starch by-products and to use them as ingredients in cosmetic, nutraceutical and food products.
2. BAKE4FUN (FP7 Capacities-SMEs, 2013-2015). The project aimed at producing spelt-based bakery products functionalized with nano-encapsulated iron.
3. BIOCLEAN (FP7, 2012-2015). The project purpose was to develop innovative, ecoefficient pilot-scale and/or field validated biotechnological solutions for degrading (and valorising) plastic wastes in terrestrial and aquatic environments.
4. NoAW (H2020, 2016-2020). The project aims at developing a circular economy approach applicable to agricultural wastes, by converting agro-waste into a portfolio of eco-efficient products, such as bio-energy, bio-fertilizers, bio-packaging and bio-molecules.
5. AgriMax (H2020, 2016-2020). Through the selection of case-scenarios previously developed to a pilot scale and their industrial transfer, the project will disclose the potential of some agro-value chains to be converted in new food additives, packaging and agricultural materials.

Relevant Patents

Berti C. *et al*. Process of making compositions comprising bio-based therephthalic acid or bio-based dimethyl terephthalate. EP 2370491 B1 (2016)

Significant Infrastructure and Equipment

Equipment for polymer production, formulation and characterization: reactors for polycondensation in small and medium scales (50 g-1 Kg of polymer), twin-screw extruder, Brabender, injection molding press; NMR, FTIR, GPC (-IR, -UV), DSC, TGA, optical microscope, DMTA, Instron for tensile tests, contact angle measuring instrument, stress-controlled rheometer, Oxygen Index Test (LOI) apparatus.

	3. Innovacio i Recerca Industrial i Sostenible S.L. IRIS	Spain

Partner Description

IRIS is an advanced engineering company specialised in process monitoring and control, and process optimisation. Our Commercial Division develops and integrates turnkey advanced process monitoring and control solutions based on: NIR chemical monitoring (handheld device and in-line solutions); hyperspectral imaging solutions for foreign body detection and chemical composition analysis; data management solutions for data processing, decision support, automation and data analysis (big production data). Our ambition is to be the first choice for industry who want to transition their product models to the Industry4.0 paradigm, whereby real-time access to data about critical process parameters and product quality attributes can contribute to improved quality, and process efficiencies.

Our Innovation Division has four core research lines that we are actively persuing as part of our Innovation Roadmap for generating new knowledge and IPRs that we can channel into new products, services and solutions for our Commercial Division in the future:

1. Optical and spectroscopic technologies for process monitoring for in/on-line and PAT applications in food, pharmaceuticals, the process industries, environmental monitoring and other applications
2. Data Processing & Information Systems
3. Novel processing technologies (ultrasound, plasma, microwave…) for extraction process intensification
4. Sustainable processing technologies: waste valorisation strategies, biocompounds and biopackaging and new bio-based materials for the Circular Economy

The company was established in 2007 and currently employs 50+ staff with a turnover of > €4 M. We are based over two sites (Barcelona and Dublin) and we are fully equipped with our engineering and production workshops, food testing and wet chemistry laboratory, optics labs, and an electronics/telecommunications lab.

IRIS is ISO 9001:2008 certified and also is certified under UNE-EN-ISO 166002: Management Systems R&D&i. IRIS is an industry member if the following sectorial networks: SPIRE, SPIE and SECPHO in additional to regional industrial collectives such as AEBALL- the industry association of Baix Llobregat and Smart Space (Association of companies and organisations that specialise in 'smartification').

Main tasks attributed in the project

IRIS will be involved in the protein extraction process intensification by means of ultrasound and microwave. Tests will be performed at bench and lab scale (TRL 3-4) in different experimental configurations. The outcomes of these activities will be validated at higher scale (TRL 5-6) in WP5, in which IRIS will provide support to the involved partners also transferring the results achieved.

IRIS will also contribute with their skills in knowledge/IPR management, as well as in commercial branding, reaching industry and business planning. They will also contribute to knowledge transfer and training activities and gearing up for the post-project phase. Finally IRIS has extensive expertise in collaborative R&D project (FP7, H2020, BBI) and industry-driven business-to-business actions. Their project management procedures are ISO and UNE certified, and they have an internal PMO that has the necessary resources and skills to ensure smooth administrative and financial reporting to the project coordinator.

Persons involved	Gender
Dr. Ing. Elodie Bugnicourt. Leader of the Eco-Materials business unit in IRIS since 2009. Coordinated several European projects in the fields of bio-based packaging materials development, process monitoring, surface treatment. Previous R&D industrial experience in chemical industry especially related to reactive polymer formulations. Led and participated in many R&D projects aimed at developing new formulations, understanding mechanisms and structure-properties relationships, developing new analytical methods and modelling (for improved quality management); as well as collaborations with academic labs and other players in the supply and value chain. Author of several papers in peer-reviewed and technical journals, and talks at numerous international conferences.	F
Dr. Gianluca Belotti has a PhD in Chemical Engineering PhD from University of Rome La Sapienza. He has a solid scientific background in the field of Biotechnology and Chemical Engineering, holding a Master Degree in Industrial Biotechnology and a PhD in Chemical Engineering. As a result of his research activities he is the first author of some peer-reviewed papers and took part as speaker to some	M

international conferences. In 2013 he moved from laboratory activities to take the role of Project Manager firstly at Parco Tecnologico Padano and then in 2015 at LEITAT - Technological center. During this period he has been involved in several FP7, H2020, Life and ERA-net projects. Currently he is the Project Coordinator of the AgriMax project (ID 720719) and the Innovation Manager of the Pulpacktion project (ID 720744).	
Dr. Laura Rodríguez Turienzo. PhD in Biological Sciences (2011) from University of Santiago de Compostela, Spain. She has participated in a number of R&D regionally and nationally funded projects aimed at developing and validating new bio-based active edible packaging, physico-chemical characterization of novel foods and valorisation of food industry wastes. She is author of several papers in peer-reviewed journals and has participated in different national and international conferences.	F
Msc. David Montes Monteserin. Polytechnic University of Catalonia, Barcelona. Prior to joining the Production Processes group at IRIS as Mechanical Design Engineer, he gained wide experience in product design and development, including dosing systems for the chemical industry and both industrial and domestic appliances for the water treatment sector. He is specialised in design for manufacturing and is proficient in SolidWorks, design software which he has taught to several companies as a freelancer.	M

Relevant publications

1. Coltelli, M.-B. et al. State of the Art in the Development and Properties of Protein-Based Films and Coatings and Their Applicability to Cellulose Based Products: An Extensive Review. Coatings 6, 1–59 (2015).
2. Bugnicourt, E. et al. Simulation and Experimental Validation of the Denaturation of a Whey Protein-Based Coating During Convection and / or Infrared Drying on a Plastic Film and Influence on its Oxygen Barrier Properties Polym. Plast. Technol. Eng. 55, 1503–1511 (2016).
3. State of the Art in the Development and Properties of Protein-Based Films and Coatings and Their Applicability to Cellulose Based Products: An Extensive Review. Coltelli, M.-B.; Wild, F.; Bugnicourt, E.; Cinelli, P.; Lindner, M.; Schmid, M.; Weckel, V.; Müller, K.; Rodriguez, P.; Staebler, A.; Rodríguez-Turienzo, L.; Lazzeri, A. *Coatings* 2016,*6*, 1.
4. Processing and Validation of Whey-Protein-Coated Films and Laminates at Semi-Industrial Scale as Novel Recyclable Food Packaging Materials with Excellent Barrier Properties"; E. Bugnicourt, M. Schmid, O. Mc. Nerney, J. Wildner, L. Smykala, A. Lazzeri, P. Cinelli.; Advances in Materials Science and Engineering, vol. 2013, Article ID 496207, 10 pages, 2013.
5. Whey protein layer applied on biodegradable packaging film to improve barrier properties while maintaining biodegradability, Patrizia Cinelli, Markus Schmid, Elodie Bugnicourt, Jessica Wildner, Agostino Bazzichi, Irene Anuillesi, Andrea Lazzeri, Polymer Degradation and Stability. 07/2014; DOI: 10.1016/j.polymdegradstab.2014.07.007

Relevant Projects:

1. DAFIA: Biomacromolecules from municipal solid bio-waste fractions and fish waste for high added value applications. H2020-NMBP-BIO-2016
2. AgriMax: Agri & food waste valorisation co-ops based on flexible multi-feedstocks biorefinery processing technologies for new high added value applications. H2020-BBI-PPP-2015-2-1
3. THERMOWHEY: Project full name: Whey protein based barrier coatings with enhanced thermoformability and agglomerated formulations for a range of packaging applications. MANUNET

Significant Infrastructure and Equipment

Spectroscopic Laboratory equipped with several instruments, namely, a FTIR-ATR, 2 UV-VIS, an Extended-SWIR and a NIR spectrophotometers as well as a SWIR Hyperspectral imager. The associated staff are properly trained and skilled in chemometrics modelling by using a commercial suite (SOLO+MIA from Eigenvector) as well as ad hoc non-commercial chemometric tools (Matlab and Python toolboxes). Electronics, mechanical and optical workshops and laboratories equipped with up-to-date hardware and tools: signal generators, wide band oscilloscopes, PCB design [Altium] and building tools, development kits for programming microcontrollers, CNC-milling machines for electronic and mechanical prototyping, and CAD software, *etc*. Food and wet chemistry lab equipped with a DSC, a Surface Energy analyser, an HPLC, a Cobb tester, microbiological equipments, microspcope and microtome for structural analysis, migration cells, colourimeter (for colour and opacity measurement), Ford cup viscometers, water vapour permeation cells, turbidimeter, climatic chambers, manual and automatic coaters, electrospray, different drying technologies, plasma surface activation, *etc*.

UNIVERSITÀ DI PARMA	4. Università degli Studi di Parma UNIPR	Italy

Partner Description

The University of Parma is one of the oldest universities in Italy. Today it hosts about 25,000 students and about 900 permanent teaching and research staff. It is currently organized in 9 different departments. The **Chemistry Group** participating in the project is based at the **Department of Food and Drug**, and it is composed by bioorganic and food chemists. The group has a long standing experience in many research areas of biomass characterization, food and feed chemistry, food waste valorization. Specific areas of expertise include gross characterization of food-derived byproducts and biomass for valorization, protein extraction and characterization in food by electrophoresis and mass spectrometry, allergen determination in foods, characterization of free and proteic amino acids by chromatographic techniques, study of the composition of food-derived peptide fractions by chromatography and mass spectrometry, biomass recovery through enzymatic reactions, study of proteolysis in food products, implementation of simulated gastrointestinal digestion protocols for proteins, study of biofunctional activities of peptides and proteins, new methods for mycotoxin determination in food.

Main tasks attributed in the project

UNIPR group will take care of all the tasks related to the characterization of proteins in the residual biomass used (WP1 and WP4), and to set up extraction procedures for proteins at the lab scale (WP2), also including enzymatic-assisted extraction using proteases. Then, it will characterize the protein material extracted in order to assess its composition in view of further valorization (WP4). Moreover, it will take care of testing the reactivity of lipid material with the nitrogen material in order to prepare and characterize new products for cosmetic applications (WP4). It will also contribute to the safety tests by determining the mycotoxin content in the obtained fractions and ingredients (WP4). UNIPR will also develop hydrolytic treatment of proteins (WP3).

Persons involved	Gender
Stefano Sforza (PhD in Chemistry) is full professor of Organic Chemistry in the Department of Food and Drug of the University of Parma and visiting professor in the Food Chemistry chair of the University of Wageningen (The Netherlands). Previous Scientific Manager and WP leader in the PROSPARE project, WP Leader and Unit Leader in the NOSHAN project, member of the Scientific Advisory Board in the iFAAM project (FP7), WP Leader and unit leader in InDIRECT (H2020). His area of expertise is food chemistry, organic chemistry, bioorganic chemistry, with particular reference to the chemistry, the analysis and the properties of peptides and proteins in food and feed biomasses.	M
Tullia Tedeschi (PhD in Chemistry) is Assistant Professor in Organic Chemistry in the Department of Food and Drug of the University of Parma. Her area of expertise is chemistry of peptides and proteins, also including chemical synthesis of amino acid and peptide derivatives, and enzymatic reactions.	F
Augusta Caligiani (PhD in Food Science) is Assistant Professor in Food Chemistry in the Department of Food and Drug of the University of Parma. Her area of expertise include food lipids profiling, biomass fractionations and analysis.	F
Barbara Prandi (PhD in Food Science) is post-doc in Food Chemistry in the Department of Food and Drug of the University of Parma. Her area of expertise is the developments of method of analysis for determining the molecular composition of food and feed biomasses.	F

Relevant publications

1. C. Anzani, B. Prandi, T. Tedeschi, C. Baldinelli, G. Sorlini, P.A. Wierenga, A. Dossena, S. Sforza. Degradation of Collagen Increases Nitrogen Solubilisation During Enzymatic Hydrolysis of Fleshing Meat . Waste and Biomass Valorization, 2017, 1-7
2. J. Muller-Maatsch, M. Bencivenni, A. Caligiani, T. Tedeschi, G. Bruggeman, M. Bosch, J. Petrusan, B. Van Droogenbroeck, K. Elst, S. Sforza. Pectin content and composition from different food waste streams. Food Chemistry, 2016, 201, 37-45.
3. I.V. Nikolaev, S. Sforza, F. Lambertini, D.Y. Ismailova, V.P. Khotchenkov, V.G. Volik, A. Dossena, V.O. Popov, O. Koroleva. Biocatalytic conversion of poultry processing leftovers: Optimization of hydrolytic conditions and peptide hydrolysate characterization. Food Chemistry, 2016, 197, 611-621
4. C.I. Butre, S. Sforza, H. Gruppen, P.A. Wierenga. Introducing enzyme selectivity: a quantitative parameter to describe enzymatic protein hydrolysis. Analytical and Bioanalytical Chemistry, 2014, 406, 5827-5841.

5. P.P. Zanello, S. Sforza, A. Dossena, F. Lambertini, C. Bottesini, I.V. Nikolaev, O. Koroleva, T. Ciociola, W. Magliani, S. Conti, L. Polonelli. Antimicrobial activity of poultry bone and meat trimming hydrolyzates in low sodium turkey food. Food & Function, 2014, 5, 220-228.

Relevant Projects:

1. INDIRECT (H2020, 2016-2019) – Indirect valorization of biomass waste streams through insects (Partner, WP Leader).
2. VALORIBIO (Emilia Romagna Region, Italy, 2016-2018): Extraction of usable biomass from insects (Partner, WP LEader)
3. NOSHAN (FP7, 2012-2016) – Recovery of food waste and valorization into feed (Partner, WP Leader)
4. PROSPARE (FP7, 2008-2011) – Recovery and valorization of byproducts from poultry industry (Coordinator)
5. iFAAM (FP7, 2013-2017) - Integrated approaches to food allergen and allergy management (Member of the Scientific Advisory Board)

Significant Infrastructure and Equipment

The UNIPR group is perfectly equipped to carry on a wide range of advanced analytical determination on complex matrices and to perform enzymatic hydrolysis and small chemical syntheses and modifications. The group has the availability of three fully equipped laboratories for chemical synthesis, extraction and purification of organic compounds from food, sample preparation. Moreover, it has two instrumental laboratories, plus one additional laboratory hosting workstations. It has the availability of 1 semi-preparative HPLC lines, 1 HPLC/UV/fluorescence, 1 UPLC/ESI-MS, 1 UPLC/ESI-MS/MS, 1 HPLC/PDA/ESI-MS/MS, 1 GC-FID, 3 GC-MS, 1 spectrophotometer, 1 spectrofluorimeter, 1 thermocycler for PCR, 2 centrifuges, 1 peptide synthesizer, three workstations for advanced computer simulations. Moreover, it has full accessibility at the Interdepartmental Center for Measurements of the University to the following instruments available there: 1 LTQ-OrbiTrap (HR-MS), 1 MALDI TOF/TOF, 3 NMR.

SSICA	5. Stazione Sperimentale per l'Industria delle Conserve Alimentari SSICA	Italy

Partner Description

SSICA is a Public Institute for applied research, established in Parma in 1922 and operating on a national scale with the specific aim of promoting the technical and technological progress in the Italian fruit, vegetable, meat and fish processing industry. The basic activity of SSICA is applied research related to the production chain of the agrifood sector, by means of which SSICA tends to supply suitable solutions to the food preserving industry in terms of product and process innovation, technology, methodology and use of raw materials, ingredients and materials (plants and packaging) and improvement of production processes and services, and of the conditions that determine the competitiveness and productivity of enterprises. SSICA has gained a wide knowledge and experience concerning the food preservation, both at laboratory scale, from an analytical point of view, and at pilot plant scale. The Groups that will be involved in this projects are: Packaging Department, Meat Department, Microbiology Department and Production Technologies Department.

Main tasks attributed in the project

In WP1 SSICA will carry out the chemical characterization in terms of nutritional composition of legumes by-products and the microbiological characterization of all residues. In WP2 SSICA will perform the pre-treatment and the laboratory extraction of proteins and fibers from legumes by-products, together with the chemical characterization of related extracts. In WP3 will optimise formulations for coating of dried cured meat. In WP4 SSICA will contribute to safety assessment of extracts by the determination of mycotoxins and pathogens and to the specific analysis of extracts to be used for meat food products. In WP5 SSICA will carry out the scale-up of legumes by-products treatment at pilot level and the meat products prototyping. Besides, chitosan films and coatings will be applied on cured meat products.

Persons involved	Gender
Dr. Angela Montanari is Responsible of Packaging Department. She coordinates the research and service activities of the packaging department regarding compatibility between packaging and food products as well as packed product safety. She is an expert on the European and National legislation on packaging material. Finally, she deals with active packaging and biopolymer film. She was University lecturer and member of the Working group of the Italian Development Ministry on the classification of chemical substances (REACH) since Dec. 2003. She has published over 100 papers in national and international journals. She is inventor of three patents. She coordinated the European project BIOCOPAC and is currently coordinating the European project BIOCOPACPLUS-LIFE.	F
Dr. Chiara Zurlini, junior research, is working in the Packaging Department of SSICA since 2001. Graduated in Food Science and Technology at the University of Parma, she studies and works in the field of plastic and metallic materials for food packaging; in particular, she studies biopolymeric materials and their application for the development of edible films and coatings of food packaging and the valorization of food industry wastes. She also has experience in the extraction on laboratory scale and pilot-scale of proteins and fibers from by-products vegetable matrix.	F
Dr. Monica Bergamaschi, Degree in Chemistry and post-graduate Master in Food Science and Technology. Researcher at SSICA (Meat Department) since 1997, she has gained experience in cooked meat technology and to interdependency studies between meat quality, processing techniques and final product properties, with special emphasis on oxidation and colour stability. A supervisor of degree theses, has authored several papers and attended scientific meetings as speaker.	F
Dr. Nicoletta Simoncini, graduated in Food Science and Technology (2003), in 2007 she became a researcher of the Meat Products Department of SSICA. She has addressed quality and safety issues in meat products, with emphasis on environmentally friendly dry-curing. Also involved in EU-funded projects, she has been committed to non-invasive technologies. Author of several scientific papers and supervisor of degree theses, she has participated in scientific congresses as speaker.	F
Dr. Mariapaola Previdi graduated in Biology at Parma University is a researcher in Microbiological department of SSICA since 1988. Her research activity mainly deals with vegetable products and acid and acidified products. In the last years, her work dealt with microbiological feature of new food products and technologies with respect to health and commercial security (study of shelf-life, microbial challenge test). She is author of publications on scientific national and international papers.	F

Dr. Andrea Brutti is PhD in Veterinary Medicine on food production and control from Parma University. He is researcher in Production Technologies Area of SSICA since 2008. Its research deals with the improvement of food safety and the extension of food shelf life through the application of mild technologies. He has experience in management of pilot plants, improvements of industrial processes, development and application of technologies for food treatments and preservation and scaling up of processes of extraction of high value substances from different matrices. He is author of publications on scientific national and international papers.	M

Relevant publications

1. M. Bertoldo, L. Ricci, E. Umiltà, C. Righetti, T. Messina, C. Zurlini, A. Montanari, S. Bronco, "On the thermal behaviour of protein isolated powder from different legumes investigated by DSC and TGA analyses", *Journal of Applied Polymer Science*, 2017, under evaluation
2. E. Umiltà, C. Zurlini, A. Montanari, "Legumes by-products as source of proteins for bioplymers production", *8th European Symposium on Biopolymers ESBP 2015*, 15-18 September 2015, Rome (Italy)
3. G. Saccani, G. Barbieri, M. Bergamaschi, C. Blasi, M. Franceschini, G. Parolari, "Bioactive natural polyphenols in reformulated meat products. Effect on quality traits", *61° International Congress of Meat Science and Technology* (ICoMST), 23-28 Agosto 2015, Clermont-Ferrand, France
4. C. Schivazappa, R. Virgili R., N. Simoncini, S. Tiso, J.M. Rodriguez, "Application of the magnetic induction technique for the non-destructive assessment of salt gain after the salting process of Parma ham", *Food Control* 80, 92-98, 2017
5. E. Berni, I. Montagna, F.M. Restivo, and F. Degola, "Ochratoxin A Control in Meat Derivatives: Intraspecific Biocompetition between *Penicillium nordicum* Strains", *Hindawi Publishing Corporation Journal of Food Quality,* https://doi.org/10.1155/2017/8370106, 2017

Relevant Patents

1. "Metodo di estrazione di un polimero poliestere o cutina dagli scarti di pomodoro e polimero poliestere così estratto", n°0001419412, Inventors: I.M. Cigognini, R. De La Torre Carreras, M. Gomez-Cardoso Bernet, 2015
2. "Extraction method of a polyester polymer or cutin from the wasted tomato peels and polyester polymer so extracted", WO 2015/028299A1, Inventors: I.M. Cigognini, A. Montanari, R. De La Torre Carreras, M. Gomez-Cardoso Bernet, 2015 (Pending)
3. "Lacquer composition having cutin based resin, method for obtaining said lacquer and said cutin based resin", n° 0001426490, Inventors: C.M. Cedri, L. Cioni, X. Gomez, A. Montanari, V. Orlandi, D. Vareckova, T. Vlcek, 2014
4. "Biopolimero da scarti dell'industria alimentare", n°Conc. 1.399.500, Inventors: A. Montanari, C. Zurlini, 2013

Relevant Projects:

1. PHYTOME, FP7, "Phytochemicals to reduce nitrite in meat products" SSICA partner
2. PROCURED, FP7-SME "Optimization of the salting process for the production of healthier and higher quality dry-cured meat products with reduced and more standardized salt content" SSICA partner
3. LEGUVAL, FP7, "Valorisation of legumes co-products and by-products for package application and energy production from biomass" SSICA partner
4. BIOCOPACPLUS, LIFE, "Sustainable bio-based coating from tomato processing by-products for food metal packaging" SSICA coordinator
5. AGRIMAX, H2020-BBI-PPP-2015-2-1"Agri and food waste valorisation co-ops based on flexible multi-feedstocks biorefinery processing technologies for new high added value applications" SSICA partner

Significant Infrastructure and Equipment

SSICA has a broad range of analytical instruments for chemical and microbiological analysis and, in particular, for the chemical and physico-chemical characterization of raw meat and foodstuffs of animal origin (Texture Analyzers, Spectrocolorimeters, UHPLC_MS and UV and Fluorimetric detectors, UPLC-Orbitrap MS, GC-FID, GC-MS, GC, ICP-ToF-MS, NIR, Spectrophotometers, Fluorometers, horizontal and 2-D gel electrophoresis, semiautomated electrophoresis on polyacrylamide minigel). SSICA has pilot structures for the application at small scale of industrial lines and transformation processes. There are equipments for separation techniques as decanter and centrifugation, membranes separation, and various drying systems, as spray-drying, drum-drying, freeze-drying. Besides, systems for skin under vacuum and modified atmosphere packaging are available. SSICA has also instrumentations for plastic materials characterization and for migration tests (TGA, DSC, permeabilimeter). A scanning electron microscope with X-ray analysis is also available.

	6. Celabor scrl CELAB	Belgium

Partner Description

Celabor is Belgian scientific and technical center based in the Petit-Rechain industrial park near Verviers (Belgium). Celabor is an SME offering scientific and technical support to companies in the field of agri-food (nutrition and extraction), packaging, environment and textiles. Celabor have 20 years of research experiences and has been involved in a number of EU and national projects and is eager to maintain its position at the cutting edge of scientific knowledge. The "Food Technologies – Extraction" department of Celabor is equipped with a technological platform unique in Wallonia boasting an ATEX zone, a test hall constituted of pilot extraction equipment, a high performing laboratory and an experienced multi-disciplinary team. This team is working on the development of extraction, fractionation and purification processes with a "green" orientation (subcritical water and supercritical CO_2), the preparation of natural plant ingredients (polysaccharides, lipids and fatty acids, carotenoids, polyphenols, alkaloids, etc.) for the food supplements, cosmetics and phytotherapy markets. One of the major activities of the "Extraction" department is focused on the biomass valorisation processes and recovery of primary and secondary metabolite from co-product and industrial agro-waste.

Main tasks attributed in the project

On the basis of its expertise in the extraction and use of bioactive compounds, CELAB will develop the protocol for the extraction of non-proteic fractions of interest from residual solid fibers-rich fraction (WP2) using conventional and innovative extraction techniques (such as supercritical CO2 or subcritical water, ultrasounds). CELAB will lead WP4 and will characterize the extracts through various testing (safety, anti-aging, anti-oxidant, antimicrobial, barrier properties and migration in packaging) for their application in cosmetic, food and packaging. CELAB will also take part to WP5 through its pilot plant for the upscaling of ultrasounds and supercritical CO2 process. CELAB will participate in WP6 on the basis of its expertise in the field of LCA. And finally, CELAB will actively disseminate the results of the project through industrial clusters and Fairs.

Persons involved	Gender
PhD. Marie-Caroline JONVILLE is a botanist specialized in chemistry and pharmacology of natural products. Her PhD in Pharmacognosy (codirection between the University of Liège - Belgium and the University of Aix-Marseille II - France) constituted in bioguided fractionation of plants for antiplasmodial activity. Dr. Jonville published 12 papers in international journals. In Celabor, Dr. Jonville focuses her activities on green extraction and characterization of plant extracts by chromatographic techniques (HPTLC, UPLC/MS-MS) and biological activities (antioxidant, anti-aging, antimicrobial).	F
Amandine BERTON is a project leader at Celabor. She holds a Master in "Biomedical sciences" from the University of Liège. She focuses her activities on green extraction using pilot plants, including, supercritical CO2 and subcritical water applied to the extraction and fractionation of bio-based materials. She is also specialist in characterization of plant extracts by chromatographic techniques (UPLC/MS-MS) and formulation of bioactives through encapsulation technique.	F
Mathilde Fortier is a project leader in the Packaging Department of Celabor. She holds a Master degree in packaging sciences from *ESIReims previously ESIEC and* focuses her activities on cellulosic packaging materials (paper, cardboard), packaging development and LCA. She is currently involved in CORNET projects.	F

Relevant publications

1. Subcritical water extraction – from lab tests to kilo batch pilot trials. Kohnen S, Beauve C., Mullender J.-P., Lefèbvre P. International Workshop on Alternative Solvents for Extraction, Purification and Formulation – WAS, Avignon, France, June 4, 2015.
2. High pressure assisted processes: supercritical CO2 and subcritical water. Kohnen S.; International Congress on Green Extraction of Natural Products – GENP Avignon, France, April 16 – 17, 2013.
3. Composition of by-products from cooked fruit processing and potential use in food products. Aguedo, M; Kohnen, S; Rabetafika, N; Vanden Bossche, S; Sterckx, J; Blecker, C; Beauve, C; Paquot, M; JOURNAL OF FOOD COMPOSITION AND ANALYSIS; Volume: 27, 2012, pp. 6
4. Walextract, a research project for the preparation, characterization, evaluation and valorization of vegetal extracts: Example of rosemary extracts. Kohnen S., Beauve C., Dessy S., Mullender J-P, Mouithys-

Mickalad A., Béatrice Robert B. and Lefebvre P.; 25th International Conference on Polyphenols; Montpelier – 2010.1-69 5. Sequential biomass valorisation of Halophytes: High pressure extraction prior to biogas production. Jonville M-C, Dröge S., Beauve C., Ballmann P., Müller M.; Green Extraction of Natural Product GENP, Torino-2016
Relevant Projects 1. WALEXTRACT (ERDF, 2009-2014) Valorisation of sustainable vegetal resources from agriculture and forestry by the extraction of high added-value natural substances for applications such as natural preservatives. 2. SUBWEX (CORNET 2013-2015) Development of Subcritical Water Extraction (SWE) as a "green" extraction and fractionation processes applied to cereals straw and fruit grains. 3. BYPROVAL (CORNET, 2016-2017) New valorisation pathway for fruit/vegetable waste by a combination of extraction and biogas production 4. AFTERLIFE (H2020-BBI-JTI-2016, 2017-2021) Advanced Filtration TEchnologies for the Recovery and Later conversIon of relevant Fractions from wastewater. The project *proposes a flexible, cost- and resource-efficient process framed in the zero-waste and circular economy approach for the recovery and valorisation of the relevant fractions from wastewater.* 5. BARBARA (H2020-BBI-JTI-2016, 2017-2020) Biopolymers with advanced functionalities for building and automotive parts processed through additive manufacturing. Valorisation of side-stream fractions and residues from agro-food production into novel polysaccharides and functional additives.
Significant Infrastructure and Equipment Green technologies: Pilot-scale supercritical fluid extractor SFE-CO_2 (two units with different capacity), laboratory and pilot-scale subcritic water extractor, microwave assisted extractor, ultrasounds extractor (up to 50L), ASE equipment. Post-treatment: laboratory and pilot-scale freeze dryer, membrane filtration, preparative chromatography (CPC, prep-HPLC-DAC, MPLC) Biological activity evaluation: DPPH, ORAC, FRAP, antimicrobial bioautography using HPTLC, anti-elastase, anti-collagenase, anti-tyrosinase. Analytical laboratory: ICP (-MS, -FID), UPLC (-DAD, -MS/MS), HPLC (-Fluo, -DAD, -ELSD), GC (-FID, -MS/MS), spectrophotometer, HPTLC system.

IGV	7. Institut für Getreideverarbeitung GmbH IGV	Germany

Partner Description

IGV GmbH is an independent SME and industry-focused service provider with experience and expertise in food processing, feed technology, microalgae biotechnology and accredited analytical laboratory services. Our service range includes scientific-technical services for product and process development, laboratory services, consulting as well as training and lifelong education. IGV GmbH offers a large number of extensive services within the following fields:

- applied product and process development for, agricultural and food industries, feed and petfood applications, pharmacy and environmental protection, as well as for nutraceuticals and cosmetic products
- development of environmental-friendly applications
- training and education
- extensive analysis of raw materials and products in our accredited labs

IGV GmbH develops innovative products, facilities and processes for crop processing and biotechnology as well as food ingredients/additives and related products. IGV is consultant for industrial partners and governmental organisations. The fields of activity are all steps of **processing** of different types of grain and all other plant materials (milling, fractionation, hydrothermal-treatment, extrusion, co-extrusion, baking). One task is the search for applications of by-products of the processing of seeds and other plant resources for food/ feed, nutraceutical manufacturing. Other plant materials and by-products that are in focus of our R&D projects are legumes, oilseeds, herbs and spices, inulin plants. Additional field of activity is the conversion into functional food, feed, pet food, aquaculture products.

Main tasks attributed in the project

The FOODTECH Unit of IGV will lead WP5 and takes part to WP2, WP3 and WP4 on the basis of its expertise in the fields of extraction, extrusion, analytical assessment and also expertise in food, feed and cosmetics formulation and prototyping. The PLANTTECH Unit of IGV will participate also in WP6 on the basis of its expertise in the field of regulatory affairs and validations, as well as in labelling and safety analysis. IGV will actively disseminate the results of the Project at well established Fairs and Exhibitions, such as ANUGA Foodtech (Köln), International Green Week (Berlin) and FIE (Europe wide).

Persons involved	Gender
Dipl. Chem.-Biol. Janos Petrusan MSc Drs. Head of Research Coordination Department with broad experience in FP6, FP7 and H2020 project scientific/technical management, but also in new product optimization, evaluation, characterization and technological transfer techniques. Will be responsible for technical and scientific support for optimization of scale- up procedures dully adhered to standard protocols.	M
Dipl.-Chem. Petra Sandau has extensive background in obtaining novel functional substances, will be responsible for the optimization of the extraction	F
Dipl.-Ing. Claudia Oestreich is familiar with national and international projects, partner for laboratory trials and industrial scale-up.	F
Dipl.-Ing. Katrin Hartung is actively working within the Processing department and participating in various projects together with industrial Partners. She will work on the technical part to identify suited vegetal sources and verify their protein content and composition, especially the content of essential amino acids.	F
Dipl.-Kaufm. Mario Krause has 2 decades years of experience in the market research, product placement strategies, product management, including marketing and sales. He will work on the market study and the financial project planning within the Business Plan.	M
Ms. Dipl.-oec.-troph (FH) Martha Walter has extensive expertise in the areas of networking, food law, product development and quality management in the food sector, she will be responsible for the regulatory part of the project.	F
Mr. Dipl.-Ing.Frank Kage has broad experience in international projects and will be responsible besides milling and fractionation, also in new product optimization, evaluation, characterization, technological transfer and validation techniques.	M

Relevant publications

1. Petrusan J., Huschek G., Rawel, H.M (2016): Protein-rich vegetal sources and trends in human nutrition. A Review, International Journal of Food Sciences and Nutrition, submitted
2. Petrusan J., Huschek G., Rawel, H.M (2015): Potentials of applying proteomics tools for characterisation of technological aspects of rapeseed protein production, Keynote Speech, YSM II, Phytochemical Society of Europe Meeting, 28 April, Murcia, Spain
3. Huschek G., Löwenstein Yv., Bönick J., Sievers S., Rawel H.M. (2015): Quantification of protein based allergenic traces in processed food products by UPLC/MS/MS and isotope marked peptides. Food Chemistry
4. Huschek G., J. Petrusan, Knapen R (2015): High purity vegetable protein concentrates for niche markets. A techno-functional approach- Keynote Speech, 8th Protein Summit, Rotterdam
5. Krause M., Knapen R. (2015): Investment Planning, Calculation basics and Marketing Opportunities for vegetal protein flakes, IGV Extrusion Workshop, August, Nuthetal, Germany

Relevant Projects

1. Project Coordinator (2016-) ProVegFlake (H2020-SME-INST 719783) High grade PROteins innovatively processed into VEGetal FLAKEs from plant source with enhanced nutritional benefits
2. WP Leader: SUPRA-BIO, FP7 Project (241640-2). Sustainable products from economic processing of biomass in highly integrated biorefineries
3. WP Leader (2012-2015): APROPOS FP7 (289170), Added value from high protein & high oil industrial co-streams
4. QA Manager (2012-2015): NOSHAN FP7 (312140) - delivering new sustainable protein formulations for pig and poultry production.
5. Project Coordinator (2011-13): *FOODSCAN*, EU-FP7-SME (286442) Biosensor platform for pesticide residue detection in foods

Significant Infrastructure and Equipment

IGV applies different procedures for follow-up and intermediary products: Aqueous extracts, Dried extracts (free from preservatives), Ethanol extracts, Lipophilic extracts of CO_2 – high pressure extracts. IGV operates also an accredited Analytical Laboratory and has several technological centers:

- Extrusion Technological Centre (single-, twin-screw and planetary roller extrusion, milling, sieving, fractionation, dry-mechanical separation, roasting, crumbling, expanding)
- Baking Technological Centre (fully equipped with ovens, fermentation chambers, kneaders, etc.)
- Extraction Technology Centre (Cell Disruption, Separation, Micro- and ultrafiltration, concentration, drying, vacuum evaporation)
- Laboratory Services Extraction (e.g.: Determination of antioxidative capacity of hydrophilic and lipophilic substances, Determination of pigments from photosynthesis)
- Closed tube PBR technology and MUTL PBR technology for microalgae cultivation
- Harvesting, dewatering and fractionation equipment

Areas of **general expertise** include food marketability (expert, reference: state of Brandenburg) cereals, trace-analysis of contaminants and residues (GC, LC/MS/MS), analytics of human and veterinary drugs (GLP like), microbiology, environmental expert (water and soil) including environmental risk assessment. Special expertise relies in regionality and authenticity identification of different cultivars via proteomics and marker identification methods validated at IGV.

- DAkkS accreditation D-PL-1402401 (since1994)
- EU GMP- certified, based on AMG, LGA BB
- QS GmbH- certified, field of FM-Monitoring of Fruit- and Vegetable (Residue Analytics)
- Governmental recognition official counter samples
- - for the Test Laboratory
- - five private cross-checking experts
- S2 Authorisation (pathogene germs)
- Official Drinking Water investigating Point

	8. Stolzenberger Bakery SB	Germany

Partner Description

STOLZENBERGER BAKERY is a modern German bakery, which from its foundation until now always tried to combine the classic and the modern. It was a successful approach, which may be easily measured by the thousands of daily customers who come to the stores each day looking for fresh bakery products. The product scale is large enough to become a complete offer for all generations and preferences. At Stolzenberger Bäckerei someone could find the whole scale of traditional German bread and fine confectionery products, but also the trendiest cakes and confectionery products, which are produced with the most modern German technology. Stolzenberger Bäckerei holds an important catering service with personalised approach to all segments of its customers, handling flexibly the special demands of specific groups. The bakery is present on a contractual basis at major events organized by the South-west German prefectures and other authorities, Main markets consist of the own store chain consisting of three modern stores and a dozen of mobile bakery stores, being also main supplier for the supermarket chains. The market chain develops everyday by contracting more and more partners, especially public institutions, groceries, supermarkets. Stolzenberger Bakery is a cradle where the traditional bread manufacturing is harmoniously combined with the most modern technologies, which are the solid basis for our future expansion.

Main tasks attributed in the project

SB Team will participate in WP4 supporting the food formulation efforts, whereas the new product development is at the forefront of the company's philosophy. In WP5 the Team of technologists will be actively involved in bringing the prototype products to industrial validation level. SB will also contribute to activities of WP6 on the basis of its expertise from previous EU Projects in the field of prototype validations, as well as in labelling. SB will spread the results of the Project at Exhibitions and Trade Fairs, such as IBA (Munich), Südback (Stuttgart) and Gastropan (Europe wide).

Persons involved	Gender
Mr. Reiner Stolzenberger, The Head of the Bakery is Mr. Reiner Stolzenberger, who is a Master Baker, with more than 35 years of experience. His work is to coordinate and to create always-fresh products with the support of well-trained professional personnel. He is also coordinating the catering activities.	M
Daniela Uhlmann: Production manager, she organises the daily work according to Production Plan. She will be in charge for innovative baked goods formulation and trials.	F
Kilian Stolzenberger: Administrative and Trade manager, he organizes the overall administrative and workflow of stores. He will be responsible for the Consumer Survey, Sensory and Knowledge Transfer at catering level.	M
Markus Kurz: Master Baker and Technological manager is the product development manager of the bakery with extensive expertise in new product development. He will be responsible for prototype product validation and technological validations.	M

Relevant references

1. Mr. Reiner Stolzenberger is the "Youth Responsible" of the German Baking Association of Baden-Württemberg.
2. The Bakery is official Training and Further Education Place for Bakers and Confectioners
3. Reiner Stolzenberger has an important track record for supporting start-ups worldwide

Relevant Projects:

1. Project Partner and WP Leader (2011-13): FOODSCAN, EU-FP7-SME (286442) Biosensor platform for pesticide residue detection in foods
2. Project Partner (2013-18): HealthyMinorCereals EU-FP7 KBBE (613609) An integrated approach to diversify the genetic base, improve stress resistance, agronomic management and nutritional/processing quality of minor cereal crops for human nutrition in Europe
3. Project Partner (2013-17) TRADEIT- FP7 KBBE (613776) Traditional Food: Entrepreneurship, Innovation and Technology Transfer.

Significant Infrastructure and Equipment

The production plant of Stolzenberger Bakery consists of a large raw material storage place, processing- dough preparation facilities, baking facilities, freezing and deep-freezing chambers. SB possesses handcrafts level for new product development and semi- industrial to industrial baking plant for prototyping & technology transfer with 8 industrial ovens, several high capacity fermentation chambers, furthermore deep-freezing capacities, processing machinery for different types of baked goods and confectionery products, as well as an important fleet of sales cars specially designed for the merchandising of the catering products.

	9. Bio Base Europe Pilot Plant vzw BBEPP	Belgium

Partner Description

The Bio Base Europe Pilot Plant (BBEPP) is an SME with 47 employees located in Ghent, Belgium. The non for profit organization was formed in 2008 to run pilot projects for international companies and research institutes. BBEPP is a diversified pilot plant for the development, scale-up and custom manufacturing of bio-based processes, products and bio-refinery concepts. The pilot plant operates at a kilogram to multi ton scale and aims at closing the critical gap between scientific feasibility and industrial application of innovative bio-based products and processes. Furthermore the personnel of BBEPP has built up decades of experience in scaling-up biorefinery and biotechnological processes. Therefore the BBEPP has the flexibility, experienced personnel and expertise to scale up and optimize a wide variety of unit operations, and to build, with industrial and scalable equipment, integrated process lines. **Website**: www.bbeu.org

Main tasks attributed in the project

Main task for BBEPP is in WP5 (**Final products prototyping and industrial up-scale of the processes).** Within this WP5 BBEPP will scale-up the selected processes from pretreatment, over (bio)chemical modification to formulation so that sufficient sample material is generated for product prototyping, and to provide input for LCA and CEA.

Persons involved	Gender
Dr. Lieve Hoflack is project manager at BBEPP. She holds a PhD in biotechnology (Ghent University, 1998) and has been working as a researcher in academia (University of TX Health Science Center, Ghent University), as well as in biotech SMEs (Plant Genetic Systems, Dyax, Vivactis), and later as project manager in SMEs (MintEurope, BBEPP). Lieve will focus on the overall coordination of the Prolific project on BBEPP's side.	F
Ir. Dieter Depuydt is process engineer. He holds a master degree in applied engineering sciences, discipline biochemistry (university college Ghent, 2008). He is an expert in the pretreatment of different types of lignocellulosic biomass. He will lead the biomass pretreatment and hydrolysis trials.	M
An Vermeulen is Head Analytical Lab. She is a professional bachelor in Chemistry, specialisation processtechnology (University College Ghent, 1998). She has extensive experience in development of analytical methods and bench scale fermentations. She will implement and coordinate the supporting analytics for the processes.	F
Ir. Emile Redant is process development engineer. He holds a master degree in applied engineering sciences, discipline biochemistry and biotechnology (university college Ghent, 2013). He is an expert in the enzymatic hydrolysis of vegetable protein, downstream process development and techo-economic assessment studies. He has will lead the protein extraction and oversee integration of the different unit operations.	M

Relevant references

BBEPP, through its experience in bio based pilot project, has provided several relevant services to industries, research institutes and public bodies:
- Technology development & demonstration based on agro residues
- Tutorial for market introduction of novel molecules produced via Industrial Biotech from renewable resources
- Policy recommendations on how to organise & support pilot activities within the industrial biotech & bio economy to bridge the valley of death

Specific awards and certifications:

- BBEPP was selected in 2014 by the European Commission as a multi-KET Pilot Lines demonstrator
- BBEPP was nominated as Regiostars 2017 finalist with the Interreg project Bio Base NWE

Relevant projects

1. PULP2VALUE: H2020-BBI-PPP-2014-1 Processing underutilised low value sugarbeet pulp into value added products
2. ERIFORE: H2020-Infradev-1-2014-1 Research Infrastructure for Circular Forest Bioeconomy
3. REHAP: H2020-SPIRE-2016 Systemic approach to Reduce Energy demand and CO2 emissions of processes that transform agroforestry waste into High Added value Products.
4. IWT-VIS-project VISIONS: 'building new values chains based on organic waste streams in Flanders'

5. 2G BIOPIC: H2020-LCE-12-2014 Second Generation Bioethanol sustainable production based on Organosolv Process at atmospheric conditions
Significant Infrastructure and Equipment BBEPP is a one-stop-shop and is equipped with a wide range of state-of-the-art industrial equipment to perform the complete production chain from the biomass green resource to the final refined bio-products, by performing biomass pretreatment, biocatalysis, (Ex proof) green chemistry, fermentation and downstream purification. BBEPP features over 2000 m² of pilot production halls, a supporting laboratory, logistics and utility infrastructure. An overview of the equipment can be found on the website www.bbeu.org/expertise-and-equipment Furthermore the personnel of BBEPP has built up decades of experience in scaling-up biorefinery and biotechnological processes. Therefore the BBEPP has the flexibility, experienced personnel and expertise to scale up and optimize a wide variety of unit operations, and to build, with industrial and scalable equipment, integrated process lines.

Conserves France	10. CONSERVES FRANCE CONS	France

Partner Description

CONSERVES FRANCE is a society for the transformation of fruit and vegetables, located in Nimes in the region of Gard. The social capital belongs for 99.96% to O.P. Apo Conerpo, located in Italy in the province of Bologna, Uniproledi, located in Bias in France in the region of Lot et Garonne, and to the cooperative Conserve Italia (member of O.P. Apo Conerpo), located in S. Lazzaro di Savena (BO) in Italy. The total sales volume of 2016 is 79.7 milions of euros, of which 35.1 millions of euros correspond to the sales of fianl products obtained from raw materials provided by Uniproledi in the facility of St Sylvestre, that include also sweet corn. In addition 4.45 millions of euros were obtained in the facility of Tarascon, which raw material was provided by local or Spanish farms. The remaining sales, around 50%, was obtained with the final products realized by Conserve Italia in Italian facilities. 28.5 millions of euros of these refer to tomato-based products made with raw material coming from the member cooperatives of Apo Conerpo. The overall sales volume of Conserves France, generated from the transformation of the raw material from Uniproledi in France and from Apo Conerpo in Italy, corresponds to around 63,6 millions of euros (80% of the total). Conserves France commercialized its products with the brands Cirio (tomato), Barbier Dauphin (ready-to-eat products), Dea (ethnic specialties), Valfrutta (vegetables) and with commercial brands (*private label*) of some French retailers (Casino, Carrefour, Auchan, Leclerc).

Main tasks attributed in the project

WP1-residues characterization :

- Provide to partners different type of legume residues (non-compliant fresh pea seeds and rehydrated beans and chickpeas seeds)
- Provide data regarding amounts of different streams, geographical provenience, biochemical and safety characteristics

WP6 – LCA and cost analysis

- Provide preliminary data for Life Cycle Assessment (LCA) and Cost Effectiveness Analysis (CEA)
- Collaboration in LCA and CEA and possibly in market analysis of future ingredients

WP7 – Dissemination/exploitation

- Dissemination of project at events to which the company takes part
- Possible inputs of exploitation strategies

Persons involved	Gender
Dr. Marika Bondi was graduated in Food Science and Technology at Agricultural Faculty of Bologna University. She works in Research and Development division of Conserve Italia, where from 2008 to 2010 she was assistant to Canned Vegetables manager. Since 2010 she works in the Fruit Juices Product Development.	F

Relevant publications, and/or products, services

- LINEE GUIDA PER L'APPLICAZIONE DEL RISK ASSESSMENT AI NIAS E ALLE SOSTANZE NON VALUTATE (Istituto italiano imballaggio 2016)
- LINEE GUIDA per l'impiego delle BIOPLASTICHE nel packaging a contatto con alimenti (Istituto italiano imballaggio 2013)
- LINEE GUIDA per la valutazione dell'idoneità al contatto con alimenti del packaging realizzato con materiale proveniente da riciclo (Istituto italiano imballaggio 2012)

Relevant Projects:

NA

Significant Infrastructure and Equipment

Conserves France has two production facilities where around 300 people, among permanent and seasonal, work.

St Sylvestre sur Lot, located in the region of Lot and Garonne, with an industrial surface of 43.000 m^2 and a production capacity of around 85.000 ton. This facility is specialized in the transformation of sweet corn, green beans, peas and tomato for the production of canned vegetables and concentrated tomato. During the 2016 campaign the following amounts of raw material, provided mainly by Uniproledi (France), were processed.

ST SYLVESTRE	Raw material (ton)	Uniproledi (%)
Sweet corn	31.382,7	99,7
Green beans	10.100,2	98,4
Tomato	15.891,6	100,0
Peas	1.818,9	100,0
TOTAL (ton.)	59.193,4	99,6

Tarascon, located in the region of Bouches du Rhone, with a surface of 45.000 m^2 and a production capacity of 30.000 ton. Beyond the production of food specialties and ready-to-eat products, it produced the ratatouille, a traditional Provencal dish, containing tomatoes, zucchini, peppers, eggplant, onions and garlic.

	11. RTD TALOS Ltd. TALOS	Cyprus

Partner Description

TALOS is an SME offering high quality services in the field of project management both in European and national level. It is being considered among the biggest consultancy firms in Cyprus. TALOS has years of demonstrated experience in project management being involved since its foundation in more than 50 European research projects and having taken the responsibility for managing administrative and financial tasks, reporting, maintenance of agreements between the consortium members and handling overall legal, ethical, financial and administrative issues. Moreover, TALOS is an expert in exploiting and disseminating project results and has prepared numerous successful Exploitation Agreements with special emphasis on proper IPR management among the partners. One of its strong assets is to organize and convey well-structured dissemination activities which help the partners to fully benefit from the project results. The quality guarantee of all projects that are undertaken is a fundamental concern of the company and to this aim, TALOS has applied a set of measures and processes in order to ensure that project implementation always represents the highest possible quality. TALOS is currently managing numerous EU-funded Research and Innovation projects which have received funding through Horizon 2020 programme. TALOS has developed a strong network with SMEs and R&D performers as well as with stakeholders involved in the exploitation of research results on a European scale.

Additionally, TALOS is the Coordinator of the Cyprus Business Angels Network (CyBAN, a full member of the European Business Angels Network) which is the only Angel Investment Network in Cyprus. CyBAN reviews up to 100 businesses per year to find the most exciting and cutting edge companies. From this around 15 businesses are selected and presented to the Business Angels though Company Presentation Events and online Company Exhibition. Already numerous start-up companies with innovative ideas were funded through CyBAN and TALOS. Being the coordinator of Cyprus Business Angels Network, TALOS maintains strong connections with the SME ecosystem and investor community in Europe.

TALOS is also the co-founder of ERMIS Research & Incubator Center with numerous success stories of incubated start-up companies. ERMIS was founded in February 2003 by TALOS and four other organisations (Cyprus College, Scientific and Technological Park of Crete, and Spidernet) and has facilitated the development of newly established high technological enterprises.

TALOS has also operated the Research and Technological Development Unit as part of the Cypriot branch of the Enterprise Europe Network during the financial framework period 2007-2013. TALOS' role focused on helping SMEs to create their business profiles and develop their ideas into effective project proposals. In doing so, TALOS worked closely with the Cyprus Chamber of Commerce and Industry as well as with the Research Promotion Foundation which operated the Information and Knowledge Transfer Unit respectively.

Main tasks attributed in the project

Given the above expertise, TALOS has been chosen to lead the Communication and Dissemination activities of the project (WP7) as well as to assist in WP6 (Task 6.3). Furthermore, given TALOS' extensive network at both national and European level and experience in business development, project management, innovation and IPR issues, its role will also be critical in assisting with all non-scientific aspects of the project execution as well as networking activities with stakeholders.

Persons involved	Gender
Dr. Alexandros Michaelides, General Manager of TALOS, has been the National Contact Point in Cyprus for SMEs and Innovation during the 5th and 6th Framework Programme between 1999-2006. Having obtained his Ph.D. degree in the field of Metallurgy and Materials Technology and being involved in a numerous international research projects he has wide experience in managing research activities and facilitating technological development. Mr. Michaelides has been the director of ERMIS Research & Incubator Center, the most successful Incubator Center in Cyprus with significant achievements in the field of innovation and technology transfer.	M
Melinda Kuthy has a Master's degree in Economics from the Corvinus University (Budapest) and joined TALOS in 2008 as Business Development Manager. Previously she worked as Chief Financial Officer and Head of the Project Management Department at the Invention and Research Center in Hungary. She has acted as the coordinator of numerous large-scale research projects funded by the European Commission under FP6, FP7 and H2020 but also for national projects funded by the Research Promotion Foundation. She has also extended experience in managing dissemination and	F

communication activities, as well as preparing exploitation plans and managing intellectual property rights. Melinda is currently a Ph.D. candidate at the Cyprus University of Technology, Faculty of Management and Economics, having her research thesis in the field of sustainable tourism development.	
Nikos Gkouveris holds a Bachelor's degree in Mineral Resources Engineering of Technical University of Crete and he is currently in the process of obtaining a Master's degree in Business Administration at European University of Cyprus and his MBA Thesis topic is "The Extent of Application of Benchmarking Practices in Process Redesign". He has actively participated and has extensive experience in managing and disseminating of H2020 and FP7 projects.	M

Relevant publications

1. Use and Diffuse handbook: Strategic Guide to Successful Use and Dissemination of the Results of research and development projects (2009): http://ec.europa.eu/research/sme-techweb/pdf/use_diffuse.pdf
2. Financing SMEs in Cyprus: no stone left unturned? (2014) – published by the Association of Chartered Certified Accountants: http://www.accaglobal.com/content/dam/acca/global/PDF-technical/small-business/pol-afb-fsic.pdf
3. Michaelides, A., Strongylopoulos, G., Smart Specialisation Report Cyprus (2013, European Commission, DG Research)
4. Winner of Lloyd's List Global Innovation Award 2014 for the LYNCEUS project (see below, Grant Agreement number 286148)
5. TALOS staff members regularly hold speeches, presentations and training seminars on dissemination, exploitation and IPR protection aspects at a national level.

Relevant Projects

1. CY-Biobank (Biobanking and the Cyprus Human Genome Project) (2015-2016) H2020 Grant Agreement number: 664561. This project was funded under H2020-WIDESPREAD-2014-1, Teaming Phase 1 call and aimed at the preparation of a business plan for the establishment of a biobank in Cyprus. TALOS was responsible for conveying market research and preparing the business plan in close collaboration with the coordinator and the advanced partners. (www.ucy.ac.cy/cybiobank/)
2. LYNCEUS (People localisation for safe ship evacuation during emergency) (2012-2015), FP7 (Capacities, Research for SME associations/groupings), Grant Agreement No. 286148. TALOS was the Coordinator of the LYNCEUS project being responsible for the overall financial, legal and administrative management of the project. TALOS also played an active role in dissemination and exploitation activities assisting the end-user partners for the best possible exploitation of the project results. (www.lynceus-project.eu)
3. WISERBAN (Smart miniature low-power wireless microsystem for Body Area Networks) (2010-2013) FP7, Challenge 3: Components, Systems, Engineering, Objective http://www.wiserban.eu/ Contract number: 257454 TALOS was the operational manager of the project, being responsible for all the financial, legal and administrative aspects of project management. (www.wiserban.eu)
4. Microflusa (Fabricating colloidal materials with microfluidics) (2015-2019) H2020 Grant Agreement number: 664823 TALOS is the Operational and Dissemination Manager of the project, being responsible for all the financial, legal, administrative aspects of project management as well as all the Dissemination activities. (www.microflusa-project.eu)
5. Lynceus2Market (An innovative people localisation system for safe evacuation of large passenger ships) (2015-2018) H2020 Grant Agreement number: 636286. TALOS is the Coordinator of the project being responsible for the overall financial, legal and administrative management of the project. It also plays an active role in the dissemination and exploitation activities assisting the end-user partners for the best possible exploitation of the project results. (www.lynceus-project.eu)

Significant Infrastructure and Equipment

TALOS has all the necessary facilities and software in order to successfully carry out all its tasks in the project related to dissemination, communication and exploitation activities.

cosmetic BEAUTY CARE EXCELLENCE	12. COSMETIC COSM	Greece

Partner Description

COSMETIC is a dynamic SME producing a great range of tailor –made cosmetics, food supplements and medical devices. It is currently employing 35 employees. Its core competence lies on the production of specialised creams. The company is also manufacturing oils, sunscreens, shampoos, gels, toothpastes, mouth rinses, bath salts etc.

COSMETIC offers tailor made products to its customers with specific properties and has solid experience in picking the most appropriate chemical species and packaging to match the desired properties. The staff routinely develops new formulas that are tested in terms of stability, sensorial evaluation and antimicrobial activity.

COSMETIC is producing different products for more than 200 brands and can offer insight on the specification of the more interesting products for the industry and to assess the technical, sensorial and economic viability parameters.

Main tasks attributed in the project

COSMETIC will be responsible to set the technical, market and regulatory specifications for the new products and will co-lead WP1 COSM will work on the stabilization of extracts in WP3. COSM will evaluate and select the best performing fractions (WP4) and develop relevant formulations. The best performing materials will be also produced as prototypes (WP5).

Persons involved	Gender
Dr. Georgios Tsatsos PhD in Biochemistry, is the founder and owner of COSMETIC. He is a chemist (AUTH, Greece) with MSc in Biotechnology at University of West of England (U.W.E.) in Bristol, UK and holds a PhD in Biochemistry from Imperial College, London, UK. He has attended a course on Managing Organisations at University of Birkbeck, London, UK. He has worked in Clariant AG in Basel, Switzerland in organic synthesis, in Zeneca LifeScience Molecules in Billingham, UK with lipases and as a post-doctoral researcher in Imperial College. He has been active in formulating cosmetics for the last 10 years. He has been involved in 4 successfully completed FP7 proposals and 2 H2020.	M
Dr. Efi Voulgari PhD in Pharmacy, is a scientist at COSMETIC. She is a Biologist (Univ. of Patras, Greece) with MSc in Pharmacy, Greece and a PhD in Pharmacy from University of Patras, Greece. She has worked in University of Patras as a Researcher in the development of magnetic nanoparticles for drug delivery.She has also worked in BIC SA in Greece.	F
Dr Georgia Tsotsou PhD in Biochemistry, she is a Senior Scientist at COSMETIC. She is a chemist (AUTH, Greece) with a PhD in Biochemistry from Imperial College, London, UK. She was granted a Marie Curie Postdoctoral Fellow and worked for 2 years in Biocatalysis and Biotransformation group, Rhodia-Centre des Recherches de Lyon, Lyon, France. She has worked as a consultant in several companies; chemicals, perfumes, pharmaceutical. She has been the author of 11 journal publications, 1 book article and 4 patents. She has participated in 10 research grants, 7 of which were granted. She is teaching in the Technological Educational Institute (TEI) of Athens.	F
Lina Lezgidou is a scientist at COSMETIC. She holds a Pharmacy degree by AUTH, Greece and a MSc degree in Cosmetology by Universite Paris-Sud. She has experience as a cosmetics safety assessor.	F
Anastasios Saravanis He works as a lab technician and has more than 20 year's experience in preparing cosmetics at several companies.	M

Relevant publications, and/or products, services

- Baccile N, Babonneau F, Banat IM, Ciesielska K, Cuvier AS, Devreese B, Everaert B, Lydon HL, Marchant M, Mitchell CA, Roelants R, Six L, Theeuwes E, Tsatsos G, Tsotsou GE, Vanlerberghe B, Van Bogaert INA, and Wim Soetaert W (2017) Development of a Cradle-to-Grave Approach for Acetylated Acidic Sophorolipid Biosurfactants ACS Sustainable Chem. Eng. 5 (1), pp 1186–1198
- Tsatsos has produced more than 5000 cosmetic formulations in the last 10 years.

Relevant Projects:

1. **BIOSURFING** (FP7, 2011-2015). The project aimed at creating new-to-nature and tailor-made biosurfactants through metabolic engineering of the unconventional yeast *Candida bombicola* and to use them as ingredients in cosmetic and cleaning products.
2. **DIBBIOPACK** (FP7, 2012-2016). The target was to develop multifunctional packaging for food, pharmaceutical and cosmetic industries, in a wide range of processes (injection & blow extrusion), which improve the structural and barrier properties, introduce smart features and increase sustainability using nanotechnology.
3. **OLI-PHA** (FP7, 2012-2015). A novel and efficient method for the production of polyhydroxyalkanoate polymer-based packaging from olive oil waste water.
4. **CITRUSPACK** (H2020, 2017-2020). CITRUSPACK aims to demonstrate that sustainability and efficiency can be applied to agricultural and industrial practices by developing the potential of waste in order to diversify and grow farmers/producers' income by means of its valorisation into a series of new value chains.

Significant Infrastructure and Equipment

The lab is equipped with a Silverson rotor-stator homogeniser, Brookfield viscometer for rheological tests, time-temperature programmable incubators, centrifuge, Lumisizer, optical microscope, UV-vis spectrophotometer, laminar flow chamber, refractometer *etc*.

	13. illycaffè S.p.A. ILLY	Italy

Partner Description

Founded in 1933 by Francesco Illy, illycaffè produces and sells worldwide a single blend of premium quality coffee made from nine varieties of pure Arabica beans. Illy products are enjoyed in more than 140 countries, on all five continents, and are served in about 100,000 establishments. Based in Trieste, the company is led by the third generation of the Illy family: illycaffè S.p.A. belongs (100%) to the Illy family's holding company – Gruppo illy S.p.A. illycaffè's approach to innovation is focused on the creation of sustainable value for its customers, consumers and stakeholders in general, by providing products, solutions and services in line with expectations, and at the highest levels of excellence and quality. The company's achievements in these areas have been made possible through investments made in technology and research and development, through relationships with strategic partners and, first and foremost, through the professionalism and expertise of the people working in the company.

Main tasks attributed in the project

- Illycaffè will supply samples of silver skin, the typical coffee roasting waste product, and possibly other coffee derived by-products in the framework of WP1 and supply information on the coffee byproduct streams (WP1)
- Provide data regarding amounts of different streams, geographical provenience, biochemical and safety characteristics (WP1)
- Test of coffee capsule prototypes (WP5)
- Information on coffee (stream volumes, logistics, etc.) to be integrated in LCA and CEA (WP6)
- Dissemination of project at events to which the company takes part (WP7)
- Co-organisation of a workshop on residual biomasses from agro-industrial processing (WP7)
- Possible inputs of exploitation strategies (WP7)

Persons involved	Gender
Dr. Luciano Navarini: Head of Research and Scientific Coordination of illycaffè since 2008. Master's degree in Chemistry at the University if Trieste. He is author of more than 85 scientific papers published on peer-reviewed international journals, of 1 book chapter and of several contributions published on conference Proceedings regarding coffee & food chemistry, foam & emulsions, polysaccharides and biopolymers chemistry, physico-chemical properties and sensory perception. Co-inventor of 15 patents. He has participation at national and international conferences, workshops and seminars (Europe, United States, Latin America and Asia) also in quality of Chairman, Supervisor and Invited Speaker .Member of the Italian Chemical Society, Board Member of the Association for Science and Information on coffee (ASIC), past Member of the Technical Advisory Committee of World Coffee Research (WCR). He was full Professor at the University Master on "Coffee Economics and Science Ernesto Illy" classes on "Coffee Characterization" – collaboration between Italian institutions: University of Udine, University of Trieste and International School of Advanced Studies and illycaffè s.p.a.	M

Relevant publications and major achievements

1. Valerio Ferrario, Harumi Veny, Elisabetta De Angelis, Luciano Navarini, Cynthia Ebert and Lucia Gardossi "Lipases Immobilization for Effective Synthesis of Biodiesel Starting from Coffee Waste Oils";, *Biomolecules* 2013, 3, 514-534.
2. Elena Zanella, Micol Della Zassa, Luciano Navarini, Paolo Canu, "Low-Temperature Co-pyrolysis of Polypropylene and Coffee Wastes to Fuels, *Energy & Fuels*, 2013, 27, 1357-1364.
3. Ernesto Illy, Francesco's son, co-founded in 1966 the Association Scientifique Internationale pour le Cafè, now Association for Science and Information on Coffee (ASIC), to fulfil the desire of the coffee scientists to dispose of a permanent organization, which would periodically bring together specialists from all over the world working in the different fields of coffee science and technology, giving them an opportunity of presenting and comparing their investigations. Thanks to this network, it has been possible to undertake several Research activities leading to develope new technologies.
4. Three of the coffee industry's eight transformational innovations were developed by illy (pressurized packaging, precursor of the modern espresso machine and pre-tamped, pre-dosed, expertly ground espresso in single serve paper pods).
5. Several specialty Arabica coffee varieties have been created by illy in the course of collaborative research projects and six of these have been donated in 2013 to the World Coffee Research, a collaborative, not-

for-profit research organization, created by the global coffee industry, including Illy, in 2010 to ensure the future of coffee in the face of threats like climate change.
Relevant projects • **IPCOS:** Imprinted polymers as coffee sensors. (H2020 research and innovation programme under the Marie Sklodowska-Curie grant agreement No. 642014). • **BREEDCAFS**: BREEDing Coffee for AgroForestry Systems (Ho2020 SFS-2016-2 grant agreement No. 727934)
Significant Infrastructure and Equipment The department of Research and Innovation – recognized internationally as a center of excellence – is composed of two research centers for quality control (one in Trieste and one in Sao Paulo, Brazil) and five laboratories, linked to promote the horizontal transfer of knowledge, and working in conjunction with universities and Research institutions: Aromalab and SensoryLab (both certified ISO 17025.2005 by Accredia, the sole Italian accreditation body), BioLab, TechLab and FoodScienceLab. The workforce within the laboratories is made up of: 5 chemists, 1 molecular biologist, 1 biologist, 2 engineers, 4 food technologists, 3 technicians.

	14. Nutrition Sciences N.V. NS	Belgium

Partner Description

Nutrition Sciences N.V. (NS) is a Belgian private company (26 collaborators) belonging to the Nuscience-group (1125 collaborators) and is well known for its R&D activities in animal production and spends much attention to the introduction of new feed ingredients, concepts, services and technologies in animal livestock production for its customers. Their innovation-driven attitude enables them to supply sustainable and economically viable solutions for the efficient production of animal feed and feed ingredients within the food value chain. Main focus today is developing concepts and services for (1) reduction of raw material depletion and (2) lowering the dependency for antibiotics in livestock production.

Main tasks attributed in the project

- Specification requirements on target compounds and compliance with existing regulations in the feed industry (WP1)
- Fractions selection for feed (WP4)
- Feed products prototyping (WP5)
- Product regulations review for the foreseen feed products and data for LCA and CBA (WP6)

Persons involved	Gender
Dr. Geert Bruggeman PhD is an expert in animal nutrition. In his current position, he is responsible for the company innovations and for its strategic research for the development of novel functional feed ingredients and concepts related to improvement of raw material usage and reduction of dependency for antibiotics in livestock production. Geert Bruggeman has profound expertise in coordinating of national and international research and innovation projects, participated as a partner or coordinator in projects in Eureka, FP4, FP5, FP6, FP7 and H2020 and acts as a member of several advisory boards.	M
Dr. Stefanie Versringe PhD has a degree in animal feed processing and has a profound knowledge on (functional) feed ingredients in terms of their economics, markets and specific quality control / quality assurance requirements for successful implementation in the market via different validation models	F

Relevant publications

1. Alakomi, H-L., Höhl, A., Horvatek Tomic, D., Thomas, M., Bruggeman, G., Tassis, P., Prukner-Radovcic, E., Tzika, E., Axelsson, L., Kneifel, W. & Saarela, M. (2016). Antimicrobial Resistance and Residues in the EU – Current Situation and Possible Countermeasures, Emphasis on *Campylobacter* and *Salmonella*. Quality Assurance and Safety of Crops & Foods, 8(3), 399-413.
2. Hoorfar, J., Bang-Berthelsen, I., Jones, F.T., Häggblom, P., Bruggeman, G. & Zentek, J. (2011). Emerging Safety and Quality Issues in Compound Feeds with Implications for Human Foods. *In:* HOORFAR, J., JORDAN, K., BUTLER, F & PRUGGER, R. (Eds.), Food Chain Integrity: A Holistic Approach to Food Traceability, Safety, Quality and Authenticity, Woodhead Publishing, 130-143
3. Jordan, K., Dalmasso, M., Zentek, J., Mader, A., Bruggeman, G., Wallace, J., De Medici, D., Fiore, A., Prukner-Radovcic, E., Lukac, M., Axelsson, L., Holk, A., Ingmer, H. & Malakauskas, M. (2014). Microbes versus Microbes: Control of Pathogens in the Food Chain, Journal of the Science of Food and Agriculture, 94(15), 3079-3089.
4. MOLATOVÁ, Z., SKŘIVANOVÁ, E., BARÉ, J., HOUF, K., BRUGGEMAN, G. & MAROUNEK, M. (2011). Effect of coated and non-coated fatty acid supplementation on broiler chickens experimentally infected with Campylobacter jejuni. Journal of Animal Physiology and Animal Nutrition, 95, 701-706.
5. Binter, C., Straver, J., Häggblom, P., Bruggeman, G., Lindqvist, P.A., Zentek, J. & Andersson, G. (2011). Transmission and Control of Salmonella in the Pig Feed Chain: A Conceptual Model. International Journal of Food Microbiology, 145, S7-S17.

Relevant Projects:

1. **InDirfect (BBI,** www.bbi-indirect.eu**)** "Direct and Indirect biorefinery technologies for conversion of organic side-streams into multiple marketable products" (responsible for in vitro testing of project outcomes in livestock production)**. In this project Nutrition Sciences is WP leader on validation of nutritional feed concepts suitable for reduction of dependency for antibiotics in livestock production as well to counteract the global depletion of raw materials.**
2. **National project LESSantibiotics** "Reducing antimicrobial use and resistance in the food chain by innovative nutritional interaction at animal feed level" (coordinator)

3. PiGutNet (www.pigutnet.eu) ""**Antibiotics as a factor of dysbiosis and spread of antibiotic resistance genes".** Nutrition Sciences N.V. is are co-chairing the WG dealing with "**Antibiotics as a factor of dysbiosis and spread of antibiotic resistance genes"**
4. NOSHAN (FP7 project, www.noshan.eu) "Upgrading by-products from vegetables and fruits in the animal feed" (responsible for in vitro testing of project outcomes in livestock production)**. In this project Nutrition Sciences is WP leader one extraction and validation of nutritional feed concepts suitable for reduction of dependency for antibiotics in livestock production.**

Relevant Patents

48 relevant patents

Significant Infrastructure and Equipment

Nutrition Sciences N.V. has its own laboratory, in order to perform and validate novel feed technologies as well as novel and sustainable feed concepts. In this respect, the laboratory contains chemical, physico-chemical as well as biotechnonological sections and is fully equiped for analysis of nutritional content (Weende parameters, mineral quantification by means of ICP, *etc.*), analysis of feed behaviuor (Holmen, particle size distribution, *etc.*) and analysis of (bio)functionality (antimicrobial, agglutination, prebiotic behaviour, *etc.*) as well as a validated gastrointestinal simulation model). In this respect, demonstration production ptential with novel feed ingredients can be performed. Finally, Nutrition Sciences N.V. has access to different trade fairs for demonstration of outcomes of the project and is (co)organising different workshops and trainings for different stakeholders in the food chain.

Pleurette	15. Pleurette SAS PLEUR	France

Partner Description

Pleurette is a startup that upcycles residues into mushrooms. They collect coffee grounds for oyster mushroom kits ready to grow at home. In addition to that they grow fresh mushrooms in recycled shipping containers and offer a gourmet assortiment of organic mushrooms. To face lots of ecological challenges, like pollution, waste production or over consumption of meat, Pleurette developed Valofungi projects in H2020 Call Katana Project with an NL consortium partner:
1) Brewery spent grains / Coffee waste turned into substrate to produce healthy delicious mushrooms
2) Enrich mushrooms with Vitamin D = functional food
3) Use the waste of (oyster) the mushroom cultivation industry in Europe to produce vegan and gluten free products to replace meat (Vegan Organic gluten-free bolognaise sauce, Vegan steak)
4) Find added value in biobased residues (waste) working on these topics with universities and research centers

Main tasks attributed in the project

WP1-residues characterization:

- Provide to partners different type of fungi residues (mycelia, leftovers, residues of fungi processing), from different type of fungi: agaricus bisporus, pleurotus ostreatus (main interesting mushroom : focus on it) and lentinus edodes (shiitake: minor interest)
- Provide data regarding amounts of different streams, geographical provenience, biochemical and safety characteristics. Data on flow at European level will be studied by Pleurette SAS and provided mainly for FR BE IT SP DE NL GB IE PL HU.

WP6 – LCA and cost analysis

- Provide preliminary data for Life Cycle Assessment (LCA) and Cost Effectivess Analysis (CEA)
- Collaboration in LCA and CEA and possibly in market analysis of future ingredients

WP7 – Dissemination/exploitation

- Dissemination of project at events to which the company takes part
- Possible inputs of exploitation strategies

Persons involved	Gender
Jurgen ENGERISSER, 2 Master Degrees in Catholic University of Louvain, worked in different fields and various functions (FMCG, sales and field marketing; Bank & Insurance, sales & advisor, Bank & Insurance, Management Program and Communication + Event organization; Industry Service & Production, Project Design, Financing & Development; Industry, Innovation and sustainable project development; Consulting, circular economy freelance consultant) giving him an insight on innovative products, processes and projects design & development on different markets in different countries in Europe (FR, BE, NL, DE, ENG). Accredited International Blue Economy Consultant, System Design Thinker & Lecturer he is now entrepreneur co-founder, Commercial Director, Strategist, Innovation & R&D Leader and Chairman of the board of an Urban Agriculture and Agro-Food innovative start up in north of France. He also work in Belgium on added value extractions and products of waste (coffee waste and spent grains from brewery mainly) with a belgian partner company "Maxiline Sprl". Collaborate as representative of NEO ECO RECYCLING (FR & BE). Own belgian consulting structure: "WINERGIES".	M
Gabrielle RADOUX, 2 Master Degrees in Catholic University of Louvain, worked in cultural area. Now is Operational Director and also in charge of the mushroom and substrate production processes in the company. She is multiskilled and organize team work.	F

Relevant publications, and/or products, services

1. Ready to grow gourmet oyster kits on coffe waste (Pleurette) 2. High tech shipping containers ready to grow mushroom full automated to manage growing parameters (Pleurette) 3. Organic Vegan Gluten-Free Bolognaise Pasta Sauce made from mushroom industry waste (oysters unsold feet) (Pleurette) 4. Sustainable development, biomimetic innovation and blue economy workshop and conferences (Winergies) 5. Cosmetics and scrub from a biowaste (pre-INCI status) (Winergies-Maxiline)
Relevant Projects 1. CABECO project: cascading benefits of coffee waste by extractions and upcycling (Winergies-Maxiline) 2. Fungi Up: turning coffee grounds into mushroom in upcycled shipping containers (Winergies) 3. Vortex cold Air Mill and Dry Innovative Technology (Winergies) 4. Vegefarm: Sustainable Systemic Circular eco vegetable and mushroom farm (Winergies) 5. Valorisation of mineral residues extractions (Winergies)
Significant Infrastructure and Equipment Pilot biobased urban and agricultural waste mushroom substrate production unit 4 upcycled shipping containers ISO high cube 40-feet as full automated growing rooms for Oysters, Shiitake, Eryngii 24,000 m^2 undergrounds to cultivate Shiitake, Nameko, Oysters Logistic 300 m^2 warehouse daily connected to Brussels, Liege, Maastricht, Eindhoven, Rotterdam, Paris, Marseille, *etc*.

FEMTO ENGINEERING	16. FTEN ENGINEERING SRL FTEN	Italy

Partner Description

Femto is a mechanical engineering and manufacturing company able to product design, prototyping, engineering, production equipment design and construction and techno-polymer component production requirements. Femto furnishes a global engineering service, providing support from the initial development phases of a new product up to complete industrialisation. With Femto, companies requiring technical and technological operating support can study a new product from the initial aesthetic/functional design phase, simulate its manufacturing processes, construct prototypes (rapid prototyping), production equipment and produce this in complete continuous and univocal partnership, without fragmentation and with fully coherent methods of operation, all to the advantage of development times and the substantial quality of products for a wide range of application sectors. Femto cooperates with a number of leading manufacturing companies operating in market sectors such as: automotive, transport, household appliances, electro-tools, biomedical, electro-medical, marine, telecommunications, sports equipment, and consumer goods in general. Femto also cooperate with several Univeristies and research institutions, in particualr with University of Pisa. Investing more than 10% of annual turnover in R&D, Femto furnishes not only technical expertise but also innovative machinery and technologies using software and hardware systems able to machine and manage all types of materials with up to concurrent 5-axis controls.

Femto's team has specific expertise in the plastic materials, metalwork construction and composite materials sectors, always based on experience acquired through studies of strictly technical, precision parts and projects in forefront technical environments. Since 2008, Femto has developed an important specialization in the production and use of bio-based polymers for a lot of different applications.

Main tasks attributed in the project

- State of the art processing technology and valorization review for packaging applications (WP1)
- Utilization of polyphenols, caffeine, phytosterols and carotenoids as well as fibers to modify the properties of polymeric matrices will be investigated in order to improve polymer performances towards applications into films and capsules (WP3)
- Fraction selection for polymers and composites (WP4)
- Prototyping polymers and composites (WP5)
- Data for LCA and CBA (WP6)
- Femto will subcontract part of research activity to Pisa University necessitating the use high-end equipments due to long lasting cooperation in research and located in Pisa University.

Person involved	Gender
Dr. Ing. Giuseppe Amato Mechanical Engineer, Professor in Materials Science and Technology in Siena University (2003/2008) and in Metallurgy In Siena University (2004/2008). CEO of Femto Engineering srl and of NE.RO srl that is an Italian biomedical Company working in neurophysiology application. He is the scientific responsible of PHOTOSYNPLAST research team (2015-2018).	M
Dr. Andrea Cavallo Degree in Geology, Researcher in Certema Laboratoies, expert in the field of the Electronic Scanning Microscopy (SEM) and Microelectronics Microscope (EMPA) for Mineralogy, Petrology, Experimental Petrology and Microstructure Geology, and Materials Science.	M
Ing. Tommaso D'Amico Mechanical Engineer, manufacturing manager in FemtoEngineering, expert of applied technology expecially in mechanical machining, multi axis machine and wire edm. with	M
Ing. Michele Zambetta Mechanical Engineer, expert in design and manufacturing with a long experience in prodct developing an manufacturing. He was senior designer in Maserati works in FemtoEngineering as responsable of the manufacturing area of our laboratory Certema, for milling and Additive Manufacturing.	M
Marco Vannini Technical manager at FemtoEngineering he has an important experience with platic materials and injection technology. He has more than ten years of experience with funded projects, during which he is involved in on board experiments.	M
Dr. Patrizia Cinelli PhD in Chemistry at Pisa University. She is assistant professor of applied chemistry and Materials Technology at Pisa University. She has over 20 years of experience in materials science, polymer processing and characterization, biodegradation and life cycle assessment.	F

She is co-author of over 50 papers in peer-reviewed journals, 9 book chapters, and 8 patents on innovative materials. She has worked in over 15 EC project from FP5, FP6, FP7 and Horizon 2020.	
Prof. Andrea Lazzeri (DICI-UNIPI) Full Professor of Materials Science and Engineering at Pisa University since 1st January 2009. He is specialized in preparation and use of micro-and nano-particles in polymeric composites, rubber toughening of plastics, deformation and fracture of materials and in ceramic matrix composites. He is author of 40 publications on international journals and 4 patents. He coordinated two European projects on materials science in FP7, and is currently coordinating a project in Horizon 2020, NMP-21-2014-IA, nanocathedral.	M

Relevant publications, and/or products, services

1. Iezzi G., Caso C., Ventura G., Vallefuoco M., Cavallo A., Behrens H., Mollo S., Paltrinieri D., Signanini P., Vetere F. (2014). First documented deep submarine explosive eruptions at the Marsili Seamount (Tyrrhenian Sea, Italy): A case of historical volcanism in the Mediterranean Sea. GONDWANA RESEARCH, Vol. 25, p. 764-774, doi:10.1016/j.g r.2013.11.001
2. Della Ventura G, Bellatreccia F, Marcelli A, Guidi MC, Piccinini M, Cavallo A, Piochi M (2010). Application of micro-FTIR imaging in the Earth sciences. ANALYTICAL AND BIOANALYTICAL CHEMISTRY, vol. 397, p. 2039-2049, doi: 10.1007/s00216-010-3811-8
3. Mollo S, Iezzi G, Ventura G, Cavallo A, Scarlato P (2012). Heterogeneous nucleation mechanisms and formation of metastable phase assemblages induced by different crystalline seeds in a rapidly cooled andesitic melt. JOURNAL OF NON-CRYSTALLINE SOLIDS, vol. 358, p. 1624–1628, doi:10.1016/j.jnoncrysol.2012.04.010
4. Femto has developed many manufacturing applications based on the use of bio-polymers, such as the cordless phone named ECO PHONE realized entirely with bio-material for the most important Italian telecommunication company TELECOM (more than 30.000 pieces sold)

Relevant Projects

- AGRIMAX: Agri & food waste valorisation co-ops based on flexible multi-feedstocks biorefinery processing technologies for new high added value applications (BBI Grant Agreement no: 720720, starting date 01.10.2016).
- PERFORMANCE KBBE-2012-6, GA n°: 312092 "Development of Personalised Food using Rapid Manufacturing for the Nutrition of elderly Consumers"
- Femto coordinates the regional project named PHA Project POR FESR 2014-2020 Regione Toscana - Bandi RSI 2014 – Bando 2 – "Produzione eco-sostenibile di polidrossialcanoati (PHAs) e loro utilizzo in materiali compositi per applicazioni in campo marino/costiero e agricolo/vivaistico "(2015-2017).
- Numerous project for different national and international companies such as Leonardo/Finemeccanica, Gilbarco Veeder Root, MBDA Myssile System, BioMerieux, Coma, *etc*.

Significant Infrastructure and Equipment

Microanalisys laboratory equipped with several instruments, among which the FE-SEM high resolution microscope Zeiss Merlin.

Mechanical department equipped with different machines : Milling system Sigma MISSION 5M, Milling system Mikron VCP 600, Milling system Hermle C600V, Milling system Fagima SPINFAST 220, EDM system Agie AGIETRON IMPACT 3, Wire EDM Sodick A 535, Grinding machine Delta SYNTH 800x500 COMPACT PLUS, Radial Drill 40x1250 416.02, Circular saw Shark 280SX Jet, Late CU 400/1000, Column drill IM115 and 35ME

Production department equipped with different machines : Injection machine 9HP300/1300 SANDRETTO, Injection machine 9HP75/285 SANDRETTO, Injection machine 9HP120/430 SANDRETTO, Injection machine BMbiraghi mod. SINTESI 100, Injection machine BMbiraghi mod. SINTESI 35, Inkection machine BMbireghi mod. SINTESI S580, Injection machine Negri Bossi serie CANBIO mod. VS250.

Through subcontract to University of Pisa:

Minilab Haake mini extruder equipped with MiniLab Injection Moulding Press; COMAC co-rotating twin screw extruder EBC 25 HT semi-industrial extruder for producing granules equipped with a AMUT flat die extrusion unit for producing films and sheets; Teknomast Blow Mouling Extruder (25 mm); Charpy Impact Tester CEAST 9050; Differential Scanning Calorimeter TA Q200; DMTA Gabo Eplexor® 100N; MTS Universal Testing Machine Criterion model C 43 (equipped with a 10KN load cell).

INNOVA COOP Sviluppo cooperativo	17. INNOVACOOP srl INNOVA	Italy

Partner Description

INNOVACOOP is an SME set in 2012 to promote innovation and internationalization processes of cooperatives associated to Legacoop Emilia-Romagna, a cooperative association representing 1,250 enterprises, 150,000 employees, 2.8 million shareholders and a total production value amounting to 32 billion Euro in 2016.

INNOVACOOP shareholders are the associations belonging to Legacoop that operate at local level and with different sector competencies, namely: Legacoop Emilia-Romagna, Legacoop Romagna, Legacoop Emilia-Ovest, Legacoop Estense, Legacoop Bologna, ANCPL Legacoop (the National Association of Industrial and Workers Cooperatives, representing 1,810 cooperatives and 4 Consortia at national level operating in the construction, manufacturing, designing/engineering and cultural sectors) and Legacoop Agroalimentare Nord Italia (347 agrifood cooperatives, among which prominent players of the "Made in Italy" Food & Beverage sector (Granarolo, Parmareggio, Cantine Riunite&CIV, Apofruit, Alce Nero).

INNOVACOOP carries out promotion, information, dissemination, support to knowledge and technology transfer activities for the benefit of regional cooperatives associated to Legacoop. INNOVACOOP also supports the participation of cooperatives into public programmes sustaining research & innovation and internationalization processes. Activities consist in technical assistance for the development of the cooperative business model at international level (*e.g.* in Argentina, Brasil and Tunisia) and for the implementation of innovation and internationalization processes. More specifically, actions are developed in order to:

- favour the access to public financing for project development and funding;
- stimulate networking, research, technical/technological expertise and knowledge scouting and sharing;
- promote Industry-Science relations and the development of innovation partnerships; cooperation with Universities, research centres, innovation clusters and platforms;
- support cooperation among cooperatives;
- foster the creation of innovative startups and spinoffs.

INNOVACOOP is fully integrated in the regional, national and European cooperative movement and operates through a widespread network consisting in the following organisations:

- Legacoop European Offices in Brussels;
- COOPERATIVES EUROPE, the European association promoting the cooperative business model in Europe, composed by 84 member organisations from 33 European countries across all business sectors;
- CECOP-CICOPA EUROPE, the European confederation of industrial and service cooperatives which affiliates 27 members in 15 European countries, including organizations promoting cooperatives and national confederations or federations of cooperatives representing 50,000 enterprises employing 1.3 million workers;
- COGECA, the General Confederation of Agricultural Cooperatives in the European Union, the main representative organization for the agricultural and fisheries cooperative sector. COGECA has currently 35 full members from the EU Member States, 4 affiliated members and 36 partner organisations, which include representatives of countries like Iceland, Norway, Switzerland and Turkey.
- INNOVACOOP cooperates with the Emilia-Romagna Regional Office in Brussels, also for dissemination activities in the framework of EU funded projects.

Main tasks attributed in the project

INNOVACOOP will take part to exploitation and dissemination activities within WP7

Person involved	Gender
Francesca Montalti - Vice-President and board member in charge of innovation development in INNOVACOOP. Master's Degree in International Relations and Industrial Policies, Post graduate Master's Degree in Economic Development and Organisation. Former professional experience as project manager in Aster, the Emilia-Romagna Consortium for Industrial Research and Innovation, carrying out activities related to technical assistance in developing countries, support to regional innovation policies and science-industry relations, management and coordination of European regional networks and Projects.	F
Marilena Raule – Project Manager in INNOVACOOP.	F

Master's Degree in Economics and Post Graduate Master's Degree in Cooperatives Economy. Main activities: Project development, management and reporting; international relations; support to cooperatives' internationalization through institutional cooperation and BtoB relationships.	
Relevant publications, and/or products, services INNOVACOOP has carried out information and awareness raising activities – workshops, meetings and seminars, focus groups, study visits to public labs, technology foresight labs - carried out for the benefit of agrifood cooperatives in different thematic areas, among the others: innovation in packaging; advanced analytics methodologies safeguarding food origins and traceability; precision agriculture; reduction of water consumption in agrifood processes; innovation in fresh ready-to eat fruits; valorization of vegetable proteins and applications in the agrifood sector. The different initiatives have been organized in cooperation with Universities, public and private research centres, innovative companies, Institutions and cooperatives of other sectors (industrial and retailers' coops). INNOVACOOP has also favored the promotion of innovative startups in the agrifood sector through dedicated Startup Programmes (Coopstartup Farmability) and matchmaking activities with the associated Legacoop cooperatives. INNOVACOOP has provided technical assistance and support to public funding for the creation of an innovative startup operating in the field of industrial valorization of vegetable proteins from alfalfa plants formerly used as animal feeding.	
Relevant Projects: Since the foundation, INNOVACOOP has developed experience in the design and management of internationalization and R&D&I projects in different sectors within regional, national and European Programmes, namely: the ERDF Regional Operational Programme, the EAFRD but also LIFE+ and European Territorial Cooperation Programmes. INNOVACOOP has supported Legacoop Emilia-Romagna in the successful implementation of AQUA -Adoption of Quality Water Use in Agro-industry sector (LIFE+). It has provided technical assistance to Legacoop Romagna in the submission of GECO2, aimed at creating and testing a regional voluntary CO2 market (under Italy-Croatia Territorial Cooperation Programme - still under evaluation). It has supported single cooperatives in the submission for funding and/or implementation of R&D projects. Among the others: ASPARTA, a three-year project on shelf-life extension in ready-to eat fruits; ALFA ALFA INNOVA, a project based on the application of protein extraction technologies to traditional alfalfa processing and VALORIZATION OF INDUSTRIAL HEMP PRODUCTION IN PIEDMONT.	
Significant Infrastructure and Equipment Dissemination, communication and exploitation activities will be carried out in the offices of the Legacoop Tower (located in the Bologna Exhibition Area), offering common spaces fully equipped for meetings, workshops and seminars. Communication activities will benefit from INNOVACOOP's website (www.innovacoop.eu), the INNOVACOOP information service on innovation opportunities called „F1rst-Innovacoop" (available via INNOVACOOP's website) and the different communication channels (websites, social media, newsletters) of the different Legacoop associations operating at local and national level.	

UNIVERSITÀ DI PISA	18. Universitat di Pisa UNIPI	Italy

Partner Description

The Department of Civil and Industrial Engineering part of Pisa University (UNIPI) is located in Largo Lucio Lazzarino 1, in the centre of Pisa. The Industrial Chemistry and Materials Science group, directly involved in PROLIFIC project activities, consists of about 30 researchers and 10 technicians. DICI was established in 2012, following a major reform of the Italian higher education system, as a merger of the former departments of Aerospace Engineering, Chemical Engineering, Industrial Chemistry and Materials Science, Mechanical, Nuclear and Production Engineering, and a part of the Department of Civil Engineering. The institute research covers a wide variety of skills in Materials Science: characterization and development of materials, polymer processing, design and modelling of mechanical properties. These research activities have a strong influence in areas of high social and economic impact, such as materials and environment, health, energy, industrial. The research activity is carried out in the DICI-UNIPI laboratories by employing advanced scientific instruments in collaboration with a variety of both national and international academic and private institutions.

UNIPI is very skilled in projecting polymers blends composition (additives selection, processing conditions) in order to tune properties of materials including biodegradability, morphology, thermal and mechanical properties as attested by papers, patent, and participation in European, National and regional projects but also by the direct cooperation with industries to bring, e.g., their patented results to the market. UNIPI staff has experience in test of compost, and biodegradability in soil and marine water, as well as reactive extrusion, copolymerization, selection of plasticizers, nucleating agents, and compatibilization of natural fibres with polymeric matrices. UNIPI staff is also expert in studying and quantifying the crystalline and amorphous fractions of a material, whose ratio affects biodegradability many physical properties, as barrier, mechanical, thermal, aging, adhesion, and, consequently, the final performance of a material.

Main tasks attributed in the project

- State of the art processing technology and valorization review for packaging applications (WP1)
- Utilization of polyphenols, caffeine, phytosterols and carotenoids as well as fibers to modify the properties of polymeric matrices will be investigated in order to improve polymer performances towards applications into films and capsules (WP3)
- Fraction selection for polymers and composites (WP4)
- Prototyping polymers and composites (WP5)
- Data for LCA and CBA (WP6)

Person involved	Gender
Dr. Patrizia Cinelli PhD in Chemistry at Pisa University. She is assistant professor of applied chemistry and Materials Technology at Pisa University. She has over 20 years of experience in materials science, polymer processing and characterization, biodegradation and life cycle assessment. She is co-author of over 50 papers in peer-reviewed journals, 9 book chapters, and 8 patents on innovative materials. She has worked in over 15 EU project from FP5, FP6, FP7 and Horizon 2020..	F
Prof. Andrea Lazzeri (DICI-UNIPI) Full Professor of Materials Science and Engineering at Pisa University since 1st January 2009. He is specialized in preparation and use of micro-and nano-particles in polymeric composites, rubber toughening of plastics, deformation and fracture of materials and in ceramic matrix composites. He is author of 40 publications on international journals and 4 patents. He coordinated three European projects on materials science in FP7.	M
Dr. Francesca Signori Researcher at Department of Civil and Industrial Engineering, Pisa University, PhD in Biomaterials at Trento University (Italy) in 2003. As PhD student, she performed part of the research (2001-2002) at the University of Twente, The Netherlands. In 2006 she received the European Biomaterials and Tissue Engineering Doctoral Award, by the European Society of Biomaterials. Her research topics focus on polymer chemistry and physics, mainly on the synthesis, the engineering and the structure-properties relationship assessment of biodegradable polymeric materials for biomedical applications. Moreover, she got interested in the use of biodegradable thermoplastic materials in the field of nanotechnology and for industrial applications. She is also active in the preparation and investigation of biodegradable bio-based polymer composites and blends. Her scientific activity is testified by co-authorship of 35 papers in peer-reviewed international journals.	F

Dr. Vito Gigante is a Chemical Engineer graduated at University of Pisa on 27 February 2015 with a thesis on 'Properties of mixed compounds and base poly (lactic acid) (PLA) / polycarbonate (PC) made from recycled polymers'. He got a PhD in Chemical Engineering cum laude, at Pisa University. He studies biodegradability, life cycle assessment and the processability and mechanical properties of sustainable multiphase polymer systems for structural applications	M

Relevant publications

- P. Cinelli, M. Schmid, E. Bugnicourt, J. Wildner, A. Bazzichi, I. Anguillesi, A. Lazzeri. "Whey protein layer applied on biodegradable packaging film to improve barrier properties while maintaining biodegradability", Polymer Degradation and Stability, 2014, 108, 151-157.
- P. Cinelli, M. Schmid, El. Bugnicourt, M.B. Coltelli, A. Lazzeri, "Recyclability of PET/WPI/PE Multilayer Films by Removal of Whey Protein Isolate based Coatings with Enzymatic Detergents" Materials, 2016, Vol. 9(6), 473; doi:10.3390/ma9060473.
- P. Cinelli, M. Seggiani, N. Mallegni, V. Gigante, A. Lazzeri "Processability and Degradability of PHA-Based Composites in Terrestrial Environments". International journal of molecular sciences 2019," DOI:10.3390/ijms20020284,
- M.Seggiani,; P.Cinelli, E. Balestri; N. Mallegni, E. Stefanelli, Eleonora; A. Rossi, C. Lardicci, Claudio; A. Lazzeri, "Novel sustainable composites based on poly(hydroxybutyrate-co-hydroxyvalerate) and sea grass beach-CAST fibres: performance and degradability in marine environments", Materials, 11(5), 772 ,2018, doi: 10.3390/ma11050772.
- P. Cinelli, M.B. Coltelli, N. Mallegni, P. Morganti, A. Lazzeri, "Degradability and sustainability of nanocomposites based on polylactic acid and nanochitin" Chemical Engineering Transaction 60, 2017, ISBN 978-88-95608-50-1, ISNN 2283-9216

Relevant Projects

- NANO-CHITOPACK: Sustainable technologies for the production of biodegradable materials based on natural chitin-nanofibrils derived by waste of fish industry, to produce food grade packaging (GA no:315233, FP7-SME, completed)
- DIBBIOPACK: Development of injection and blow extrusion molded biodegradable and multifunctional packages by nanotechnology: improvement of structural and barrier properties, smart features and sustainability (GA no: 280676, FP7-NMP-2011-LARGE-5, completed)
- OLI-PHA: A novel and efficient method for the production of polyhydroxyalkanoate polymer-based packaging from olive oil waste water. (GA no: 280604-2, Completed)*
- AGRIMAX: Agri & food waste valorisation co-ops based on flexible multi-feedstocks biorefinery processing technologies for new high added value applications (BBI Grant Agreement no: 720720, starting date 01.10.2016).
- PERFORMANCE KBBE-2012-6, GA n°: 312092 "Development of Personalised Food using Rapid Manufacturing for the Nutrition of elderly Consumers"

Significant Infrastructure and Equipment

- Minilab Haake mini extruder equipped with MiniLab Injection Moulding Press;
- COMAC co-rotating twin screw extruder EBC 25 HT semi-industrial extruder for producing granules equipped with a AMUT flat die extrusion unit for producing films and sheets;
- Teknomast Blow Mouling Extruder (25 mm);
- Charpy Impact Tester CEAST 9050; Differential Scanning Calorimeter TA Q200;
- DMTA Gabo Eplexor® 100N;
- MTS Universal Testing Machine Criterion model C 43 (equipped with a 10KN load cell).

4.2 Third parties involved in the project (including use of third party resources)

With the exceptions of FHNW and BBEPP (see tables below), all participants declare that they do not involve third parties.

FHNW

Does the participant plan to subcontract certain tasks (please note that core tasks of the project should not be sub-contracted)	Y
If yes, please describe and justify the tasks to be subcontracted **Synthesis of a few kg batch of the protected nanobiocatalyst by a contract manufacturing organisation according to the protocol developed by FHNW.** WP3 subtask 3.2.2. Immobilized and nano-protected proteases hydrolytic treatment of protein fractions. Subcontracting sum: 20,000 €	
Does the participant envisage that part of its work is performed by linked third parties[94]	N
If yes, please describe the third party, the link of the participant to the third party, and describe and justify the foreseen tasks to be performed by the third party	
Does the participant envisage the use of contributions in kind provided by third parties (Articles 11 and 12 of the General Model Grant Agreement)	N
If yes, please describe the third party and their contributions	

BBEPP

Does the participant plan to subcontract certain tasks (please note that core tasks of the project should not be sub-contracted)	Y
If yes, please describe and justify the tasks to be subcontracted **External spray drying of protein extracts.** This belongs to Task 5.1 (in WP5), for formulating the selected fractions according to the required specifications of the end-users, that will use these fraction in Task 5.2, 5.3 and 5.4. y Subcontracting sum: 30,000 €	
Does the participant envisage that part of its work is performed by linked third parties	N
If yes, please describe the third party, the link of the participant to the third party, and describe and justify the foreseen tasks to be performed by the third party	
Does the participant envisage the use of contributions in kind provided by third parties (Articles 11 and 12 of the General Model Grant Agreement)	N
If yes, please describe the third party and their contributions	

INNOVA

Does the participant plan to subcontract certain tasks (please note that core tasks of the project should not be sub-contracted)	N
If yes, please describe and justify the tasks to be subcontracted	
Does the participant envisage that part of its work is performed by linked third parties	N
If yes, please describe the third party, the link of the participant to the third party, and describe and justify the foreseen tasks to be performed by the third party The ASSOCIAZIONE NAZIONALE COOPERATIVE DI PRODUZIONE, LAVORO E SERVIZI (ANC-PLS) will act as linked third party to INNOVA COOP. There is a capital link between INNOVA COOP and ANC-PLS, as the latter is a member of INNOVA COOP with a 4 % capital share. Personnel of ANC-LPS will be pivotal for establishing the contact to a number of networks and cooperations within Italy and Europe, and to bring in know-how on their positions and demands for PROLIFIC output/products.	
Does the participant envisage the use of contributions in kind provided by third parties (Articles 11 and 12 of the General Model Grant Agreement)	N
If yes, please describe the third party and their contributions	

[94] A third party that is an affiliated entity or has a legal link to a participant implying a collaboration not limited to the action. (Article 14 of the Model Grant Agreement).

5 Ethics and Security

5.1 Ethics

5.1.1 Ethics issues checklist

Does your research involve animals? YES

- Page 47 (WP5, Task 5.5)
- Details on species and rationale for their use, numbers of animals to be used, nature of the experiments, procedures and techniques to be used:
  - Species: piglet (*Sus scrofa domesticus*) & chicken (*Gallus domesticus*)
  - For screening feed ingredients in WP5: 160 piglets, for collecting zootechnical and health parameters
  - For screening feed ingredients in WP5: 160 chickens, for collecting zootechnical and health parameters
- Justification of animal use (including the kind of animals to be used) and why alternatives cannot be used.
  - Justification of use of piglets and chickens: both are representative of animal livestock species in Europe. The efficiency and acceptance of new functional feed ingredients using novel protein sources necessitate animal feed trials.
- Documents to be provided: ethical self-assessment, trial protocol and permission by ethical committee. The trial protocols and permission by ethical committee will be planned 3 weeks before starting the trials.

Does your research involve research procedures that may cause pain, suffering, distress or lasting harm to live non-human vertebrate animals ? NO

5.1.2 <u>How to deal with the issue</u>

Animal trials have been designed in accordance with existing guidelines in the respective countries, and each study will be approved by legal, relevant ethical commissions for animal care and well-being. Proper actions will be taken to ensure that any *in vivo* test comply with the Helsinki declaration in its latest version as well as relevant EU legislations, such as EU Directive 2010/63/EU. In addition, partner Nutrition Sciences N.V. agrees that the storage of information about biological material in their data management system will be coded completely anonymously and the processing of data and biological material within the consortium will strictly follow the directive 95/46/EC of the European Parliament.

Nutrition Sciences N.V. will be involved in work using (farm) animals. All such work will as above be approved by local legal authorities and in accordance with the Amsterdam protocol on animal protection and welfare, great care will be taken to ensure minimal suffering of the animals. Therefore, application of the 3 R's will be considered: **R**eplacement, when *in vitro* models are available; **R**efinement, by optimal planning of the measurements within the animal trials and **R**eduction, by optimising the statistical approach. The animal trials will be always conducted according this 3xR approach. Since feeds are feed-grade (GMP+, HACCP) and interactions with the animals do not create pain, extensive stress, the animals are grown – in relatively normal daily conditions, also supplying them diets with xenobiotic concentration below legal norms.

The ethical, legal, social and safety issues raised by the subject and activities of the proposal will be continuously monitored by the steering committee of the PROLIFIC project and dealt with adequately in accordance with the existing national and international regulations. The coordinator of the consortium will be responsible for critical assessment of compliance to the ethical issues in the project. Moreover, dr. ir. Geert Bruggeman (leading the Nutrition Sciences work) has obtained a European FELASA D degree entitling him to conduct and supervise animal trials.

5.1.3 What do you need to provide?

At this stage, partner Nutrition Sciences performed an ethics self-assessment.

- See above for how the proposal meets the national legal and ethical requirements of the country or countries where the tasks raising ethical issues are to be carried out;
- Details on how we intend to address the issues in the ethical issues table, in particular as regards:
  - research objectives: study of vulnerable populations of piglets (*Sus scrofa domesticus)* and chickens *(Gallus domseticus)*
  - research methodology: standard piglet and poultry trial, providing diets in compliance with EU legislation.
  - the potential impact of the research: piglets and poultry can re-enter the food chain after the trials.

In the case of Prolific , 4 post grant ethics requirements were identified as follows:

Ethics Requirement 1: Humans

- The procedures and criteria that will be used to identify/recruit research participants must be submitted as a deliverable.
- The informed consent procedures that will be implemented for the participation of humans must be submitted as a deliverable.
- Templates of the informed consent/assent forms and information sheets (in language and terms intelligible to the participants) must be submitted as a deliverable.
- The applicant must clarify whether children and/or adults unable to give informed consent will be involved and, if so, justification for their participation must be included in the grant agreement before signature.
- In case children and/or adults unable to give informed consent are involved, details on how the consent of the legal representatives (and assent, when applicable) will be acquired must be included in the grant agreement before signature.
- The applicant must clarify whether vulnerable individuals/groups will be involved, and the measures to protect them and minimise the risk of their stigmatisation must be included in the grant agreement before signature. The applicant must clarify whether invasive physical procedures will be used and what measures will be taken to minimise possible pain, and include this information in the grant agreement before signature. Details on incidental findings policy must be included in the grant agreement before signature. Copies of opinions/approvals by ethics committees and/or competent authorities for the research with humans must be kept on file.

This ethics requirement will be addressed in deliverable D9.2 in month 6

Ethics Requirement 2: Human cells/tissues

- In case of use of human cells/tissues available commercially, details on cell/tissue types and provider must be kept on file.
- In case human cells/tissues are obtained within the project, details on cell/tissue types must be kept on file.
- In case human cells/tissues are obtained from another project, details on cell/tissue types must be kept on file.
- In case human cells/tissues are obtained from a biobank, details on the cell/tissue types and on the biobank and access to it must be kept on file.
- Copies of relevant documents for using, producing or collecting human cells or tissues (e.g., ethics approval, import licence, accreditation/designation/authorisation/licensing) must be kept on file.

This ethics requirement will be addressed in deliverable D9.4 in month 6

Ethics Requirement 3: Animals

- The beneficiary must confirm that relevant authorisations for animal experiments (covering also the work with genetically modified animals, if applicable) have been obtained, and kept on file.
- General information on the nature of the experiments and the procedures to ensure animal welfare and adherence to the Three Rs principle must be provided.
- The beneficiary must confirm that training certificates/personal licenses of the staff involved in animal experiments have been obtained and are kept on file.
- Number of months to fulfill the requirement after the project starts

This ethics requirement will be addressed in deliverable D9.3 in month 6

Ethics Requirement 4: Environment

- Further information about the possible harm to the environment caused by the research and the measures that will be taken to mitigate the risks must be provided
- The beneficiary must confirm that authorisations for relevant facilities (e.g. security classification of laboratory, GMO authorisation) have been obtained, and are kept on file.
- The beneficiary must confirm that appropriate health and safety procedures conforming to relevant local/national guidelines/legislation are followed for staff involved in this project.

This ethics requirement will be addressed in deliverable D9.1 in month 2

5.2 Security[95]

Not applicable

The PROLIFIC project will NOT perform or handle

- activities or results raising security issues,
- 'EU-classified information' as background or results

[95] See article 37 of the Model Grant Agreement

Confirmed members of the Stakeholder Advisory Board

EuropaBio
The European Association for Bioindustries

To: Prof. Dr. Habil Philippe F.-X- Corvini
University of Applied Sciences and Arts Northwestern Switzerland
School of Life Sciences

17 July 2017

LETTER OF INTENT

I, the undersigned, hereby confirm the commitment of Joanna Dupont-Inglis, Industrial Biotech Director at EuropaBio as an Advisory Board member to the PROLIFIC project, in case it is accepted for funding by the European Commission. The project proposal will be submitted under the call: "BBI 2017.R4 – PROTEINS AND OTHER BIOACTIVE INGREDIENTS FROM SIDE STREAMS AND RESIDUES", at the latest on the 7th September 2017.

The objective of PROLIFIC is to apply a range of processing technologies, to recover a significant amount of proteins/peptides and other added-value compounds (fibers, lipids, phenols) from industrial processing residues of legumes, fungi and coffee and to demonstrate and validate several high-value applications of the extracted fractions in the food, feed, packaging and cosmetic sectors.

As an Advisory Board member, I will be informed of the results obtained by PROLIFIC developments.
As an Advisory Board member, during the project lifetime, Joanna Dupont-Inglis will:

- Advise from industrial insights from biofuels stakeholders.
- Advise the consortium on the directions and future steps of the project to undertake for successful outcomes.
- Ensure that the developments foreseen throughout the project are in line with industrial requirements of biofuel refining industries and needs.

The PROLIFIC project, if successful, is scheduled to start in 2018 and will last for 4 years.

Joanna Dupont-Inglis is willing to take part to the PROLIFIC Advisory Board. This letter is not intended to be a legally binding document, but we assume that the scope and terms of co-operation may be later negotiated, defined and signed in a mutual agreement.

Yours sincerely,

.....
(signature)

Joanna Dupont-Inglis, Industrial Biotech Director, EuropaBio

.....
(stamp)

EuropaBio aisbl
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+32 2 735 03 11 | +32 2 735 49 60
www.europabio.org | VAT BE 0477 520 310

• COOPBOX GROUP S.P.A.
42021 BIBBIANO (RE) VIA VITTORIO VENETO N°1 A/B TEL.0522/251511 FAX 0522/251570
CAP. SOC. € 6.825.000 I.V. P.IVA. N. 01872130909 . C.F. REG. IMPRESE RE N.12300480154 REA RE 254773
SOCIETA' SOGGETTA ALLA' ATTIVITA' DI DIREZIONE E COORDINAMENTO DI CCPL S.C.

LETTER OF INTENT

I, the undersigned, hereby confirm the commitment of COOPBOX GROUP S.p.A. as an Advisory Board member to the PROLIFIC project, in case it is accepted for funding by the European Commission. The project proposal will be submitted under the call: "BBI 2017.R4 – PROTEINS AND OTHER BIOACTIVE INGREDIENTS FROM SIDE STREAMS AND RESIDUES",
at the latest on the 7th September 2017.

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As an Advisory Board member, I will be informed of the results obtained by PROLIFIC developments.

As an Advisory Board member, during the project lifetime, COOPBOX GROUP S.p.A. will:

- Advise from industrial insights from bio-based industries stakeholders.
- Advise the consortium on the directions and future steps of the project to undertake for successful outcomes.
- Ensure that the developments foreseen throughout the project are in line with industrial requirements of bio-based industries and needs.

The PROLIFIC project, if successful, is scheduled to start in 2018 and will last for 4 years.

COOPBOX GROUP S.p.A. is willing to take part to the PROLIFIC Advisory Board. This letter is not intended to be a legally binding document, but we assume that the scope and terms of co-operation may be later negotiated, defined and signed in a mutual agreement.

Yours sincerely,

Bibbiano, 08/08/2017

Dott. Claudio Fornaciari

R&D Project Manager

www.coopbox.com • info@coopbox.it

STABILIMENTI
STABILIMENTO DI BIBBIANO* • 42021 BIBBIANO (REGGIO EMILIA) • ITALY • VIA VENETO,1/A/B • TEL. 0522 251511 • FAX 0522 889007
STABILIMENTO DI FERRANDINA* • 75010 FERRANDINA (MATERA) • ITALY • LOCALITÀ BORGO MACCHIA • TEL. 0835 757163 • FAX 0835 757160

fruttagel®
scelta naturale

From:
FRUTTAGEL S.C.P.A.
VIA NULLO BALDINI 26
48011 ALFONSINE RA

To: INNOVACOOP
VIALE ALDO MORO 16
40127 BOLOGNA BO
info@innovacoop.eu

LETTER OF INTENT

I, the undersigned, hereby confirm the commitment of FRUTTAGEL S.C.p.A. as an Advisory Board member to the PROLIFIC project, in case it is accepted for funding by the European Commission. The project proposal will be submitted under the call: "BBI 2017.R4 – PROTEINS AND OTHER BIOACTIVE INGREDIENTS FROM SIDE STREAMS AND RESIDUES",
at the latest on the 7th September 2017.

The objective of PROLIFIC is to apply a range of processing technologies, to recover a significant amount of proteins/peptides and other added-value compounds (fibers, lipids, phenols) from industrial processing residues of legumes, fungi and coffee and to demonstrate and validate several high-value applications of the extracted fractions in the food, feed, packaging and cosmetic sectors.

As an Advisory Board member, I will be informed of the results obtained by PROLIFIC developments.

As an Advisory Board member, during the project lifetime, FRUTTAGEL S.C.p.A. will:

- Advise from industrial insights from bio-based industries stakeholders.
- Advise the consortium on the directions and future steps of the project to undertake for successful outcomes.
- Ensure that the developments foreseen throughout the project are in line with industrial requirements of bio-based industries and needs.

The PROLIFIC project, if successful, is scheduled to start in 2018 and will last for 4 years.

FRUTTAGEL S.C.p.A. is willing to take part to the PROLIFIC Advisory Board. This letter is not intended to be a legally binding document, but we assume that the scope and terms of co-operation may be later negotiated, defined and signed in a mutual agreement.

Yours sincerely,

22/08/17
(date)
Prot. m u 1702464

FRUTTAGEL S.C.p.A.
Il Presidente
Stanislao Giuseppe Fabbrino

(stamp)

FRUTTAGEL® Società Cooperativa Agricola per Azioni
Sede: Via Nullo Baldini, 26 - 48011 ALFONSINE (RA) - C.P. n. 136 - Tel. 0544 866511 - Fax Servizi Generali: 0544 83609 - Fax Uff. Ricezione Ordini: 0544 866557
C.F. e Iscr. Reg. Impr. RA 01529420356 - P. IVA 01271980391 - REA RA: N. 132485 - C. ISO: IT 01271980391 - Albo Soc. Coop.ve a mutualità prevalente N° A104475
E-mail: segreteria@fruttagel.it - sito internet: www.fruttagel.it

ETH
Eidgenössische Technische Hochschule Zürich
Swiss Federal Institute of Technology Zurich

Health Sciences and Technology D-HEST
Institute of Food, Nutrition and Health IFNH

ETH Zurich
Prof. Dr.-Ing. Alexander Mathys
Head of Sustainable Food Processing Laboratory
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CH-8092 Zurich, Switzerland

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alexander.mathys@hest.ethz.ch
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TO: Dipl.-Chem. János-István Petrusán MSc, Drs.
Head of Research and Product Management
Administration/R&D and PM Department
IGV Institut für Getreideverarbeitung GmbH
Arthur-Scheunert-Allee 40/41
14558 Nuthetal OT Bergholz-Rehbrücke

Zurich, 28 August 2017

LETTER OF INTENT

I, the undersigned, hereby confirm the commitment of the ETH Zurich Sustainable Food Processing Laboratory with me as an Advisory Board member to the PROLIFIC project, in case it is accepted for funding by the European Commission. The project proposal will be submitted under the call: "BBI 2017.R4 – PROTEINS AND OTHER BIOACTIVE INGREDIENTS FROM SIDE STREAMS AND RESIDUES",
at the latest on the 7th September 2017.

The objective of PROLIFIC is to apply a range of processing technologies, to recover a significant amount of proteins/peptides and other added-value compounds (fibers, lipids, phenols) from industrial processing residues of legumes, fungi and coffee and to demonstrate and validate several high-value applications of the extracted fractions in the food, feed, packaging and cosmetic sectors.

As an Advisory Board member, I will be informed of the results obtained by PROLIFIC developments.

As an Advisory Board member, during the project lifetime, the ETH Zurich Sustainable Food Processing Laboratory will:

- Advise from industrial insights from bio-based industries stakeholders.
- Advise the consortium on the directions and future steps of the project to undertake for successful outcomes.
- Ensure that the developments foreseen throughout the project are in line with industrial requirements of bio-based industries and needs.

ETH
Eidgenössische Technische Hochschule Zürich
Swiss Federal Institute of Technology Zurich

The PROLIFIC project, if successful, is scheduled to start in 2018 and will last for 4 years.

The ETH Zurich Sustainable Food Processing Laboratory is willing to take part to the PROLIFIC Advisory Board. This letter is not intended to be a legally binding document, but we assume that the scope and terms of co-operation may be later negotiated, defined and signed in a mutual agreement.

Should you need any additional information please feel free to contact me at +41 44 632 97 63 or by email at alexander.mathys@hest.ethz.ch anytime.

Yours sincerely,

Prof. Dr.-Ing. Alexander Mathys

ETH Zurich
Head of Sustainable Food Processing Laboratory
Institute of Food, Nutrition and Health IFNH
Department of Health Science & Technology D-HEST

davines S.p.A.

via Ravasini 9/A 43100 Parma – Italy
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601 West 26th street suite 1733
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T: +1 212 9242470 F: +1 212 924229

From:
DAVINES S.p.A
VIA Ravasini, 9a
43126 Parma

To:
Philippe Corvini, University of Applied Sciences and Arts Northwestern Switzerland
Annalisa Tassoni, University of Bologna

LETTER OF INTENT

I, the undersigned, hereby confirm the commitment of DAVINES S.p.A. as an Advisory Board member to the PROLIFIC project, in case it is accepted for funding by the European Commission. The project proposal will be submitted under the call: "BBI 2017.R4 – PROTEINS AND OTHER BIOACTIVE INGREDIENTS FROM SIDE STREAMS AND RESIDUES",
at the latest on the 7th September 2017.

The objective of PROLIFIC is to apply a range of processing technologies, to recover a significant amount of proteins/peptides and other added-value compounds (fibers, lipids, phenols) from industrial processing residues of legumes, fungi and coffee and to demonstrate and validate several high-value applications of the extracted fractions in the food, feed, packaging and cosmetic sectors.

As an Advisory Board member, I will be informed of the results obtained by PROLIFIC developments.

As an Advisory Board member, during the project lifetime, DAVINES S.p.A. will:

- Advise from industrial insights from biofuels stakeholders.
- Advise the consortium on the directions and future steps of the project to undertake for successful outcomes.
- Ensure that the developments foreseen throughout the project are in line with industrial requirements of biofuel refining industries and needs.

The PROLIFIC project, if successful, is scheduled to start in 2018 and will last for 4 years.

Codice fiscale e Partita IVA 00692360340 – R.E.A. PR 156502 – Capitale soc. € 2.500.000,00

davines S.p.A.
via Ravasini 9/A 43100 Parma – Italy
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New York NY 10001 - U.S.A.
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DAVINES S.p.A. is willing to take part to the PROLIFIC Advisory Board. This letter is not intended to be a legally binding document, but we assume that the scope and terms of co-operation may be later negotiated, defined and signed in a mutual agreement.

Yours sincerely,

..... 05/09/2017
(signature) (date)

Dr. Paolo Goi,
Technical Director
(name and position in the organisation)

.....
(stamp) Davines S.p.A. Cap. Soc. 2.500.000,00 € i.v. 43126 PARMA / ITALIA - Via Ravasini, 9/A Tel. +39 0521/965611 - Fax +39 0521/292597 C.F. - P.I. 00692360340

ASSICA
Associazione Industriali delle Carni e dei Salumi
aderente a Confindustria
www.assica.it

Milan, August 29th 2017

EXPRESSION OF INTEREST

We are writing to confirm that the ASSICA is willing to provide external support to the *BBI 2017.R4 - PRO-LiFiC -Integrated cascade PROcesses for the extraction of proteins and bioactive molecules from Legumes, Fungi and Coffee agro-industrial side streams*, coordinated by Dr. Philippe CORVINI (FHNW) with the scientific guidance of Dr. Annalisa TASSONI (UNIBO).

We agree with the aims and scope of the project, as specified in the proposal, and we believe that such an undertaken project will be beneficial to for the take-up of innovative sustainable technologies to recover proteins and bioactive ingredients from feedstock or to convert residual biomass and industry side streams into ingredients/additives for food, feed and other high-value markets.

If the project is funded, it will be a pleasure for us to get involved in the project activities carried out by SSICA, partner of the project, by testing meat products obtained, and by taking part in the dissemination activities aimed at acquiring new skills, tools and methodologies and to contribute to share a sustainable knowledge-based economy capable of deliver eco-innovation and consumer protection.

Yours sincerely,

ASSICA
Associazione Industriali delle Carni e dei Salumi
Milanofiori Strada 4 - Pal. Q8
20089 ROZZANO MI
C.F. 80074350150

Davide Calderone, ASSICA Director

***ASSICA - Associazione Industriali delle Carni e dei Salumi**, is the national trade association established within Confindustria in 1946 to represent companies involved in the production of Italian deli meats, pork slaughtering, and in the processing of other meat-based products (corned beef, fats and lard, etc.). ASSICA co-operates with the national and international public institutions to create a legal framework aimed at promoting business and sector's development. Headquarters are in Rozzano (Milan) and two other operative offices are located in Rome and Brussels.*

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DUMOULIN
expert in animal feed

From:
Dumoulin sa
Parc Industriel 18
B-5300 SEILLES

To:
Celabor scrl
Av. Du Parc 38
4650 CHAINEUX

LETTER OF INTENT

I, the undersigned, hereby confirm the commitment of Dumoulin sa as an Advisory Board member to the PROLIFIC project, in case it is accepted for funding by the European Commission. The project proposal will be submitted under the call: "BBI 2017.R4 – PROTEINS AND OTHER BIOACTIVE INGREDIENTS FROM SIDE STREAMS AND RESIDUES", at the latest on the 7th September 2017.

The objective of PROLIFIC is to apply a range of processing technologies, to recover a significant amount of proteins/peptides and other added-value compounds (fibers, lipids, phenols) from industrial processing residues of legumes, fungi and coffee and to demonstrate and validate several high-value applications of the extracted fractions in the food, feed, packaging and cosmetic sectors.

As an Advisory Board member, I will be informed of the results obtained by PROLIFIC developments.

As an Advisory Board member, during the project lifetime, Dumoulin sa will:

- Advise from industrial insights from bio-based industries stakeholders.
- Advise the consortium on the directions and future steps of the project to undertake for successful outcomes.
- Ensure that the developments foreseen throughout the project are in line with industrial requirements of bio-based industries and needs.

The PROLIFIC project, if successful, is scheduled to start in 2018 and will last for 4 years.

Dumoulin sa is willing to take part to the PROLIFIC Advisory Board. This letter is not intended to be a legally binding document, but we assume that the scope and terms of co-operation may be later negotiated, defined and signed in a mutual agreement.

Yours sincerely,

.....
(signature)

.....06/09/2017
(date)

Werner REUTER R&D Manager
.....
(name and position in the organisation)

.....
(stamp)

DUMOULIN SA/NV
PARC INDUSTRIEL B-5300 SEILLES

ANNEX 1. List of tentative members of PROLIFIC Stakeholder Advisory Board

Type	Company/ institution	Country	Type of activity	Contact person	Confirmed	Partner
Feedstocks producers/ suppliers (waste handlers)	Renelux Ltd.	Cyprus	Feedstock producer	Yiannis Daskalakis		TALOS
	RENEWI	Belgium	Waste handler	Didier Van Weyenberge		BBEPP
	Sopred soc. coop.	Italy	Production of alfaalfa	Italino Babini		INNOVA
	COAMS- Consorzio delle Organizzazioni di Agricoltori Moltiplicatori di Sementi	Italy	Seed multiplication	Edmo Tersi		INNOVA
	Agribologna soc. coop.	Italy	Fruit and vegetable production	Emiliano Cocci		INNOVA
	INALCA SpA	Italy	Large industry. Meat processing and meat food production	Giovanni Sorlini		UNIBO
Feedstock processors	Biogaz du Haut-Geer	Belgium	Side-streams of legumes valorised in biogas production, creation of an extraction Unit	Gaëtan de Seny, Gérant		CELAB
	AP-BIO- Portugal's Biotechnology Industry Organization	Portugal	Bioprocess Engineering	Sara Monteiro		UNIBO
	DECHEMA e.V.	Germany	Promotion of Research on Chemical Technology and Biotechnology	Jochen Michels		UNIBO
	Alfavita srl	Italy	Processors of alfalfa	Matteo Lamborghini		INNOVA
	Salute Sativa soc. coop.		Hemp processing	Ornella Palladino		INNOVA
	Cevico soc. coop.	Italy	Wine production			INNOVA
Food industry	Barilla Spa	Italy	Large industry. Cereal-based, pasta, breakfast items	Giancarlo Addario		UNIBO
	Indulleida S.A.	Spain	Large industry. Food and beverage based on fruits and vegetables	Miguel Angel Cubero		UNIBO IRIS
	Food Ingredient Technologies	Belgium	Functional food company	David Landau		CELAB
	Fruttagel Soc coop.	Italy	Production of fruit and vegetable enriched juices	Stanislao Fabbrino	Yes	INNOVA
	De.Co Industrie soc. coop.	Italy	Bakery production	Vito di Chiaro		INNOVA
	Granarolo Spa	Italy	Food production (dairy, vegetable juices and vegan products, pasta)	Vittorio Zambrini		INNOVA UNIBO

Type	Company/ institution	Country	Type of activity	Contact person	Confirmed	Partner
	Salumificio Golfera in Lavezzola S.p.A.	Italy	Minced meat and dry-meat food products	Emanuela Zavaglia		SSICA
	Perdomini-IOC S.p.A.	Italy	Produces and markets aids for food, biotechnology for the oenological industry, filtration and separation products for all stages of the oenological and food chain.	Mirco Costalunga		SSICA
Cosmetic industry	Davines Spa	Italy	Csometic manufacturer	Paolo Goi	Yes	UNIPR
	Henkel AG	Germany	Cosmetics formulas (licensing)	Rolf Bayersdörfer		IGV
	BioExcellia Ltd.	Cyprus	Cosmetics manufacturer	Niki Christoforou		TALOS
	Eco Cosmetics Ltd.	Cyprus	Cosmetics manufacturer	Savvas Basilagas		TALOS
	Labiocos	Switzerland	Cosmetic producer	Malvin Richard		IRIS
	Acos Visoderm	Belgium	Natural cosmetic company	Geneviève Vidrequin		CELAB
	CO.Ind soc. coop.	Italy	Production of cosmetics and coffee	Susanna Tarozzi		INNOVA
Packaging industry	Novamont Spa	Italy	Bio-based packaging large industry	Giulia Gregori		UNIBO
	Roquette	France	Large Industry. Packaging	Camille Burel		IRIS
	Coopbox Spa	Italy	Production of food packaging	Claudio Fornaciari	Yes	INNOVA
	Alcas Spa	Italy	Food packaging producer	Diodati Elena		FTEN
	Brandani Gift Group	Italy	Appliances producer	Enrico Brandani		FTEN
	Tosti Srl	Italy	Plastic parts manufacturer	Ilaria Tosti		FTEN
	Fass Spa	Italy	Plastic parts manufacturer	Matteo Niccolai		FTEN
	Planet Bioplastics Srl	Italy	Bioplastic manufacturer	Andrea Lazzeri		FTEN
Feed industry	Dumoulin	Belgium	Animal feed producer	Werner Reuter	Yes	CELAB
	Agromedica Ltd.	Cyprus	Feed manufacturer	Demetris Lapathiotis		TALOS
	Panchris Group	Cyprus	Feed manufacturer	Demetris Sakkas		TALOS
	Vitatrace Nutrition Ltd.	Cyprus	Feed manufacturer	Christodoulos Angastiniotis		TALOS
	Champion Pet Food Ltd.	Cyprus	Feed manufacturer	George Tsappis		TALOS
	Dumoulin	Belgium	Animal nutrition company	Werner Reuter		CELAB

Type	Company/ institution	Country	Type of activity	Contact person	Confirmed	Partner
	Progeo Soc. Coop.	Italy	Feed Industry	Marco Pirani		INNOVA
	Spezialfutter Neuruppin	Germany	Special feed formulations (licensing)	Nils Osnowski		IGV
Possible end users of PROLIFIC fractions	Novadelta	Portugal	Coffee Production, Interest in Packaging Applications			UNIBO
	BioActor BV	Netherlands	Functional Food company	Sam Possemier		BBEPP
	Synaco	Belgium	Polyphenols incorporated in Functional food	Pascale Azzam		CELAB
	Extense-Pharma	Belgium	Plant proteins incorporated in functional food	Christelle Willaert		CELAB
	Conpalux	Belgium	Full service subcontracting for cosmetic and dermo-cosmetic products	Jean-Pierre Gribomont		CELAB
	Aethra Food Ingredients	Italy	Commercialisation of food ingredients and additives	Monica Saccani		INNOVA
	AM Nutrition	Norway	Supplier of pea products for food, feed, pet food and technical applications	Janne L. Arsvoll		INNOVA
	Ernteband Fruchtsaft GmbH, Winnenden	Germany	Protein smoothies	Knud Schulz		IGV
	Uckerkaas GmbH	Germany	Fortified dairy products	Jürgen Schöber		IGV
	Kampffmeyer Mühlen GmbH	Germany	Cereal based product fortification	Franz Engelke		IGV
Associations of bio-based industries, Clusters,	Asso BioPlastiche	Italy	Bioplastics	David Robert Newman		UNIBO
	ASSICA, Associazione Industriali delle Carni e dei salumi	Italy	Meat food association	Davide Calderone	Yes	SSICA
	ETP-SusChem	Belgium	Chemistry, Biotechnology and Chemical Engineering	Gernot Klotz		UNIBO
	SusChem-IT	Italy	Chemistry, Biopolymers and Biotechnology	Luigi Busetto		UNIBO
	CLIB 2021	Germany	cluster			IRIS UNIBO
	BIOVALE	United Kingdom	cluster	Margaret Smallwood		IRIS
	Europe's Bioeconomy Intercluster	Germany	cluster	Henning Mertens		IRIS
	French *Bioeconomy Cluster*	France	cluster	Dumange Boris		IRIS
	BioEconomy Cluster	Germany	cluster	Horst Mosler		IRIS
	Vlaanderen circulair	Belgium	Circular economy	Dimitri Strybos		BBEPP

Type	Company/ institution	Country	Type of activity	Contact person	Confirmed	Partner
	Flanders BioBased Valey	Belgium	Biobased industry	Stefan Ruyters		BBEPP
	EuropaBio	Belgium	EuropaBio represents 80 corporate and associate members and bio regions, and 17 national biotechnology associations in turn representing over 1800 biotech SMEs	Joanna Dupont-Inglis	Yes	FHNW
	Wagralim	Belgium	Walloon competitiveness cluster in agro-food industry	Sophie Bourez		CELAB
	German Institute for Food Technology	Germany	Food chain stakeholders	Sergiy Smetana		IGV
Federations Foundations	European Polymer Federation	France	RTD Promotion in polymeric field	*Jean-François Gerard*		UNIBO
	Stichting Zero Waste Europe (ZWE)	Netherlands	Foundation gathering 20 organizations and around 300 municipalities committed to work to eliminate waste in Europe.	Enzo Favoino		UNIBO
	Fundacion Centro Tecnologico de Acuicultura de Andalucia	Spain	A non-profit, private foundation, which aims at developing applied research, technology, innovation, sustainability and training within the aquaculture sector	Rocio Robles		UNIBO
	European Feed Manufacturers' Federation (FEFAC)	Europe	Federations of associations from EU Member States and of an increasing number of observer members from non-EU countries. FEFAC is the only independent spokesman of the European Compound Feed Industry at the level of the European Institutions			NS
Policy makers	Nova Institute	Germany	SME. Techno-economic and environmental evaluation, market research, dissemination, project management and policy for a sustainable bio-based economy.	Michael Carus		IRIS
Technology Transfer and innovation professionals	Cy.R.I.C. Cyprus Research and Innovation Center	Cyprus	Research Center	Panayiotis Philimis		TALOS
	Campden BRI Magyarország Nonprofit Kft (CBHU)	Hungary	Independent industrial, non-profit food research and development organisation providing R+D, knowledge transfer and training services to a large number of food businesses.	András Sebők		UNIBO
	Bridge2Food	The Netherlands	Food industry	Gerard Klein Essink		BBEPP
	ETH Zurich Sustainable Food Processing Laboratory	Switzerland	Developing technology for food processing	Alexander Mathys	Yes	IGV
	Entex Rust& Mitschke GmbH	Germany	Extrusion processing technology optimizations	Thomas Malzahn		IGV

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Investors	Cyprus Business Angels Network	Cyprus	Investors	Andreas Koupparis		TALOS
	European Investment fund	EU	Investors			IRIS
	European investment bank	EU	Investors			IRIS
	Vanguard Initiative	EU	Investors	Filippo Martinelli		IRIS
Consumer associations	Cyprus Consumers Association	Cyprus	Association	George Stylianou		TALOS
	European Food Information Council (EUFIC)	Belgium	Non-profit organization which provides science-based information on food safety and quality and health and nutrition to the media, health and nutrition professionals, educators and opinion leaders, in a way that promotes consumer understanding.	Laura Fernández Celemín		TALOS
	Coop Italia	Italy	Consumers' cooperative operating in the retail sector	Chiara Faenza		INNOVA
Cooperative associations	Legacoop Agroalimentare	Italy	Association of agri-food cooperatives	Cristian Maretti		INNOVA
	ANCPL Legacoop	Italy	Association of industrial cooperatives	Maurizio de Santis		INNOVA
	COGECA	Belgium	General Confederation of Agricultural Cooperatives in the European Union	Leonardo Pofferi		INNOVA
	CECOP-CICOPA EUROPE	Belgium	The European confederation of industrial and service cooperatives	Elisa Terrasi		INNOVA
	Federcanapa	Italy	Industrial hemp association	Beppe Croce		INNOVA
	Federation of Agricultural Cooperatives of Catalonia	Spain	Association of agri-food cooperatives	Marius Simon		IRIS
	International Research Association of Feed Technology	Germany, Europe	Dissemination activities among members Europe wide	Rolf-Michael Blume		IGV
Association and initiatives related to sustainability assessment	Zero Waste Europe	Europe	Network	Different per country		FHNW
	Ecoinvent	Switzerland	Promote good practice	Gregor Wernet		FHNW
	European Platform on Life Cycle Assessment	Europe	Platform, Knowledge-base	EU JRC		FHNW
	Life Cycle Initiative	France	Advisory, stewardship, knowledge consensus	Feng Wang		FHNW